ABC PHARM LP
CLINICAL STUDY PROTOCOL

Protocol Title:	An Open-label, Phase 1, Multicenter Study to Evaluate the Safety and Preliminary Anti-tumor activity of ABC-01 in Human Leukocyte Antigen -A*02:01-Positive Adult Subjects with Unresectable, Advanced and/or Metastatic Solid Tumors That Are Positive for the *TP53* XYZ Mutation
Protocol Number:	ABC-01-001
ABC Pharm LP Investigational Product:	ABC-01
IND Number:	29294
NCT Trial Identifier	NCT0000000
Sponsor:	ABC Pharm LP 1800 Undefined Pike USA
Protocol Version/Date:	Amendment 5 11 Dec 2024

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LIST OF ABBREVIATIONS

Abbreviation	Definition
ADA	anti-drug antibodies
AE	adverse event
AFP	alpha fetoprotein
ALC	absolute lymphocyte count
ALCOA+C	attributable, legible, contemporaneous, original, accurate, complete
ALT	alanine transaminase
ANC	absolute neutrophil count
aPTT	activated partial thromboplastin time
AST	aspartate aminotransferase
ASTCT	American Society for Transplantation and Cellular Therapy
ATG	antithymocyte globulin
BiPAP	bilevel positive airway pressure
BOIN	Bayesian Optimal Interval
BOR	best objective response
BUN	blood urea nitrogen
CA 15-3	Cancer antigen 15-3
CA 19-9	Carbohydrate antigen 19-9
CA 27-29	Cancer antigen 27-29
CA-125	Carbohydrate antigen 125
CAR	chimeric antigen receptor
CBC	complete blood count
CBR	clinical benefit rate
CEA	Carcinoembryonic antigen
CNS	central nervous system
CPAP	continuous positive airway pressure
CR	complete response
CrCl	creatinine clearance
CRISPR	clustered regularly interspaced short palindromic repeats
CRP	C-reactive protein
CRS	cytokine release syndrome
CSF	cerebral spinal fluid
CT	computed tomography
CTA	clinical trial assay

Abbreviation	Definition
CTCAE	Common Terminology Criteria for Adverse Events
cfDNA	cell-free DNA
DL1	dose level 1
DL2	dose level 2
DL3	dose level 3
DLT	dose-limiting toxicity
DOR	duration of response
ECG	electrocardiogram
ECHO	echocardiogram
ECOG	Eastern Cooperative Oncology Group
eCRF	electronic Case Report Form
EDC	electronic data capture
EEG	electroencephalogram
EOS	end of study
eTCR	engineered T cell receptor
EU	European Union
F1CDx	FoundationOne® CDx
FDA	U.S. Food and Drug Administration
FDG-PET	fluorodeoxyglucose-positron emission tomography
FFPE	formalin-fixed paraffin-embedded
GCP	Good Clinical Practice
G-CSF	granulocyte-colony stimulating factor
HLA	human leukocyte antigen
HNSCC	head and neck squamous cell carcinoma
IB	Investigator's Brochure
ICANS	immune effector cell-associated neurotoxicity syndrome
ICE	immune effector cell-associated encephalopathy
ICF	informed consent form
ICH	International Conference on Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
ICP	intracranial pressure
IEC	Independent Ethics Committee
IL	interleukin
IND	US Investigational New Drug

Abbreviation	Definition
INR	international normalized ratio
IP	Investigational Product
IPM	Investigational Product Manual
IRB	Institutional Review Board
IUD	intrauterine device
IV	intravenous
IVF	intravenous fluid
IVIG	immunoglobulin
KO	knock out
LDH	lactate dehydrogenase
MMP	medical monitoring plan
MRI	magnetic resonance imaging
MTD	maximum tolerated dose
NCI	National Cancer Institute
NSCLC	non-small cell lung cancer
ORR	objective response rate
OS	overall survival
PBMC	peripheral blood mononuclear cell
PCR	polymerase chain reaction
PFS	progression free survival
PD	pharmacodynamic
PK	pharmacokinetic
PO	oral
PR	partial response
PSA	Prostate specific antigen
PT	prothrombin time
QTc	corrected QT
RECIST	Response Evaluation Criteria in Solid Tumors
rIL-2	recombinant interleukin-2
RP2D	recommended Phase 2 dose
RT-PCR	real-time reverse transcription-polymerase chain reaction assay
SAE	serious adverse event
SCT	stem cell transplant
SRT	Safety Review Team

Abbreviation	Definition
SC	Subcutaneous
SD	Stable Disease
SOA	Schedule of Assessments
SOC	standard of care
SSI	site-specific integration
SSO	sequence-specific oligonucleotide
SUSAR	suspected unexpected serious adverse reaction
TCR	T cell receptor
TEAE	treatment-emergent adverse event
TGFBR2	transforming growth factor beta receptor 2
TGF-β	transforming growth factor-beta
TTR	time to response
UK	United Kingdom
US	United States
WBC	white blood cell

1. SYNOPSIS

Sponsor: ABC Pharm LP	**Protocol number:** ABC-01-001 Amendment 5
Name of Investigational Product: ABC-01	**Phase:** 1

Title: An Open-label, Phase 1, Multicenter Study to Evaluate the Safety and Preliminary Anti-tumor activity of ABC-01 in Human Leukocyte Antigen-A*02:01-Positive Adult Subjects with Unresectable, Advanced and/or Metastatic Solid Tumors That Are Positive for the *TP53* XYZ Mutation

Short Title: A Study to Investigate Treatment with ABC-01 in Adult Subjects with Unresectable, Advanced, and/or Metastatic Solid Tumors That Are Positive for HLA-A*02:01 and the *TP53* XYZ Mutation

Investigational Product: ABC-01 is an autologous T cell therapy product genetically engineered *ex vivo* to express an HLA-A*02:01-restricted T cell receptor (TCR), targeting *TP53* XYZ mutant tumors, and to knock out the endogenous human TCR and the gene encoding transforming growth factor beta receptor 2 (TGFBR2).

ABC-01 will be administered via intravenous (IV) infusion at dose levels up to 10×10^9 engineered (e)TCR+ T cells.

Number of Study Centers: Part 1: up to 25 sites in the United States and Canada; Part 2: up to 80 sites in the United States, Canada, UK, EU, and Australia.

Number of Subjects: Up to approximately 162 subjects may be enrolled and treated in the study as detailed below.

Part 1 (Dose Escalation and Disease Histology Evaluation): Up to approximately 64 subjects.

- **Dose Escalation**: the aim of Dose Escalation is to determine the maximum tolerated dose (MTD); up to 24 subjects may be enrolled and treated.
  - A minimum of 3 subjects will be evaluated for each dose escalation decision;
  - If a dose limiting toxicity (DLT) occurs in the first 3 subjects at any dose level, a minimum of 6 subjects will be evaluated for the dose escalation decision at that dose level;
  - MTD will be determined according to Bayesian Optimal Interval (BOIN) methodology (Section 10).
- **Disease Histology Evaluation:** the aim of the Disease Histology Evaluation portion of the study is to further evaluate safety and preliminary anti-tumor activity in disease specific histologies and determine the recommended phase 2 dose (RP2D); up to 40 subjects may be enrolled and treated.
  - This portion of the study starts once an MTD has been identified by the Safety Review Team (SRT). Additional subjects may be enrolled and treated at the MTD or a lower dose.

In Part 1, when a specific disease histology accumulates 6 subjects treated at the same dose level, the SRT will convene to assess safety and preliminary anti-tumor activity within that histology at that dose level and may make a recommendation on whether or not to dose additional subjects in Part 2.

Part 2 (Disease Cohort Expansion): Up to approximately 98 subjects (comprising of up to 7 cohorts of up to 14 subjects each).

The aim of the Disease Cohort Expansion portion of the study is to further evaluate safety and preliminary anti-tumor activity in disease specific histologies at the RP2D. Part 2 will enroll subjects in up to 7 disease-specific cohorts to further evaluate the preliminary anti-tumor activity and safety of ABC-01 in disease specific settings at the RP2D.

In total, up to approximately 20 subjects may be enrolled and treated from Part 1 and Part 2 combined for each disease specific setting.

Rationale:

Despite recent increase in anti-cancer treatment approvals, patients with recurrent and/or metastatic solid tumors continue to represent a high unmet-need population with very limited effective treatments available. *TP53* is the most commonly mutated gene across different cancer types, and shared neoantigens resulting from *TP53* mutations are broadly expressed across tumors and individuals. To date, no approved therapies targeting *TP53* mutated cancers are available. ABC-01, an autologous engineered T cell product expressing a TCR specific for the *TP53* XYZ mutation, offers the potential to treat a sizable number of patients across different histologies. T cells in ABC-01 incorporate a clustered regularly interspaced short palindromic repeat (CRISPR)-Cas9-mediated genetic knock out of the TCR alpha and beta genes, to prevent endogenous TCR and the TGFBR2 gene expression, rendering them resistant to immunosuppressive effects of transforming growth factor-beta (TGF-β) in the tumor microenvironment.

Target Population:

The study will enroll subjects with unresectable, advanced and/or metastatic cancer, including non--small cell lung cancer (NSCLC), colorectal adenocarcinoma, head and neck squamous cell carcinoma (HNSCC), pancreatic adenocarcinoma, breast cancer, ovarian cancer, or any other solid tumor histologies that are positive for the *TP53* XYZ mutation.

Subjects must present progressive disease, stable disease, or lack of clinical benefit from previous treatment at the time of study treatment.

Refer to the full study inclusion and exclusion criteria in Section5.3.

Key Inclusion Criteria

- At least 18 years of age at the time of signing the informed consent.
- Diagnosed with one of the histologies below:
  - NSCLC
  - Colorectal adenocarcinoma
  - HNSCC
  - Pancreatic adenocarcinoma
  - Breast cancer
  - Ovarian cancer
  - Any solid tumor histology positive for *TP53* XYZmutation will also be considered.
- Subject has advanced solid cancer, defined as unresectable, advanced, and/or metastatic disease (Stage III or IV) after at least 1 line of systemic standard of care (SOC) treatment regimen and for which there are no available curative treatment options.
- Tumors must harbor a TP53 XYZvariant mutation, and subject must be HLA-A*02:01 positive (at least 1 allele) confirmed by a CLIA-accredited laboratory-based assay described in Appendix F.
- Subject has at least 1 measurable lesion per computed tomography (CT) scan or magnetic resonance imaging (MRI) per Response Evaluation Criteria in Solid Tumors (RECIST) v1.1.

Study Design:

This is a Phase 1, open-label, multicenter study to evaluate the safety and preliminary anti-tumor activity of ABC-01 in HLA-A*02:01 subjects with unresectable, advanced, and/or metastatic NSCLC,

colorectal adenocarcinoma, HNSCC, pancreatic adenocarcinoma, breast cancer, ovarian cancer or any other solid tumor histologies that are positive for the *TP53* XYZmutation.

The study is divided into two parts, Part 1 (Dose Escalation and Disease Histology Evaluation) followed by Part 2 (Disease Cohort Expansion).

- **Part 1 Dose Escalation** will investigate escalating doses of ABC-01 in adult subjects with eligible solid tumor histologies to evaluate safety and determine MTD using the BOIN methodology.
- **Part 1 Disease Histology Evaluation** will further evaluate the safety and preliminary anti-tumor activity at or below the MTD in disease specific histologies and will determine the RP2D.

In Part 1 (Dose Escalation and Disease Histology Evaluation), when any specific disease histology accumulates 6 subjects treated at the same dose level, the SRT will convene to assess safety and preliminary anti-tumor activity within that histology at that dose level and may make a recommendation on dosing additional subjects at RP2D in Part 2 (Disease Cohort Expansion).

- **Part 2 Disease Cohort Expansion** will further evaluate the preliminary anti-tumor activity and safety of ABC-01 at the RP2D in disease specific settings.

Part 2 Disease Cohort Expansion may enroll specific histologies concurrently with Part 1 Disease Histology Evolution after the MTD is determined.

Subjects in the **Disease Histology Evaluation** and **Disease Cohort Expansion** portions ofthe study will be enrolled in up to seven disease-specific cohorts. Requirements for prior lines of therapy in each cohort are specified in the inclusion criteria Section5.3.1.

Part 1 (Dose Escalation and Disease Histology Evaluation)

In Part 1, up to a maximum of 64 subjects may be enrolled and treated.

In Dose Escalation, the number of subjects enrolled and treated per dose level will be determined by the incidence of DLTs observed and will depend on the number of dose levels that are tested. The dose levels of ABC-01 to be evaluated are presented inFigure 1. DLTs are defined in Section9.6.

Up to 24 subjects will contribute to the MTD determination.

A DLT evaluation period of 28 days (Day 28) after ABC-01 infusion (Day 0) will be used. Only DLTs that occur during the DLT evaluation for each DLT-evaluable subject will be used for the DLT rate calculation. The “sentinel” subject approach will be followed in the Dose Escalation portion of the study so that the first subject at a given dose level will complete the DLT assessment period (28 days post infusion) prior to treatment of the second subject. Treatment of the third subject will be staggered by at least 7 days.

During Dose Escalation, a model-assisted BOIN design will be used to find the MTD of ABC-01 with 3 subjects assigned as a cohort and treated at the specific dose level; then, depending on the DLT rate, another 3 subjects may be assigned. The MTD will be evaluated and determined after one of the dose levels accumulates 9 subjects who are enrolled, treated, and had the opportunity to be followed for the DLT period without an escalation/de-escalation. The target toxicity rate for the MTD is 30%, and the maximum sample size is up to 24 subjects enrolled, treated, and had the opportunity to be followed for the DLT evaluation period.

The BOIN design uses the following rule, optimized to minimize the probability of incorrect dose assignment, to guide dose escalation/de-escalation:

- If the observed DLT rate at the current dose is ≤ 23.6%, escalate the dose to the next higher dose level;
- If the observed DLT rate at the current dose is > 35.9%, de-escalate the dose to the next lower dose level;
- Otherwise, stay at the current dose.

Part 1 Disease Histology Evaluation may enroll up to 40 subjects with a starting dose at or below the MTD to further assess safety and evaluate the RP2D. Once 6 subjects have been dosed in one dose level and in a specific histology, the data will be reviewed by the SRT to evaluate the risk-benefit and make recommendations for expanding disease specific cohorts in Part 2.

With up to approximately 64 subjects treated in Part 1, there is at least 95% probability to observe any adverse events with more than 5% incidence rate.

Refer to Section9.8.2for stopping rules for Disease Histology Evaluation (Part 1).

The RP2D will be assessed based on the totality of data collected in Part 1, including safety, anti-tumor activity, pharmacokinetics (PK), pharmacodynamics, and biomarker data.

Part 2 (Disease Cohort Expansion)

Part 2 may begin after approval from the SRT and upon determination of RP2D to evaluate the preliminary anti-tumor activity, and further evaluate the safety of ABC-01 at the RP2D for each of the Disease Expansion Cohorts.

With up to approximately 20 subjects total in each cohort, the margin of error for a 90% confidence interval is less than 20%.

Refer to Section9.8.2for stopping rules for Disease Cohort Expansion (Part 2).

Study Periods:

The study will be conducted in 6 periods: the Prescreening Period, Screening Period, Enrollment/Leukapheresis Period, Lymphodepletion Period, Treatment Period, and Follow-up Period (which includes Post-treatment, Extended, and Long-Term Follow-up Periods).

Prescreening Period

Documented evidence of both *TP53* XYZmutation and HLA-A*02:01 positivity is required to enter the Screening Period.

Evidence of *TP53* XYZ and/or HLA-A*02:01 can be demonstrated by historically available data, through prior tests performed using any of the CLIA-accredited laboratory-based assays described in Appendix F,or by central testing.

Subjects required to complete the Prescreening Visit (ie, subject is not known to be positive for *TP53* XYZmutation and/or HLA-A*02:01 by the assays defined in Appendix F) will sign a Prescreening ICF, and the following tests below will be performed, as applicable:

- Central testing for the *TP53* XYZmutation will be based on a Food and Drug Administration (FDA)-approved assay for the detection of genetic mutations (FoundationOne® CDx [F1CDx] solid tumor assay) performed on tumor tissue material sent to a specialty laboratory.

And/or

- Central testing for HLA-A*02:01 will be conducted on blood sample(s) collected and sent to a central and/or specialty laboratory for testing.

Refer to Section 8.1for further details.

Screening Period

The Screening Period starts when the subject signs the Main informed consent form (ICF) to participate in the study. During the Screening visit, procedures are performed to assess eligibility for study enrollment.

Once the eligibility is confirmed by the Investigator and reviewed by the Sponsor Medical Monitor as described in the Sponsor Medical Monitoring Plan (MMP), the subject can be enrolled and proceed to leukapheresis.

If, at any time prior to enrollment, the subject fails to meet the eligibility criteria, the subject will be considered a screen failure.

Refer to Section8.2 for further details.

Enrollment/Leukapheresis

Subjects are considered enrolled once leukapheresis commences.

Refer to Section8.3 for further details.

Lymphodepletion Period

Lymphodepleting chemotherapy will be administered from Day -7 to Day -3 from day of ABC-01 treatment (Day 0).

Refer to Section8.4 for further details.

Treatment Period

The Treatment period begins on the date of ABC-01 infusion (Day 0). During the Treatment period, the subject will be hospitalized for a minimum of 8 days (Day 0 to Day 7). Recombinant interleukin-2 (rIL-2) will be administered subcutaneously (SC) at 300,000 IU/kg once daily.

Refer to Section8.5 for further details.

Follow-up Period

Post-treatment Follow-up period will begin on Day 14 and continue to the Day 84 visit.

The Extended Follow-up period will begin after Day 84 with the first study visit at Week 18 and will continue to the Month 24 visit.

The Long-Term Follow-up period will begin after Month 24 with the first study visit at Month 30 and will continue to the Month 180 visit.

Refer to Section8.6 for further details.

Length of Participation:

The duration of the study will vary depending on the subject's response to treatment and survival. For subjects who complete the entire protocol from date of initial informed consent through the completion of the Long-Term Follow-up period, the duration will be approximately 15 years.

Duration of treatment is 8 days, starting on Day 0, when subjects receive ABC-01 and first dose of rIL-2. Each subject will remain in the hospital for a minimum of 8 days (Day 0 to Day 7).

Objectives	**Endpoints**
Primary	
Part 1 (Dose Escalation and Disease Histology Evaluation)	
• Evaluate the safety of ABC-01 in subjects with unresectable, advanced, and/or metastatic solid tumors • Evaluate MTD and RP2D(s)	• Incidence of DLTs, treatment-emergent adverse events (TEAEs), and serious adverse events (SAEs)
Part 2 (Disease Cohort Expansion)	
• Evaluate the preliminary anti-tumor activity of ABC-01 at the RP2D in subjects with unresectable, advanced, and/or metastatic solid tumors	• Objective response rate (ORR) per RECIST V1.1 determined by Investigator assessment • Best objective response (BOR) • Duration of response (DOR)

	• Clinical benefit rate (CBR) (complete response [CR], partial response [PR], stable disease [SD]) • Time to response (TTR) • Progression-free survival (PFS) • Overall survival (OS)
• Further evaluate the safety of ABC-01 at the RP2D in subjects with unresectable, advanced, and/or metastatic solid tumors	• Treatment-emergent adverse events, and serious adverse events
Secondary	
Part 1 (Dose Escalation/Disease Histology Evaluation)	
• Evaluate preliminary anti-tumor activity of ABC-01 in subjects with unresectable, advanced, and/or metastatic solid tumors	• ORR per RECIST V1.1 determined by Investigator assessment • BOR • DOR • CBR (CR, PR, SD) • TTR • PFS • OS
Exploratory Objectives (Part 1 and Part 2)	
• Evaluate PK and pharmacodynamics • Evaluate parameters associated with blood and/or tumor tissue and association with clinical outcomes • Evaluate the manufacturing feasibility of ABC-01	• Expansion, persistence, migration, phenotype, and functionality of the engineered T cells in peripheral blood and/or tumor • Investigation offactors and/or biomarkers in blood and/or tumor tissue associated with ABC-01 mechanism of action and possible correlation with clinical responses, toxicity, and/or changes in tumor and/or neoantigen expression • Manufacturing success rates
Safety Review Team An SRT will be appointed for this study. • In Part 1 Dose Escalation, the SRT will meet after each cohort of 3 subjects are treated and followed for 28 days, or at a decision point based on the BOIN design to assess DLTs and make recommendations on the conduct of the study based on the totality of the data. • In the Part 1 Disease Histology Evaluation portion of the study, the SRT will convene approximately every 4 months and/or every time 6 subjects have been treated and followed for tumor response in a single histology, whichever occurs first.	

- In Part 2 (Disease Cohort Expansion), the SRT will continue to meet every 4 months to review the accruing safety and anti-tumor activity data.

Ad hoc SRT meetings may be organized throughout the study based on emerging data or upon request of any SRT member.

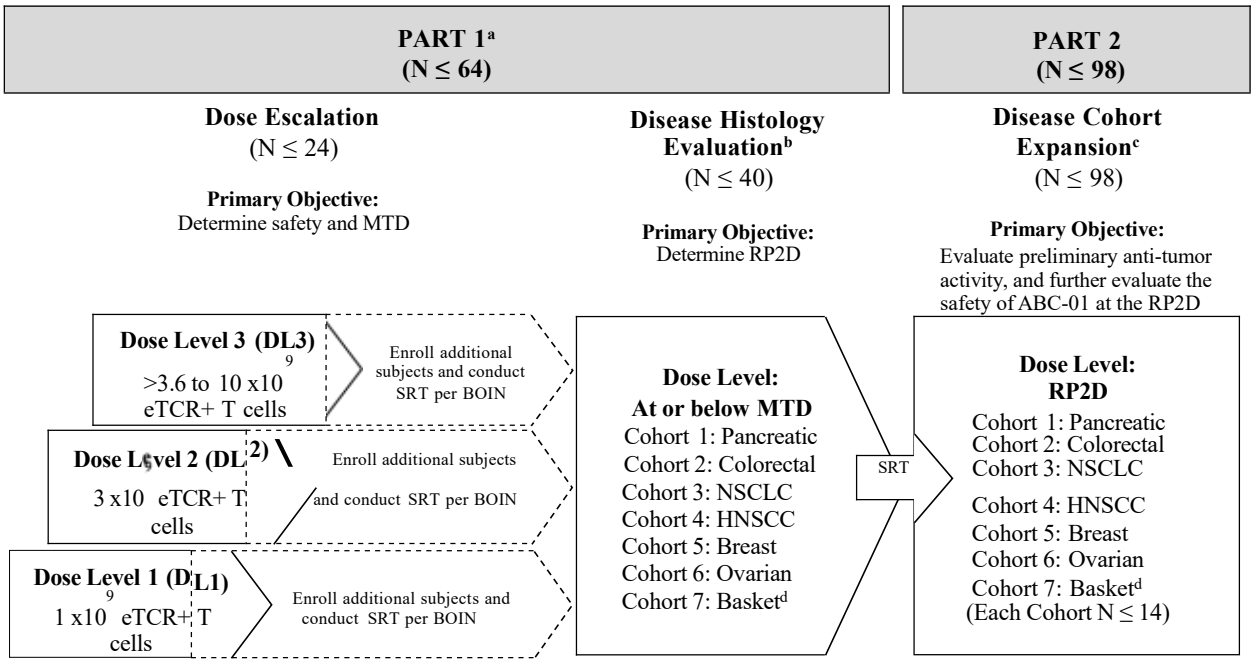
Refer to Section9.8.1and SRT charter for additional information regarding the SRT.

1.1. Study Schematic

The study schematic is presented inFigure 1.

Figure 1: ABC-01-001 Study Schema

Figure 1a. ABC-01-001 Study Treatment Schema



Abbreviations: HNSCC, head and neck squamous cell carcinoma; NSCLC, non-small cell lung cancer

[a]A minimum of 3 subjects at each dose level with at least 9 for MTD determination. If a DLT occurs in the first 3 subjects at any dose level, a minimum of 6 subjects will be evaluated for the dose escalation decision at that dose level.

[b] In Part 1, if a disease histology accumulates N=6 subjects at the same dose level, the SRT will convene to assess safety and preliminary anti-tumor activity within that histology and make a recommendation on dosing additional subjects in Part 2.

[c] Disease-specific expansion to further evaluate safety, and preliminary anti-tumor activity at the RP2D

[d] Basket cohort will enroll histologies not specified in Cohorts 1 through 6.

Figure 1b. ABC-01-001 Study Periods

Abbreviations: rIL-2, recombinant interleukin-2; IV, intravenous; SC, subcutaneous; eTCR, engineered T cell receptor

[a] Subjects are required to complete the Prescreening Visit if Subject is not known to be positive for *TP53* XYZ mutation and/or HLA-A*02:01 (ie, prior test was not done or a CLIA-accredited laboratory result by the assays defined in Appendix F is not available). Subjects who are not known to be positive for *TP53* XYZ mutation or HLA-A*02:01 (ie, prior test was not done, or the result is not available) must wait for the required prescreening results prior to signing the Screening ICF.

[b] Subjects are enrolled in the study upon initiation of leukapheresis for ABC-01 manufacture.

[c] Lymphodepleting chemotherapy should not be initiated prior to confirmation of availability of ABC-01 for subject treatment by the Sponsor.

[d] Treatment period starts on Day 0 and ends on Day 13. Subject hospitalization is required for 8 days from Day 0 through Day 7. ABC-01 is infused on Day 0. rIL-2 is administered on Day 0 up to Day 7.

[e] ABC-01 is administered at the dose level defined for the Phase 1 dose level the subject is assigned to.

[f] Post-treatment assessment period starts with Day 14 and ends on Week 12.

[g] Extended Follow-up period starts after Day 84 with first study visit at Week 18 and continue to Month 24.

[h] Long-term Follow-up period will begin after Month 24, with the first study visit at Month 30 and continue to Month 180.

2. INTRODUCTION

2.1. Background

2.1.1. Target Indication and Population

Worldwide, an estimated 19.3 million new cancer cases, and around 10 million cancer deaths occurred in 2020 (Sung, 2021). Breast cancer in females has surpassed lung cancer as the most commonly diagnosed cancer, with an estimated 2.3 million new cases, followed by lung, colorectal, prostate, and stomach cancers. In the United States (US), there will be an estimated 1.9 million new cancer cases diagnosed and over 600,000 cancer deaths in 2022 with cancers of the lung, breast, colorectum, and prostate as the most frequently diagnosed cancers, and lung cancer remains the leading cause of cancer mortality (https://seer.cancer.gov/). The great majority of patients that present with metastatic and/or recurrent solid tumors have short duration of responses to currently available treatments, highlighting the need for new and effective therapies.

TP53 is a key tumor suppressor gene that represents the most commonly mutated gene across different cancer types (Lane, 1992). Oncogenic driver mutations in *TP53* are common across different cancer types and tend to occur in hotspot locations within the genome. Mutations on these genes can encode aberrant immunogenic peptides, also known as neoantigens, that are often shared between tumor histologies and individuals and that can elicit anti-tumor immune responses when recognized by T cells. These highly tumor-specific neoantigens can be presented to T cells via the major histocompatibility complex on the surface of tumor cells and can be harnessed as therapeutic targets for adoptive T cell therapies. *TP53* mutations are highly prevalent across a broad variety of histologies, including breast, colorectal, lung, head and neck, pancreatic, and ovarian cancers among others. The prevalence of *TP53* mutations ranges from 30% in breast cancer patients to up to 70% in individuals diagnosed with colorectal cancer, with the XYZ mutation being the most commonly recurrent across indications (https://www.genome.gov/Funded-Programs-Projects/Cancer-Genome-Atlas).The incidence of the R175 H variant ranges from 6.7% in breast cancer, 6% in colorectal, 4.2% in pancreatic, 3.7% in head and neck to 1.2% in lung cancer (Chiang, 2021). Across these tumor types, no therapy is yet Food and Drug Administration (FDA)-approved for treatment directly targeting p53, nor the specific XYZmutation; successful development of such a treatment will address a significant unmet medical need for patients with a variety of cancer histologies.

2.1.2. Treatment Options

The use of chemotherapy regimens, radiotherapy, and surgery has been for many decades the cornerstone treatment for patients with solid tumor malignancies. Within the past decade, the emergence of immunotherapies and novel targeted therapies has revolutionized the field of oncology. Immune checkpoint inhibitors, including antibodies targeting cytotoxic T-lymphocyte antigen-4 and programmed cell death protein-1, have been approved in first- or -second line -treatment of a broad spectrum of solid tumors, including melanoma and non-small cell lung cancer (NSCLC) (KEYTRUDA United States Package Insert, 2022; YERVOY United States Package Insert, 2022). Despite the clinical benefits provided by these new treatment modalities with some durable responses observed in patients who achieve a complete response, the majority of patients fail to respond, and duration of response is limited (Darvin, 2018).

Cancer mutations are found in tumor cells and are absent in nontumorous cells. Some of the mutations are expressed, processed, and presented on tumor human leukocyte antigen (HLA) molecules to T cells. Those mutation-containing peptides that are recognized by T cells are neoantigens. A goal of immuno-oncology is to induce recognition of tumor cells by -tumor specific T cells. The number of mutations, and particularly the number ofclonal immunogenic mutations, predicts tumor response to immune-strengthening therapeutics, such as checkpoint inhibitors (Carbone, 2017;McGranahan, 2016). In addition, a high mutational burden and predicted neoantigen tumor load was associated with improved clinical outcomes in melanoma patients treated with tumor-infiltrating lymphocytes (Sarnaik, 2021), highlighting the importance of neoantigens in response to immunotherapy and adoptive T-cell therapy.

Neoantigens are tumor-specific mutations that can be successfully targeted by adoptive cellular therapy using autologous T cells, either by recognizing neoantigens arising from patient-specific mutations (private neoantigens) or by recognizing systemically recurrent mutations shared across individuals (shared neoantigens). Recently, the use of adoptive T cells engineered to express T cell receptors (TCRs) targeting shared neoantigens, including *TP53* and *KRAS*, has shown clinical responses in heavily pretreated patients with advanced breast and pancreatic cancer (Kim, 2022; Leidner, 2022). To date, no therapies targeting the *TP53* mutations are approved; however, these clinical data demonstrate the potential of adoptive T-cell therapies targeting shared neoantigens to bring clinical benefit to a patient population with significant clinical unmet need.

2.1.3. Description of ABC-01

ABC Pharm LP has developed a proprietary platform that enables efficient engineering of autologous T cell products. The genetic modification ofT cells used to make ABC-01 consists of clustered regularly interspaced short palindromic repeat (CRISPR)-Cas9-mediated, site-specific integration of a DNA repair template encoding the neoantigen-reactive TCR sequences at the endogenous TCRα constant domain locus. The double-stranded DNA repair template encodes an HLA-A*02:01-restricted TCR targeting the XYZ mutation in *TP53* and contains a modified dihydrofolate reductase with mutations in 2 amino acids that confer resistance to methotrexate, which is used for enrichment of engineered T cells during manufacturing of ABC-01.
CRISPR Cas9 ribonucleoprotein system is also utilized to disrupt the endogenous TCRβ locus and hence prevent mispairing of the exogenously introduced TCR with endogenous TCR chains.

In addition, ABC Pharm LP 's platform enables additional editing of genes important for T cell biology and T cell functions. The objective is to generate T cell products capable of overcoming limitations derived from the immunosuppressive tumor microenvironment. ABC-01 has been designed to incorporate a genetic knock out of the transforming growth factor beta receptor 2 (*TGFBR2)* gene, rendering the T-cell product insensitive to the inhibitory effects of transforming growth factor beta (TGF-β), whose presence in the tumor microenvironment has been correlated with poor prognosis in advanced cancers such as NSCLC, head and neck squamous cell carcinoma (HNSCC) or colorectal adenocarcinoma, promoting tumor progression and metastasis as well as driving the development of resistance to cytotoxic targeted immunomodulatory therapies.

2.1.4. Supportive Nonclinical Data

Traditional nonclinical in vivo pharmacology (pharmacodynamics and pharmacokinetics [PK]) and toxicology studies that are routinely applied to targeted therapies, such as kinase inhibitors or monoclonal antibodies, are not usually feasible for TCR engineered T cells, because of the following:

- Non-human species do not express human HLA molecules or human *TP53* (wild type or mutated), which is a requirement for antigen presentation and recognition by the *TP53* XYZ TCR.
- It is not feasible to engineer a mouse model to express HLA, human mutated *TP53,* and all other human antigens needed to establish safety of the TP53 XYZ TCR.
- It is not feasible to engineer an animal model to express human TGF-β (and TGF-β receptor) to assess the safety of *TGFBR2* knockout (KO) T cells.
- Non-human species amenable to adoptive cell transfer of human cell products are immune deficient and lack a complete and competent immune system necessary to evaluate complex safety outcomes such as cytokine release syndrome (CRS) or neurological events.
- Clinical data generated at the National Cancer Institute NCI (NCT01174121 and NCT03412877) support the safety of targeting *TP53* XYZwith the *TP53* XYZ TCR, as no adverse events related to off-target recognition were observed in either patient treated.

Therefore, ABC Pharm LP does not consider any data generated in an animal model informative or predictive of safety. The nonclinical development plan of ABC-01 is based on:

- In vitro evaluation of product selectivity, specificity, and reactivity of the *TP53* XYZ TCR against cell lines expressing the *TP53* XYZmutation and HLA-A*02:01, and lack of activity when either the *TP53* XYZmutation and/or HLA-A*02:01 are not expressed.
- In vitro evaluation of ABC-01 product functionality using well-established bioassays to measure proliferation, cytokine production, and cytotoxicity upon co-culture with either cancer cell lines or peptide-loaded target cells.
- In vitro evaluation of product functionality in the context ofTGFBR2 KO engineered T-cell products.
- In vivo evaluation of anti-tumor efficacy of the *TP53* XYZ-TCR in a xenograft model.
- Assessment of potential genotoxicity resulting from gene-editing with CRISPR-Cas9.

Refer to the current version of the Investigator's Brochure (IB) for a summary of the nonclinical data that have been generated for ABC-01.

2.1.4.1. Pharmacology

The XYZ-TCR has been evaluated for specificity and cross-reactivity at the Surgery Branch of NCI, Bethesda Maryland (Lo, 2019) and internally at ABC Pharm LP through extensive in vitro and in vivo characterization. The TCR (AV6/BV11) was identified as a result of screening tumor- infiltrating lymphocytes recognizing mutation-derived shared neoantigens in a patient with metastatic colorectal cancer (Lo, 2019). The TCR was identified as HLA-A*02:01-restricted with recognition of the *TP53* Xyz "HMTEVVRHC" minimal epitope. TCR specificity and reactivity were measured against cell lines loaded with mutated and wild-type *TP53* peptides, as well as against cell lines expressing endogenous levels of the mutation and/or wild-type proteins. ABC-01 T cell products only showed recognition of tumors when both HLA-A*02:01 and *TP53* XYZ mutation were present, as measured by proliferation, cytotoxicity, and cytokine production. In vitro data are supported by in vivo experiments in xenograft mouse models demonstrating the anti-tumor potential of ABC-01.

Experiments were also carried out in the presence of TGF-β, added exogenously or secreted by tumor cell lines. ABC-01 cells carrying the *TGFBR2* KO lacked expression of the receptor on the cell surface, were unable to phosphorylate Smad2/3 as part of the downstream signaling cascade for TGF-β and were rendered insensitive to TGF-β inhibition, as shown by their ability to kill target cells through multiple rounds, as opposed to T cells with intact TGFBR2 (ABC Pharm LP data on file).

Detailed information can be found in the current version of the IB.

2.1.4.2. Toxicology

Formal toxicology studies were not conducted. However, the following biomarkers related to possible single-dose toxicities will be evaluated during the Phase 1 ABC-01-001 clinical study:

- Expansion, persistence, and clearance of engineered T cells will be monitored by digital- polymerase chain reaction (dPCR) and flow cytometry to document PK, lack of expansion post-infusion, or abnormal expansion of gene-modified cells in peripheral blood;
- Induction, elevation, and clearance of cytokines and chemokines to document the origin and reversibility of potential toxicities;
- Recovery of normal blood lineages and immune cell types in peripheral blood by flow cytometry to document normalization of cell populations post-conditioning chemotherapy;
- Product immunogenicity using a cell-based flow cytometry assay to detect anti-drug antibodies (ADA) and enzyme-linked immunosorbent spot analysis to evaluate T cell-mediated responses against product T cells will be conducted to monitor potential immunotoxicities or impact on product performance.

2.1.4.2.1. Genotoxicity

The ABC-01 manufacturing process utilizes CRISPR-Cas9 nuclease-mediated in-frame site-specific integration (SSI) of a DNA repair template encoding a neoantigen-specific TCR at the

endogenous TCRα locus. Concomitant CRISPR-Cas9 nuclease-mediated disruption ofthe endogenous TCRβ and the TGFBR2 loci is conducted.

The use of gene-editing tools such as CRISPR-Cas9 that enable more precise, targeted modifications has emerged as a safer alternative than retroviral vectors to minimize potential genotoxicity driven by random insertion of viral sequences in the genome. While the number of clinical studies utilizing nuclease-mediated gene editing of T cells is limited, there are no public reports to date of adverse events resulting from genotoxicity in these studies (Lu, 2020) (Stadtmauer, 2020). While any process affecting DNA integrity or causing DNA or chromosomal damage has the potential to result in genotoxicity, mature T cells appear to be quite resistant to transformation (Foy, 2022;Hu, 2021;Newrzela, 2008; Ottaviano, 2022; Scholler, 2012;Wang, 2021;Zhang, 2022), with only one case of a non-transformative temporary clonal expansion reported to date ofCD19 chimeric antigen receptor (CAR) T cells generated using a lentiviral vector (Fraietta, 2022).

To directly assess potential genotoxicity caused by the gene editing process in ABC-01, ABC Pharm LP has performed the following characterization ofT cell products manufactured from 3 healthy donors, using clinical scale lots with representative material:

- In silico analysis of potential off-target cleavage sites ofthe T cell receptor alpha constant (TRAC), T cell receptor beta constant (TRBC), and Transforming growth factor beta receptor 2 (TGFBR2) CRISPR-Cas9 RNA-guided nucleases.
- Genome-wide profiling of off-target cleavage activity ofthe CRISPR-Cas9 RNA-guided nucleases in an established sequencing assay (GuideSeq).
- Confirmation of off-target cleavage activity by targeted amplicon sequencing. Analysis of potential translocations occurring at the edited loci and significant recurrent off-target sites using a dPCR assay.
- Measurement of potential for cytokine-independent growth and time-to-senescence using an established in vitro bioassay.
- Assessment of chromosomal aberrations by G-banded karyotyping.
- Assessment of double-stranded DNA repair template integration in on- and significant recurrent off-target sites.

Additional information is available in the current version of the IB.

2.1.4.2.2. Immunogenicity

To enable quantification of engineered T cells in ABC-01 products, the constant domain of the TCRβ chain is modified to express a murine “Mur6” epitope, comprising 6 amino acids, that can be detected using a murine TCRβ constant domain-specific antibody (H57-597). The presence of murine components poses a risk for immunogenicity. However, human TCRs with murine components, fully murine TCRs, and CARs with single chain variable fragment domains of murine origin have been used in several clinical studies, with no adverse events (AEs) attributed to these murine components (Doran, 2019; Johnson, 2009;Locke, 2019;Lu, 2017;Nagarsheth, 2021). Therefore, the risk of AEs resulting from immunogenicity of recognition of murine components in ABC-01 is considered to be low. However, an enzyme-linked immunosorbent spot

assay has been developed and qualified at BioAgilytix to monitor T-cell immunogenicity against the Mur6 region of the double-stranded DNA repair template as well as against the modified human *DFHR*, which is mutated at positions L23F and F32S. In addition, an ADA assay has been developed and qualified internally at ABC Pharm LP to monitor antibody responses against ABC-01 products containing the Mur6 epitopes. Samples for immunogenicity studies will be collected throughout the clinical trial at various timepoints.

2.1.4.3. Pharmacokinetics

Studies to evaluate the PK profile of ABC-01 have not been conducted but are supported by PK data from NCI (Lo, 2019;Kim, 2022). The PK profile of ABC-01 will be evaluated as part ofthe schedule of assessments in the Phase 1 portion of the ABC-01-001 study.

2.1.5. Supportive Clinical Data

Study ABC-01-001 is the first-in-human study of ABC-01.

To date, no drugs targeting mutations in *TP53* have been approved for the treatment of cancer, but clinical evidence exists supporting the rationale for targeting *TP53* neoantigens utilizing cell therapies (Lo, 2019) reported observing a partial response (PR) in a colorectal carcinoma patient after receiving therapy with tumor-infiltrating lymphocytes containing a TCR reactive against the XYZmutation in TP5*3* (Lo, 2019).Similarly, a second PR was observed when utilizing the same TCR reported by Lo and colleagues to treat a breast cancer patient using an engineered T cell product (Kim, 2022). In this report, the patient experienced a 6-month PR after relapsing to 10 lines of prior hormonal and chemotherapy regimens.

2.2. Study Rationale

The planned Phase 1 clinical study will investigate the use of genetically engineered TCR+ cells (eTCR) in HLA-A*02:01-positive adult patients with unresectable, advanced, and/or metastatic tumors harboring the XYZ mutation of *TP53*. HLA-A*02:01 is expressed by approximately 50% of the white population. The estimated frequency of *TP53* XYZ mutation in various cancers suggests a potential of ABC-01 to treat a sizable number of patients across multiple tumor histologies. Further, there are no currently approved therapies that target the *TP53* mutations, and preliminary clinical data using adoptive T cell therapies targeting the XYZ mutation have demonstrated the potential for clinical benefit in these patient populations with continued unmet need. Utilizing a TCR that specifically recognizes mutations within the tumor and lacks reactivity against wild-type proteins is expected to minimize the occurrence of off-target/off-tumor AEs that have been observed with other engineered TCR T -cell therapies targeting non-tumor restricted antigens. In addition, knocking out the *TGFBR2* gene is expected to enable ABC-01 to overcome the limitations derived from the immunosuppressive tumor microenvironment and render a more efficacious T cell product.

2.2.1. Rationale for Study Design Elements

The current study focuses on evaluating safety and preliminary anti-tumor activity in HLA-A*02: 01 patients with solid tumors that are positive for the *TP53* XYZ mutation.

Part 1 Dose Escalation will investigate escalating doses of ABC-01 in adult subjects with eligible solid tumor histologies to evaluate safety and determine MTD.

The model-assisted Bayesian Optimal Interval (BOIN) design is a recognized tool (FDA, 2021) for drug development to determine MTD (Liu, 2015;Yuan, 2016) and will be used to determine the MTD of ABC-01 in a dose escalation design.

Part 1 Disease Histology Evaluation will further evaluate the safety and preliminary anti-tumor activity at or below the MTD in disease specific histologies and will determine RP2D. Once a disease histology accumulates 6 subjects treated at the same dose level, the SRT will convene to assess safety and preliminary anti-tumor activity within that histology at that dose level and may make a recommendation on whether or not to dose additional subjects in Part 2.

Part 2 Disease Cohort Expansion will further evaluate the preliminary anti-tumor activity and safety of ABC-01 at the RP2D in the disease specific histologies through expansion of up to a total of approximately 20 subjects in each disease-specific cohort (including subjects dosed in Part 1 and Part 2).

The statistical rationale for up to approximately 20 subjects in each indication is that the margin of error for a 90% confidence interval is less than 20%.

The results from this study will form the basis for decisions for future studies.

2.2.1.1. Justification for Dosing Strategy

ABC-01 is a personalized engineered T-cell product that targets the *TP53* XYZ neoantigen present in a particular tumor when presented on HLA-A*02:01 molecules. As the *TP53* XYZ mutation is exclusively expressed by tumor cells, the risk of off-tumor recognition by ABC-01 is considered low. Nonclinical data support *TP53* XYZTCR's specificity for its cognate antigen and lack of off-target recognition, as well as specific restriction by HLA-A*02:01.

Nonetheless, extensive nonclinical characterization of ABC-01 has been performed (see Section2.1.4), and the Phase 1 clinical study ABC-01-001 will employ gradual dose escalation starting at 1×10^9 engineered T cells, which is approximately 2 logs lower than the dose utilized at the NCI for the treatment of patient 4349 using an engineered T cell product containing 3.39×10^{10} TCR+ T cells (Kim, 2022). In addition, engineered T cells administered at comparable dose ranges have been utilized in other adoptive T cell clinical trials, with no major safety concerns observed specifically related to the T cell doses used (Lu, 2017;Nagarsheth, 2021).

The doses of ABC-01 selected for this study are based on prior clinical experience in studies of TCR-T cell therapies for solid tumors. In these studies, IV doses of TCR-T cells between 1×10^9 and 53×10^9 have been used and shown to be safe and well tolerated (Doran, 2019;Kebriaei, 2018;Nagarsheth, 2021). More recently, studies specifically targeting shared neoantigens with engineered TCR-T cells, particularly *TP53* or *KRAS* variants, have utilized doses of 16×10^9 and 53×10^9, respectively (Kim, 2022;Leidner, 2022). These studies have not only demonstrated the manufacturing feasibility of these TCR T cell doses but also the tolerability and efficacy of these products. The safety profile of these multiple TCR-T cell products utilizing doses of more than 50×10^9 TCR T cells supports the dose escalation being proposed for the ABC-01 study of doses up to 10×10^9 TCR T cells.

If the minimum dose required for treatment at an assigned dose level is not met during manufacturing, ABC-01 may be remanufactured from a repeat leukapheresis. The Sponsor Medical Monitor and Investigator will discuss the options for remanufacture and assess the need

for additional bridging therapy during remanufacturing based on the subject's clinical condition. Yields from different ABC-01 manufacturing runs from the same subject may be utilized to reach the final target dose.

Given the potential benefits and tolerable safety profile of lymphodepletion with DrugC/DrugF combined with recombinant interleukin-2 (rIL-2), subjects will receive the following regimen:

- DrugC, 30 mg/kg intravenous (IV) on Day -7 and Day -6
- Mesna (administered per institutional guidelines and/or investigator discretion, unless contraindicated) on Day -7 and Day -6
- DrugF, 25 mg/m^2 IV on Day -7 through Day -3
- Two-day rest before ABC-01 administration
- rIL-2, 300,000 IU/kg once daily, subcutaneously (SC) for a maximum of 8 doses, with first dose being administered approximately 4 hours following ABC-01 treatment. rIL-2 may be administered up to 24 hours after ABC-01 treatment due to AEs or other clinical reasons.

2.2.1.2. Rationale for Lymphodepleting Chemotherapy

Before administration of ABC-01, subjects will receive a nonmyeloablative lymphodepleting regimen consisting of DrugC and DrugF to induce lymphocyte depletion and create an optimal environment for expansion of ABC-01 in vivo. Lymphodepletion may function by eradicating cytokine sinks for the transferred cells, eliminating T regulatory cells, and enhancing antigen-presenting cell activation (Dafni, 2019). In addition, lymphodepletion has been demonstrated to provide adoptively transferred T cell products, such as ABC-01, with the ideal environment for expansion, persistence, and migration to the tumor site (Kochenderfer, 2017; Turtle, 2016).

Lymphodepletion with DrugC (30 to 60 mg/kg/day) and DrugF (25 mg/m^2/day) has been utilized in many clinical trials investigating tumor-infiltrating lymphocytes and TCR T-cell therapy and has been well tolerated in subjects with solid tumors (Doran, 2019; Nagarsheth, 2021;Nguyen, 2019). Specifically, DrugC treatment for 2 days with DrugF for 5 days has been used as lymphodepletion in several clinical trials of TCR-transduced T cells (Morgan, 2013;Robbins, 2015;Dudley, 2008) and tumor-infiltrating lymphocytes (Dudley, 2008). Increasing levels of lymphodepleting chemotherapy have been shown to correlate with clinical response in adoptive cellular therapy (Dudley, 2008). The depth and duration of the lymphodepletion in nonclinical models correlates with antitumor activity of the adoptively transferred tumor-specific CD8$^+$ T cells (Wrzesinski, 2010).

2.2.1.3. Rationale for Recombinant Interleukin-2 Treatment and Dose

Recombinant IL-2 has been utilized in many clinical trials investigating tumor-infiltrating lymphocyte and TCR T cell therapies. Doses of rIL-2 between 125,000 IU/kg/day and 720,000 IU/kg every 8 hours have been administered and demonstrated to enhance T cell activity with manageable side effects (Nguyen, 2019). The dose selected for this study of 300,000 IU/kg

SC daily could possibly minimize the AEs and risks associated with rIL-2 infusion and still provide support for ABC-01 expansion and persistence.

2.3. *TP53* XYZ and HLA-A*02:01 Testing

2.3.1. *TP53* XYZ Next Generation Sequencing-Based In-Vitro Diagnostic and HLA-A*02:01 Typing Assays

The *TP53* XYZTCR expressed on ABC-01 T cell products specifically recognizes the *TP53* XYZmutation when presented on HLA-A*02:01. HLA-A*02:01 is the most common MHC class I allele in the white population, with a frequency of approximately 50% of individuals bearing this allele (http://allelefrequencies.net/).It is anticipated that most subjects will have existing *TP53* XYZmutation status results from prior tests performed as part of their routine care. Subjects with tumors harboring *TP53* XYZ mutation, as detected by one of the protocol-defined tests described inAppendix F(Foundation Medicine F1CDx, Caris Tissue NGS assay, Tempus xT CDx, Dana Farber OncoPanel Test, MD Anderson Mutational Analysis Precision Panel, BostonGene Tumor Portrait™ Test, Providence St. Joseph Health Molecular Genomics Laboratory ProvSeq test or HopeSeq Solid tumor comprehensive panel (City of Hope) and demonstrated HLA-A*02:01 positivity using the assay described in Appendix F(American Red Cross) will be eligible to proceed to screening. Subjects who have documented HLA-A*02:01 positivity from an assay other than those described in Appendix Fmay proceed to Screening, however a sample must be provided during the Screening Period for central HLA testing.

During the Prescreening Period, subjects who are not known to be positive for *TP53* XYZ mutation and/or HLA-A*02:01 (ie, prior test was not done, the result is otherwise not available, or testing was not performed using one of the protocol-defined CLIA certified laboratory assays described inAppendix F) will undergo applicable testing during Prescreening and must wait for the test results prior to signing the Screening ICF and beginning study-related screening procedures. Subjects whose prescreening test results do not confirm *TP53* XYZ mutation and/or HLA-A*02:01 positivity will not proceed to Screening.

Those subjects for whom archival samples are not available will be requested to undergo a biopsy to obtain a specimen to assess mutation status. Biopsy collection is a clinical standard procedure with an expected morbidity or mortality rate of <2%. Therefore, tumor sampling for the purposes of this investigation should not pose meaningful additional risk. In addition, these biopsies are routinely performed and are considered to carry manageable risks that are outlined in the informed consent form.

Based on reported performance of the selected devices, the risk of misidentifying a subject as eligible for enrollment is considered to be low. Nevertheless, the risk of obtaining a false positive or false negative remains. The risk of a false positive is that a subject would be enrolled into the study without the presence of a *TP53* XYZ alteration or HLA-A*02:01 and would not be expected to benefit from treatment with ABC-01. This risk is considered low because the IVDs to be used for identification of the *TP53* XYZ mutation are routinely used in clinical practice to facilitate treatment decisions and have undergone robust validation for the detection of genetic alterations of interest and are therefore unlikely to generate incorrect results. The risk of a false negative is that a subject would not be enrolled into the study when they could potentially benefit

from treatment with ABC-01. This risk remains low based on the assay characteristics for the proposed tests.

Further, per the study eligibility criteria, subjects eligible for enrollment in the ABC-01-001 study will not be foregoing or delaying a treatment that is known to be curative. Subjects have previously received at least one line of approved standard of care therapy and have no available curative treatment options. Subjects with stable disease or who lack clinical benefit from previous therapy may be eligible for enrollment in consultation with the Sponsor Medical Monitor. Additional rounds of previous required therapy are detailed in Section5.3.1. Furthermore, there are no current FDA approved therapies that directly target *TP53* alterations, nor the specific XYZ alteration presented in this study.

The investigational therapy being administered in this trial is not expected to expose the subjects to excess risk. The doses of ABC-01 selected for this study are based on prior clinical experience in studies of TCR-T cell therapies for solid tumors. The safety profile of these multiple
TCR-T cell products utilizing doses of up to more than 50 x 10^9 TCR-T cells supports the dose escalation being proposed for the ABC-01 study of doses up to 10 x 10^9 TCR-T cells. All subjects will be closely monitored for safety, as described in Section9, and will undergo numerous safety assessments at each planned study visit, as described in Section7.9and Section 8, respectively.
Furthermore, because this targeted immunotherapy specifically acts against tissue bearing the *TP53* XYZmutation, treatment on this study would not preclude subjects from receiving future standard-of-care treatments that act against other oncologic targets.

The test characteristics and other supporting documentation for the IVDs that may be used to support eligibility determination are described inAppendix F.

Overall, based on demonstrated assay performance and relative low risk of AEs associated with a biopsy collection procedure, if required, and the use of the use of proposed IVDs to confirm the presence of *TP53* XYZ mutation should not pose a significant risk to subjects.

2.4. Benefit: Risk Assessment

ABC-01 has not been tested in humans. To date, there are no clinical data demonstrating safety or efficacy of ABC-01; however, as experience with ABC-01 progresses, a better understanding of the benefit-risk will emerge for subjects that could be treated with this novel therapy. Section9.7.1describes the potential risks of ABC-01 (CRS and immune effector cell-associated neurotoxicity syndrome [ICANS]), which have been associated with adoptive T cell therapies. The potential risks associated with CRISPR/Cas-9 mediated gene editing (Section2.1.3) are off-target toxicities, cellular immune responses to Cas9, and other unknown long-term toxicities.

During informed consent, all subjects will be informed of the anticipated and potential risks of ABC-01, rIL-2, lymphodepleting chemotherapy regimen (DrugC, DrugF), leukapheresis, and other study-related procedures.

Common side effects of DrugC include myelosuppression, infections, temporary hair loss, nausea, vomiting, diarrhea, and hemorrhagic cystitis as described in the DrugC INJECTION United States Package Insert (USPI) 2021 available at https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/212501s002lbl.pdf.

Common side effects of DrugF include fever, increased risk of infections, cytopenia, anemia, diarrhea, lethargy, vomiting, nausea, and neurological damage as described in the FLUDARA USPI 2010 available at https://www.accessdata.fda.gov/drugsatfda_docs/label/2010/020038s033lbl.pdf.

Common side effects of rIL-2 are difficulty breathing, confusion, irregular heart rate, hypotension, weight gain or loss, anemia, diarrhea, fast heart rate, fatigue, fever, rigors, headache, renal impairment, mouth sores, vomiting, nausea. Refer to Appendix D for a description of the potential risks of rIL-2 as described in the PROLEUKIN USPI 2012 available at https://www.accessdata.fda.gov/drugsatfda_docs/label/2012/103293s5130lbl.pdf.

A potential risk of leukapheresis is the sensation of tingling in the face and lips due to medications administered to prepare for the procedure. Lightheadedness or dizziness is rarely experienced.

Refer to the current version of the IB for a summary of findings from nonclinical studies and known potential benefits and risks associated with ABC-01 eTCR+ T cells.

2.5. Compliance

The study will be conducted in accordance with this clinical study protocol, Good Clinical Practices (GCP) as outlined by International Council for Harmonisation (ICH) of Technical Requirements for Pharmaceuticals for Human Use guideline E6(R2), and all applicable local and national regulatory requirements. Enrollment at any clinical study site may not begin prior to that site receiving approval from the ethics committee of record for the protocol and all materials provided to potential subjects.

Any amendments to the protocol or changes to the consent document will be approved before implementation of that amendment. Reconsent of previously enrolled subjects may be necessary depending on the nature of the amendment.

The Principal Investigator will ensure that changes to the study plan as defined by this protocol will not be made without prior agreement from the Sponsor and documented approval from the ethics committee of record, unless such a change is necessary to eliminate an immediate hazard to the study subjects.

All personnel involved in the conduct of this study will have completed GCP training as outlined by their governing institution or approved by the Sponsor.

3. OBJECTIVESAND ENDPOINTS

Table 1: Objectives and Endpoints for Study ABC-01-001

Objectives	Endpoints
Primary	
Part 1 (Dose Escalation and Disease Histology Evaluation)	
• Evaluate the safety of ABC-01 in subjects with unresectable, advanced, and/or metastatic solid tumors • Evaluate MTD and RP2D(s)	• Incidence of DLTs, TEAEs, and SAEs
Part 2 (Disease Cohort Expansion)	
• Evaluate the preliminary anti-tumor activity of ABC-01 at the RP2D in subjects with unresectable, advanced, and/or metastatic solid tumors	• ORR per RECIST V1.1 per Investigator assessment • BOR • DOR • CBR (CR, PR, SD) • TTR • PFS • OS
• Further evaluate the safety of ABC-01 at the RP2D in subjects with unresectable, advanced, and/or metastatic solid tumors	• Treatment-emergent adverse events, and serious adverse events
Secondary	
Part 1 (Dose Escalation/Disease Histology Evaluation)	
• Evaluate preliminary anti-tumor activity of ABC-01 in subjects with unresectable, advanced, and/or metastatic solid tumors	• ORR per RECIST V1.1 per Investigator assessment • BOR • DOR • CBR (CR, PR, SD) • TTR • PFS • OS

Table 1: Objectives and Endpoints for Study ABC-01-001 (Continued)

Objectives	Endpoints
Exploratory Objectives (Part 1 and Part 2)	
• Evaluate PK and pharmacodynamics • Evaluate parameters associated with blood and/or tumor tissue and association with clinical outcomes • Evaluate the manufacturing feasibility of ABC-01	• Expansion, persistence, migration, phenotype, and functionality of the engineered T cells in peripheral blood and/or tumor • Investigation of factors and/or biomarkers in blood and/or tumor tissue associated with ABC-01 mechanism of action and possible correlation with clinical responses, toxicity, and/or changes in tumor and/or neoantigen expression • Manufacturing success rates

Abbreviations: BOR, best overall response; CBR, clinical benefit rate; CR, complete response; DLT, dose-limiting toxicity; DOR, duration of response; MTD, maximum tolerated dose; ORR, objective response rate; OS, overall survival; PFS, progression-free survival; PK, pharmacokinetics; PR, partial response; RECIST, Response Evaluation Criteria in Solid Tumors; RP2D, recommended Phase 2 dose; SAE, serious adverse event; SD, stable disease; TEAE, treatment-emergent adverse event; TTR, time to response

4. STUDY DESIGN

Study ABC-01-001 is a Phase 1, open-label, multicenter study to evaluate the safety and preliminary anti-tumor activity of ABC-01 in HLA-A*02:01 subjects with unresectable, advanced, and/or metastatic cancer, including NSCLC, colorectal adenocarcinoma, HNSCC, pancreatic adenocarcinoma, breast cancer, ovarian cancer, or any other solid tumor histologies that are positive for the *TP53* XYZ mutation. Data collected in this study will support the evaluation of the potential doses that may be studied in a Phase 2 clinical research study.

The study will be conducted in 6 periods: the Prescreening Period (Section8.1), the Screening Period (Section8.2), Enrollment/Leukapheresis (Section8.3), the Lymphodepletion Period (Section8.4), the Treatment Period (Section8.5), and the Follow-up Period (Section8.6), which includes Post-treatment, Extended, and Long-Term Follow-up Periods.

The study is divided into two parts, Part 1 (Dose Escalation and Disease Histology Evaluation) followed by Part 2 (Disease Cohort Expansion).

- Part 1 Dose Escalation will investigate escalating doses of ABC-01 in adult subjects with eligible solid tumor histologies and will evaluate the safety and MTD. Within the 3 distinct treatment dose levels that are part of this study, a dose-escalation design will be employed using a model-assisted BOIN design, which is described in Section4.1. DLTs will be assessed within a 28-day DLT evaluation period. The DLT evaluation period and the definition of DLTs are provided in Section9.6.

- Part 1 Disease Histology Evaluation will further evaluate the safety and preliminary anti-tumor activity at or below the MTD in disease specific histologies and determine the RP2D. Once a disease histology accumulates 6 subjects treated at the same dose level, the SRT will convene to assess the safety and preliminary anti-tumor activity within that histology at that dose level and may make recommendation on whether or not to dose additional subjects in Part 2.

- The Part 2 Disease Cohort Expansion will further evaluate the preliminary anti-tumor activity and safety of ABC-01 at the RP2D in disease specific settings.

Part 2 Disease Cohort Expansion may enroll concurrently with Part 1 Disease Histology Evaluation after MTD is determined.

Part 1 Disease Histology Evaluation and Part 2 Disease Cohort Expansion subjects will be enrolled in up to seven disease-specific cohorts, as outlined below. Requirements for prior lines of therapy in each cohort are specified in the inclusion criteria Section5.3.1.

- Cohort 1: Pancreatic adenocarcinoma
- Cohort 2: Colorectal adenocarcinoma
- Cohort 3: NSCLC
- Cohort 4: HNSCC
- Cohort 5: Breast Cancer
- Cohort 6: Ovarian Cancer
- Cohort 7: Basket cohort (excluding histologies specified in Cohort 1 to Cohort 6)

A Safety Review Team (SRT) will make recommendations on the conduct of the study, as described in Section9.8.

Enrollment in one or more histologies (cohorts) may be prioritized based on 1) the prevalence of *TP53* R175; 2) the maturing risk benefit of the investigational product; and 3) highest unmet medical need. Sponsor will take into consideration SRT's recommendation and make a final decision.
For details on study visits, refer to Section8 for each study period, and Section8.6.7 for the Disease Progression Visit, and Section8.6.5 for Early Exit Follow-up. Refer to Section7 for a description of each procedure.

A study schematic is provided Section1.1.

4.1. Part I (Dose Escalation and Disease Histology Evaluation)

In Part 1, up to approximately 64 subjects may be enrolled and treated.

Up to 24 subjects will contribute to MTD determination.

The number of subjects enrolled and treated per dose level in the Dose Escalation part of the study will be determined by the incidence of DLTs observed and will depend on the number of dose levels that are tested. The dose levels of ABC-01 to be evaluated are presented in Table 2. DLTs are defined in Section9.6. The subject population considered for DLT analysis is described in Section10.2.1.

Table 2: Dose Level Groups

Dose Level Group	Target Dose ABC-01 TCR-Engineered T cells	Acceptable Range
Dose Level 1	1 x 10^9 eTCR+ T cells	0.8 to 1.2 x 10^9 eTCR+ T cells
Dose Level 2	3 x 10^9 eTCR+ T cells	2.4 to ≤ 3.6 x 10^9 eTCR+ T cells
Dose Level 3	>3.6 to 10 x 10^9 eTCR+ T cells	N/A

Abbreviations: TCR, T cell receptor; eTCR, engineered T cell receptor

A Dose Level -1 (de-escalation) of 5 x 10^8 eTCR+ T cells with a ±20% acceptable range may be explored as determined by the SRT.

Additional subjects may be dosed at the current dose level after the SRT makes recommendation to escalate to the next higher dose level.

Additional and/or intermediate dose levels may be explored based on the totality of the data and the SRT recommendation.

A DLT evaluation period of 28 days (Day 28) after ABC-01 infusion (Day 0) will be used. Only DLTs that occur during the DLT evaluation period for each DLT-evaluable subject will be used for the DLT rate calculation. The "sentinel" subject approach will be followed so that first subject at a given dose level will complete the DLT assessment period (28 days post infusion) prior to treatment of the second subject; treatment of the third will be staggered by at least 7 days from the previous subject's treatment.

During Dose Escalation, a model-assisted BOIN design will be used to find the MTD of ABC-01 with 3 subjects assigned as a cohort and treated at the specific dose level; then, depending on the DLT rate (see Figure 2), another 3 subjects will be assigned. The MTD will be evaluated and determined after one of the dose levels accumulates 9 subjects who are enrolled, treated, and had the opportunity to be followed for the DLT period without an escalation/de-escalation as the next action. The target toxicity rate for the MTD is 30%, and the maximum sample size is up to 24 subjects enrolled, treated, and had the opportunity to be followed for the DLT evaluation period.

As shown in Figure 2 the BOIN design uses the following rule, optimized to minimize the probability of incorrect dose assignment, to guide dose escalation/de-escalation:

- If the observed DLT rate at the current dose is ≤ 23.6%, escalate the dose to the next higher dose level;
- If the observed DLT rate at the current dose is > 35.9%, de-escalate the dose to the next lower dose level;
- Otherwise, stay at the current dose level.

Part 1 Disease Histology Evaluation may enroll an additional 40 subjects with a starting dose at or below the MTD to further assess safety and evaluate the RP2D. Once 6 subjects have been dosed in one dose level and in a specific histology, the data will be reviewed by the SRT to evaluate the risk-benefit and make a recommendation for expanding disease specific cohorts in Part 2.

With up to approximately 64 subjects treated in Part 1, there is at least 95% probability to observe any adverse events with more than 5% incidence rate.

The RP2D may be assessed based on the totality of clinical data collected in Part 1 including safety, anti-tumor activity, PK, pharmacodynamics, and biomarker data.

Figure 2: Flowchart for Study Conduct Using the BOIN Design

Abbreviations: BOIN, Bayesian Optimal Interval; DLT, dose limiting toxicity; MTD, maximum tolerated dose
Footnotes: a DLT rate = Total number of subjects who experience DLT at the current dos / Total number of DLT evaluable subjects treated at the current dose
Refer to Section10.2.2for a description of the BOIN design.

4.2. Part 2 (Disease Cohort Expansion)

Part 2 will begin after approval from the SRT and upon determination of RP2D to evaluate the preliminary anti-tumor activity, and further evaluate the safety of ABC-01 at the RP2D for each of the Disease Expansion Cohorts.

With up to a total of approximately 20 subjects in each indication, the margin of error for a 90% confidence interval is less than 20%.

Refer to Section9.8.2for stopping rules for Disease Histology Evaluation (Part 1) and Disease Cohort Expansion (Part 2).

The recommended Phase 2 dose (RP2D) will be assessed based on the totality of data collected in Part 1 and Part 2, including safety, anti-tumor activity, pharmacokinetics (PK), pharmacodynamics, and biomarker data.

4.3. End of Study Definition

End of study (EOS) is defined as the time the last subject completes the last study visit in the Long-Term Follow-up Period of the study, is considered lost to follow-up, withdraws consent, or dies.

A subject is considered to have completed the study if the subject has completed all periods of the study including the last scheduled procedure shown in the Schedule of Assessments (SOA) (Appendix B).

4.4. Study Duration

The duration of the study will vary depending on the subject's response to treatment and survival.

For subjects who complete the entire protocol from date of initial informed consent through the completion of the long-term follow-up period, the duration will take approximately 15 years to complete.

Duration of treatment is 8 days, starting on Day 0, when subjects receive ABC-01 and rIL-2. Each subject will remain in the hospital for a minimum of 8 days (Day 0 to Day 7) and will receive rIL-2 administered SC at 300,000 IU/kg once daily from Day 0 to Day 7 for maximum of 8 doses.

5. POPULATION

5.1. Recruitment

Up to approximately 162 subjects may be enrolled and treated in the study.

Part 1 (Dose Escalation and Disease Cohort Evaluation): Up to approximately 64 subjects.

- Dose Escalation:
  - A minimum of 3 subjects will be evaluated for each dose escalation decision.

If a DLT occurs in the first 3 subjects at any dose level, a minimum of 6 subjects will be evaluated for the dose escalation decision at that dose level

- Disease Histology Evaluation:
  - Up to 7 disease-specific cohorts of subjects each infused at, or below MTD.
  - Subjects treated at the highest dose level from dose escalation will be included in the RP2D determination.

Part 2 (Disease Cohort Expansion): Up to approximately 98 subjects in total in up to 7 disease-specific cohorts.

5.2. Definitions

Subjects enter the Screening Period following provision of written informed consent either directly or via a legally authorized representative and the completion of at least one Screening Visit procedure (see Section8.2).

Subjects are enrolled when leukapheresis is initiated for the manufacture of ABC-01.

A screen failure occurs when a subject has signed the Screening ICF at the Screening Visit, completed at least one Screening Visit procedure, and subsequently does not meet eligibility criteria prior to enrollment (eg, start of leukapheresis).

For subjects who are required to prescreen for the study (see Section8.1), a prescreen failure occurs when a subject has signed the prescreening ICF at the Prescreening Visit, completed at least one Prescreening Visit procedure, and subsequently did not continue to the Screening Visit (ie, did not provide informed consent at the Screening Visit and did not complete at least one screening procedure) because the subject does not have a *TP53* XYZ variant mutation and/or HLA-A*02:01 confirmed by a CLIA-accredited laboratory-based test described inAppendix F or because he or she withdrew from the study for one of the reasons described in Section5.5.

5.3. Enrollment Eligibility Criteria

Prior to enrollment, the investigator is responsible for ensuring that all enrolled subjects meet the eligibility criteria, described within this section. Throughout the Pre-Screening and Screening period, the Sponsor Medical Monitor may provide guidance and answer questions about eligibility.

Once the eligibility is confirmed by the Investigator and reviewed by the Sponsor Medical Monitor as described in the Sponsor Medical Monitoring Plan (MMP), the subject can be enrolled and proceed to leukapheresis.

5.3.1. Inclusion Criteria

To be included in this study, each individual must satisfy all the following criteria:

1. Subject must be at least 18 years of age, at the time of signing the informed consent.
2. Subject must be capable of giving signed informed consent, which includes compliance with the requirements and restrictions listed in the ICF and in this protocol.
3. Subject must be diagnosed with one of the histologies below:
   - NSCLC
   - Colorectal adenocarcinoma
   - HNSCC
   - Pancreatic adenocarcinoma
   - Breast cancer
   - Ovarian cancer
   - Any solid tumor histology positive for *TP53* XYZmutation will also be considered
4. Subject has advanced solid cancer, defined as unresectable, advanced, and/or metastatic disease (Stage III or IV) after at least 1 line of systemic standard of care (SOC) treatment regimen and for which there are no available curative treatment options. Subjects with stable disease (SD), or that present lack of clinical benefit from previous therapy (including treatment suspension due to toxicity) may be considered eligible for enrollment in consultation with the Sponsor Medical Monitor. Subjects with targetable mutations should be considered for an approved targeted agent therapy. Additional prior therapy requirements are as follows:

Tumor histology (Cohort if applicable)	Eligibility Criteria
Pancreatic adenocarcinoma	Must have received treatment with FOLFOX, FOLFIRINOX, a gemcitabine-based, or platinum-based therapy.
Colorectal adenocarcinoma	Must have received treatment with oxaliplatin or irinotecan, with or without an anti-angiogenic agent. If tumor is MSI-Hi, subject must have received a checkpoint inhibitor.

NSCLC	Must have received systemic chemotherapy agent and checkpoint inhibitor, either in combination or sequentially.
HNSCC	Must have received systemic chemotherapy treatment and/or checkpoint inhibitor, with or without cetuximab.
Breast Cancer	Must have received: HER2-positive must have received systemic chemotherapy agent and anti-HER-2 therapy HR-positive: must have received treatment with endocrine therapy and CDK4/6 inhibitors Triple negative breast cancer: must have received systemic chemotherapy agent and/or Trop-2-directed antibody/ topoisomerase inhibitor. If PD-L1 positive and eligible, must have received checkpoint inhibitor
Ovarian Cancer	Must have high grade ovarian cancer (carcinosarcomas are allowed) and have received at least one line of platinum-based treatment.
Other solid tumor histology positive for *TP53* XYZ	Must have received at least 1 prior line of systemic therapy.

5. Tumors must harbor a *TP53* XYZvariant mutation, and subject must be HLA-A*02:01 positive (at least 1 allele) as confirmed by an CLIA-accredited laboratory-based test with one of the assays described in Appendix F.
6. Subject has at least 1 measurable lesion per computed tomography (CT) scan or magnetic resonance imaging (MRI) per RECIST version 1.1.
7. Females of childbearing potential must have a negative pregnancy test (serum or blood) at the time of enrollment. Contraceptive use by men and women should be consistent with local regulations regarding the methods of contraception for those participating in clinical studies. Refer to Section9.4.1for definition of childbearing potential and criteria for acceptable birth control methods.
8. Females of childbearing potential or fertile males with female partners of childbearing potential must agree to practice adequate contraception for at least 12 months after

completion of lymphodepleting chemotherapy or after ABC-01 treatment, whichever is longer (or longer as required by local requirement).

9. Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1 at the time of enrollment.
10. Adequate hematological, renal, hepatic, pulmonary, and cardiac function indicated by the parameters below at the time of enrollment:
    - Hemoglobin ≥ 8g/dL (independent of growth factors or infusion) within 2 weeks prior to enrollment)
    - Platelet count ≥ 100.000/µL (independent of growth factors or infusion) within 2 weeks prior to enrollment)
    - Absolute lymphocyte count ≥ 100/µL
    - Absolute neutrophil count ≥ 1000/µL (independent of growth factors) within 2 weeks prior to enrollment)
    - Creatinine clearance (CrCl) ≥ 50 mL/min as estimated by Cockcroft Gault. If the Investigator determines that Cockcroft Gault is not an appropriate assessment of subject renal function, other methods may be used (eg, 24-hour urine creatinine clearance test) with the approval of the Sponsor Medical Monitor.
    - Alanine aminotransferase/aspartate aminotransferase < 3x ULN range or <5x ULN if known liver metastasis (by local lab reference ranges)
    - Total bilirubin < 1.5 mg/dl, unless the subject has Gilbert's syndrome
    - Albumin ≥ 3g/dl
    - Left ventricular ejection fraction ≥ 50% assessed by echocardiogram (ECHO). If the Investigator determines that ECHO is not an appropriate assessment of ejection fraction, other methods may be used (eg, MRI, radionuclide angiography) with the approval of the Sponsor Medical Monitor.
    - No clinically significant arrhythmias
    - Baseline resting oxygen saturation > 92% on room air
    - No clinically significant ascites, pericardial or pleural effusion per Investigator judgement. Subjects with any evidence of ascites, pericardial or pleural effusion, even if not clinically significant per Investigator, should be discussed with the Sponsor Medical Monitor.
11. Subject is likely to complete study visits and/or procedures per the protocol and comply with study requirements for study participation per Investigator judgement.
12. Willing to provide archival and/or undergo biopsy procedure to obtain fresh tissue per study requirement detailed in Section7.11.5.

5.3.2. Exclusion Criteria

If an individual meets any of the following criteria, he or she is ineligible for this study:

1. Subject has another primary malignancy within the 3 years prior to enrollment, except for non-melanoma skin cancer, carcinoma in situ (eg, cervix, bladder, breast) treated with radical intent, or low-grade prostate cancer for which the watch-and-wait approach is SOC.
2. Subject has a history of stroke or transient ischemic attack within the 12 months prior to enrollment.
3. Subject has a history of myocardial infarction, cardiac angioplasty or stenting, unstable angina, cardiac arrythmia requiring antiarrhythmic or procedure, or other clinically significant cardiac disease within the 6 months prior to enrollment or heart failure (New York Heart Association Class III or IV functional classification) at any time prior to enrollment.
4. Subject has a history of allogeneic stem cell transplant or solid organ transplantation.
5. Subject has primary central nervous system (CNS) malignancy. Asymptomatic and stable CNS metastases are permitted with the approval of the Sponsor Medical Monitor. History of CNS disease is allowed if the subject received prior definitive therapy (eg, surgical resection or radiotherapy).
6. Subject has liver metastasis that involve > 50% of the liver (by Investigator assessment) or prior liver embolization within the last 6 months.
7. Subject has uncontrolled or symptomatic electrolyte abnormality.
8. Subject has prior adoptive cell and gene therapy. Subjects treated with specific cell therapy products that would not interfere with ABC-01 safety and efficacy could be eligible for this study and these cases must be discussed and agreed with the Sponsor Medical Monitor.
9. Subject has had radiotherapy or systemic anti-cancer therapy within at least 2 weeks or 3 half-lives, whichever is shorter, prior to enrollment.
10. Subject has had a major surgical procedure within the 4 weeks prior to enrollment or is planning to have at least one major surgical procedure at any point during study participation.
11. Subject has fungal, bacterial, viral, or other infection requiring antimicrobials for management within the 2 weeks prior to enrollment. Simple urinary tract infection and uncomplicated bacterial pharyngitis are permitted. Presence of any active bacterial, viral or fungal infection should be discussed with the Sponsor Medical Monitor.
12. Subject has active hepatitis B, or C. History of treated hepatitis B or C is allowed if the viral load is undetectable per quantitative PCR and/or nucleic acid testing.
13. Subject has a history of human immunodeficiency virus (HIV) 1 or 2, tuberculosis, syphilis, or human t-lymphocyte virus I or II infection.
14. Subject has a history of severe immediate hypersensitivity reaction to DrugC, DrugF, or rIL-2; or known sensitivity or allergy to methotrexate, gentamicin, or other aminoglycosides; discuss with the Sponsor Medical Monitor if known hypersensitivity to DMSO.

15. Subject has primary immunodeficiency.
16. Subject has had a live vaccine ≤ 4 weeks prior to enrollment or plans to have a live vaccine prior to planned lymphodepleting chemotherapy and/or ABC-01 treatment.
17. Subject has an active immune-mediated disease requiring systemic steroids or other immunosuppressive treatment (except if related to prior checkpoint inhibitor therapy); therefore, subjects taking > 10 mg/day of prednisone or equivalent doses of other corticosteroids are excluded. Topical or inhaled corticosteroids in standard doses and physiological replacements for subjects with adrenal insufficiency are allowed.
18. Subjects must have recovered from prior therapy-related toxicities to prior baseline or to Grade ≤ 1 (per Common Terminology Criteria for Adverse Events [CTCAE], v5.0) except for alopecia or vitiligo. Other or higher-grade prior therapy-related toxicities will be discussed with the Sponsor Medical Monitor. Subjects with immunotherapy-related endocrinopathies stable for > 6 weeks (eg, hypothyroidism) and controlled with hormonal replacement are allowed.
19. Subject is a female of childbearing potential who is lactating or breast feeding at the time of enrollment.
20. Subject is known to have Li-Fraumeni syndrome or is known to have relatives who are diagnosed with Li-Fraumeni syndrome.
21. Per Investigator judgement, subject has an unstable or clinically significant medical condition that would jeopardize the safety of a subject. The Investigator will consult with the Sponsor Medical Monitor if subject has a medical condition that is unstable or a condition that would potentially jeopardize the safety of the subject.

5.4. Prohibited Therapies

Subjects must abstain from taking prescription or nonprescription drugs (including vitamins, recreational drugs, and dietary or herbal supplements) within 7 days before the start of leukapheresis until completion of the follow-up visit, unless, in the opinion of the Investigator and Sponsor, the medication will not interfere with the study.

The therapies below are prohibited for use during the specified time periods:

- Live vaccine for at least 4 weeks prior to lymphodepleting chemotherapy and up to 12 weeks after ABC-01 administration until hematologic toxicity has recovered to the baseline or ≤ CTCAE Grade 1.
- Corticosteroids at a pharmacological dose (> 10 mg/day prednisone or equivalent dose of other corticosteroids) and other immunosuppressive drugs are prohibited during the time periods specified below but may be continued if required as replacement therapy.
    - 7 days prior to leukapheresis
    - 5 days prior to ABC-01 administration
    - 3 months after ABC-01 administration, unless used to manage ABC-01-related toxicities or other reasons as discussed with the Sponsor Medical Monitor.

Topical or inhaled corticosteroids in standard doses and physiological replacements for subjects with adrenal insufficiency are allowed.

- Therapeutic doses of systemic anticoagulants, such as unfractionated heparin and low molecular weight heparin, should be avoided whenever possible for subjects who are at risk of bleeding due to thrombocytopenia.
- Treatments for the primary disease (eg, chemotherapy, immunotherapy, targeted agents, radiation, high-dose corticosteroids) and other investigational agents are prohibited, except as needed for bridging therapy after enrollment, or as needed for disease progression after ABC-01 treatment.

5.5. Subject Withdrawal

5.5.1. Reasons for Removal from Treatment or Study Procedures

Subjects may be withdrawn from treatment (eg, not able to receive or complete treatment with ABC-01) or from completing study visits/procedures for any of the reasons listed below.

- Subject no longer meets enrollment eligibility criteria defined in Section5.3
- Product not available (eg, Sponsor is unable to manufacture ABC-01 or is unable to release ABC-01 for subject treatment under this protocol)
- Subject is not in need of therapy for their primary disease
- Subject is not able to complete the lymphodepleting regimen per protocol
- Adverse event
- Withdrawn consent (see Section5.5.3)
- Lost to follow-up
- Disease progression per RECIST v1.1
- Disease progression- clinical deterioration
- Death
- Investigator decision
- Termination of study by Sponsor
- Other (eg, noncompliance)

Enrolled subjects who received ABC-01 should be asked to complete an Early Exit visit assessment and continue to be followed per study follow-up assessment schedule. See Section8.6.5for further details regarding Early Exit visit and subsequent follow-up.

5.5.2. Reasons for Removal from Study

Subject may be removed from participation in the study for the reasons below:

- Full withdrawal of consent (see Section5.5.3.2)

- Lost to follow-up
- Death
- Investigator decision
- Termination of study by Sponsor
- Other (eg, noncompliance)

5.5.3. Withdrawn Consent

Subjects have the right to withdraw from the study at any time, for any reason, with no prejudice to their medical care by the physician or at the institution. The Investigator will inform the Sponsor of a subject's decision to withdraw consent. The withdrawal and the type of withdrawn consent (partial withdrawal, full withdrawal, withdrawal for use of samples, withdrawal from optional studies) will be documented in the study source documents.

Subject data collected up to withdrawal of consent will be retained and included in the analysis of the study and, where permitted by local regulations, publicly available data (eg, death records) may be included after withdrawal of consent.

5.5.3.1. Partial Withdrawn Consent

Subjects may withdraw consent but agree to continue to be followed by the procedures outlined in Section8.6.5. Subjects have the right to withdraw from any study procedure at any time which will be discussed with the Investigator and documented in the study source documents.

5.5.3.2. Full Withdrawn Consent

Subjects may fully withdraw consent from any follow-up for the study. Subjects who fully withdraw consent will not follow any study-related procedures from the date of fully withdrawn consent.

5.5.3.3. Withdrawn Consent for Use of Samples

If the subject withdraws from the study, Sponsor may retain and continue to use any samples collected before such a withdrawal of consent for the purposes the participant originally consented (ie, future research in line with the original consent, including but not limited to biomarker development, assay bridging or validation, and further understanding of the disease) unless the subject withdraws consent for use of samples already collected. If the subject only withdraws consent for use of their samples for future research, the samples will be destroyed after they are no longer needed for the clinical study. The Sponsor is not obligated to destroy data that has been generated from samples analyzed prior to the withdrawn consent.

Certain samples must be retained by regulation or law (eg, manufacturing retain samples). The Sponsor will inform the Investigator of samples that may not be destroyed or returned.

The Investigator will inform the Sponsor of the type of withdrawn consent for sample storage (eg, future storage and/or including destruction of previously stored samples). Withdrawn consent specific to sample storage for future study analysis or for future research is expected to be documented in the site study records.

5.5.3.4. Withdrawn Consent for Optional Studies

The Investigator will inform the Sponsor ofthe withdrawn consent along with the date of withdrawn consent.

5.5.3.5. Replacement of Subjects

Subjects will be enrolled and treated at each dose level of Phase 1 until the specified number of subjects required for the DLT-evaluable set is reached (see Section10.2.1). If a subject was enrolled and dosed but did not receive the full intended dose (ie, lower dose than the assigned dose level or not treated with ABC-01), an additional subject may be enrolled as replacement and dosed at the same dose level.

5.5.3.6. Subjects Lost to Follow-Up

A subject will be considered lost to follow-up if the subject repeatedly fails to return for scheduled visits and the study site is unable to contact the subject.

6. STUDY INTERVENTIONS

Study interventions are defined as all prespecified, investigational and non-investigational medicinal products, medical devices, and other interventions (eg, surgical and behavioral) intended to be administered to the study subjects during the study conduct.

6.1. Description of Products

ABC-01 is a sterile cell suspension consisting of autologous T cells that have been genetically engineered ex vivo with CRISPR-Cas9 to express a HLA-A*02:01-restricted TCR specific for p53, refer to Section2.1.3.

The active substance is the genetically engineered T cells (eTCR+ T cells), which are formulated in a cryopreservation medium suitable for infusion and supplied cryopreserved at a temperature of ≤ -150°C in cryostorage bags.

ABC-01 will be administered via IV infusion at doses up to 10×10^9 eTCR+ T cells.

A subject's leukapheresed cells are processed to isolate and activate T cells. The synthetic DNA repair template encoding the XYZneoantigen-specific TCR is introduced into the isolated and activated T cells by electroporation and site-specifically integrated into the endogenous TCRα locus using CRISPR-Cas9, which is also used to knock out endogenous TCR and TGFBR2. The eTCR+ T cells are enriched and expanded. The manufactured product lot will undergo final formulation and testing for release.

ABC-01 will be supplied for each subject by the Sponsor. The product will be labeled and delivered with documentation specific to the subject to be infused.

Refer to the current version of the Investigational Product Manual (IPM) for additional information on procurement of leukapheresed cells for ABC-01 manufacturing. Refer to the current version of the IB for additional information on ABC-01 eTCR cells.

Additional details on dosing and administration of ABC-01, as well as details on pre- and post-medications are provided in Section8.5.1.2.

6.1.1. Leukapheresis

Subjects will undergo leukapheresis to obtain leukocytes (white blood cells [WBCs]) for the manufacture of ABC-01. The apheresis material bag is labeled with protocol number, subject identifier(s), and date of leukapheresis.

Refer to the (IPM) for more information regarding procedures for the preparation of apheresis material for the manufacture of ABC-01 at the manufacturing site. Refer to Section8.3 for more information regarding leukapheresis study visit procedures.

6.1.2. Lymphodepleting Chemotherapy

Lymphodepleting chemotherapy will be administered on Day -7 through Day -3 to induce lymphocyte depletion and create an optimal environment for expansion of ABC-01. Lymphodepleting chemotherapy must be repeated if ABC-01 administration is delayed for more than 14 days after termination of lymphodepletion unless agreed between Investigator and

Sponsor Medical Monitor. In all cases of ABC-01 infusion delays, contact the Sponsor Medical Monitor for guidance.

The lymphodepleting chemotherapy regimen will consist of:

- DrugC, 30 mg/kg IV administered on Day -7 and Day -6 infused over approximately 60 minutes. DrugC dose may be adjusted for renal insufficiency per institutional guidelines after discussion with the Sponsor Medical Monitor.
- Mesna (per PI discretion or institutional guidelines, unless contraindicated) administered on Day -7 and Day -6
- DrugF, 25 mg/m^2 IV on Day -7, Day -6, Day -5, Day -4, and Day -3 infused over approximately 30 minutes. DrugF dose may be adjusted for renal insufficiency per institutional guidelines after discussion with the Sponsor Medical Monitor.

Premedication for nausea and vomiting can be administered per institutional guidelines.

After at least 3 subjects are dosed at a given dose level, alternative lymphodepletion regimens combining DrugF and DrugC may be explored after review of initial data, in consultation with the SRT. Alterations to the lymphodepleting chemotherapy regimen may include modification of the DrugC and/or DrugF dosage and timing, or duration.

Any changes in the dose or other aspects of the lymphodepleting chemotherapy regimen must be discussed with the Sponsor Medical Monitor prior to administration.

Refer to Section8.4.1 for information on study procedures related to lymphodepleting chemotherapy administration.

6.1.3. Optional Bridging Therapy

Subjects may receive bridging therapy after leukapheresis and before initiation of lymphodepletion, during ABC-01 manufacturing. This bridging therapy will be administered per Investigator's discretion (excluding immunotherapy such as checkpoint inhibitors or agents with half-life longer than 1 week), only if clinically required, and should be discussed and agreed by the Sponsor medical monitor. Subjects must have recovered from treatment-related toxicities (≤ Grade 1) and must have had appropriate washout period (at least 2 weeks or 3 half-lives, whichever is shorter) prior to initiating lymphodepleting chemotherapy for the study. If the subject has higher grade treatment-related toxicities (> Grade 1) following bridging therapy, a discussion with the Medical Monitor must occur prior to initiating lymphodepleting chemotherapy.

If bridging therapy is administered, baseline disease assessment should be scheduled after completion of bridging therapy and prior to initiating lymphodepleting chemotherapy on Day -7.

Investigators should refer to the current product labels for guidance on packaging, storage, preparation, administration, and toxicity management of the bridging therapy.

6.1.4. ABC-01

6.1.4.1. Packaging, Labeling, Storage, and Handling

After manufacture and release, ABC-01, supplied frozen in cryo-storage bags, will be shipped to the study center, where it will be stored in the vapor phase of liquid nitrogen (eg, liquid nitrogen freezer) until the subject is ready to receive treatment. The Investigator ensures that verification of the product and subject-specific labels and documents to the subject's information (eg, subject study ID number) is complete at initial receipt of ABC-01.

For cases where ABC-01 is manufactured but does not meet certain release specification criteria, the Sponsor will notify the Investigator to discuss treatment options, which may include product remanufacture. Relevant country regulations will be followed, and notifications to applicable regulatory agencies and Institutional Review Board (IRB)/Independent Ethics Committee (IEC) will be done as necessary per local requirements.

Refer to the IPM for more information on ABC-01 packaging and labeling, instructions for receipt, handling, and storage, and supplies.

6.2. Dosage and Administration

ABC-01 will be administered in an inpatient setting and must not be infused without verification that the subject information on the label and accompanying documents matches the intended subject (eg, subject ID number).

Once the subject is confirmed to be eligible to receive treatment with ABC-01, the cryopreserved product will be thawed and infused IV utilizing a central venous catheter. Total infusion time should not exceed 30 minutes.

Refer toTable 2for the planned dose levels. Refer to Section8.5.1.1for eligibility criteria for ABC-01 administration and to the SOA (Appendix B) for study procedures and assessments related to ABC-01 administration.

Refer to the IPM for more information on the procedures for investigational product (IP) thawing and other preparation steps for administration.

6.3. Recombinant IL-2 Treatment

Recombinant IL-2 will be administered to support T cell expansion and persistence.
Recombinant IL-2 will be administered SC starting on Day 0, approximately 4 hours following ABC-01 treatment at a dose of 300,000 IU/kg. Recombinant IL-2 administration may be delayed for up to 24 hours after ABC-01 treatment due to an AE or other clinical reasons. Recombinant IL-2 should subsequently be administered once daily for a maximum of 8 doses. Acetaminophen may be administered during rIL-2 administration to prevent pyretic reactions.

Administration of rIL-2 may be discontinued or interrupted as clinically necessary per institutional guideline. Investigator will inform the Sponsor Medical Monitor if rIL-2 administration is discontinued or interrupted.

After at least 3 subjects are dosed at a given dose level, rIL-2 doses, timing, mode of administration (SC or IV) and/or duration may be adjusted after review of initial data, in

consultation with the SRT. Refer to Section8.5.1.3for information on rIL-2 administration and study procedures related to rIL-2 administration.

6.4. Accountability and Return of Investigational Product

6.4.1. Accountability of Investigational Product

The Investigator is responsible for ensuring adequate records of accountability of all used and unused IP is maintained. Records must be made available to the study monitor for review.

Refer to the IPM for more information on IP accountability.

6.4.2. Return or Disposal of Investigational Product

If IP is unused (eg, subject is not eligible for treatment), the IP must remain in proper storage conditions (eg, ≤ 150°C in liquid nitrogen vapor) until further instruction from the Sponsor. The IP may be disposed as permitted by the Sponsor or used for research purposes related to the clinical trial as approved by the IRB/IEC.

Refer to the IPM for more information on IP return and/or disposal.

7. STUDY ASSESSMENTS

Study assessments and procedures described in this section will be performed at the visits and timepoints presented in the SOA tables provided inAppendix B.

Protocol waivers or exemptions are not allowed.

Refer to the electronic Case Report Form (eCRF) guidelines for more information on data expected to be captured in the eCRFs.

7.1. Informed Consent

All subjects will provide written informed consent to participate in the study. Subjects must be consented by a process that is in accordance with ICH/GCP and in compliance with any local regulation or law. Subjects should be consented with, and sign, the most current study ICF that is approved by the local IRB/IEC. Refer to Section11.3for information regarding the informed consent process.

7.2. Demographics

Demographic data will be collected with the intent to assess a possible association of demographics with subject safety and treatment effectiveness. The following information will be obtained for each subject: age, sex, race, and ethnic origin. Demographic data will be collected per country and local regulations and guidelines.

7.3. Medical History

Medical conditions current at the time of enrollment and any relevant past medical history prior to enrollment will be recorded.

7.4. Diagnosis History

Information about the subject's primary disease for this study will be collected. Primary disease data will include but may not be limited to histology, additional data related to the histology (as applicable), disease history, and disease stage. As applicable and per institutional guidelines, disease-specific tumor markers (eg, carcinoembryonic antigen for colorectal adenocarcinoma) will be collected as part of diagnosis history.

7.5. Prior Therapy for Primary Disease

Information about prior therapies (eg, chemotherapy, immunotherapy, targeted therapy, radiotherapy, surgery) for the primary study disease data will be collected. Additional data related to the type of therapy may be collected as applicable.

7.6. Prior and Concomitant Therapies

Any medication and therapies, including medications, oxygen, and blood products, current at the time of screening, will be reported as prior concomitant medications. Therapies initiated after screening that are related to any SAE or to AEs related to study intervention will be reported as concomitant medications.

The Sponsor Medical Monitor should be contacted if there are any questions regarding concomitant or prior therapy.

During the study, the Investigator may also prescribe concomitant therapies that are necessary to provide adequate supportive care or used to treat AEs. List of prohibited or excluded medications are described in Section5.4.

All therapies initiated after the date of screening informed consent, or changes to therapies that occurred after screening informed consent, will be reported from the date of informed consent through 3 months after completing treatment with ABC-01. For subjects who enroll but are not treated with ABC-01, the reporting period for concomitant medications is from date of screening informed consent through 30 days through the last study visit completed. For subjects who screen fail, only concomitant medications related to SAEs are reported.

For enrolled and treated subjects, targeted concomitant medications will also be reported between 3 months and up to 24 months after completing treatment with ABC-01. Criteria for targeted concomitant medications are below:

- Immunosuppressive drugs
- Anti-infective drugs
- Vaccinations

7.7. Prior and Concomitant Procedures

Therapeutic procedures (eg, intubation, dialysis), current at the time the screening ICF was signed will be reported as prior procedure.

Procedures initiated after date of screening informed consent, or changes to procedures that occurred after screening informed consent, will be reported from the date of informed consent through 3 months after completing treatment with ABC-01. For subjects who enroll but are not treated with ABC-01, the reporting period for procedures is from date of screening informed consent through 30 days through the last study visit completed. For subjects who screen fail, only procedures related to SAEs are reported.

For enrolled and treated subjects, procedures related to targeted AEs will also be reported after 6 months and up to 24 months after completing treatment with ABC-01. Refer to Section9.1 for targeted AE criteria.

7.8. Subsequent Therapy After ABC-01 Treatment

"Subsequent therapy" refers to treatment administered for disease progressions after ABC-01 treatment, which includes chemotherapy, immunotherapy, targeted agents, radiation therapy, surgery, and other investigational agents.

Subsequent therapy data will be collected until the subject completes follow-up, withdraws consent, is lost-to-follow-up, or dies.

7.9. Safety Assessments

Planned timepoints for all safety assessments are provided in the SOA tables (Appendix B).

7.9.1. Adverse Events

Adverse events will be monitored from the time of screening informed consent through Month 24 after ABC-01 treatment, disease progression, or early study exit (see Section5.5 for early study exit criteria), whichever occurs first. The Investigator is responsible for ensuring AEs are reported and assessed per protocol requirements.

Refer to Section9for more information on AE definition, AE assessment, and reporting requirements.

7.9.2. Local Clinical Laboratory Tests

Local laboratory assessments include the panels and tests inTable 3. Abnormal laboratory values or clinically significant changes in values will be assessed by the Investigator to determine whether findings that are deemed clinically significant should be reported as medical history (if prior to enrollment), or an AE (if after enrollment). Refer to Section7.3for more information on medical history reporting and Section9.1for more information on AE reporting. Local laboratory assessments may be repeated if clinically indicated or for other reasons permitted by the Sponsor Medical Monitor.

Table 3: Local Laboratory Tests

Chemistry Panel	CBC with Differential	Other
Albumin ALT Alkaline phosphate AST Bicarbonate (CO2) total (only if clinically indicated) Total bilirubin Direct bilirubin BUN or urea (if blood urea nitrogen cannot be assessed locally) Calcium total Chloride Creatinine Glucose Inorganic phosphorous Lactate (only if clinically indicated) LDH Magnesium total Potassium Sodium Uric acid	WBC count Neutrophil ANC Lymphocyte ALC Monocyte (%) Absolute monocyte count Eosinophil Absolute eosinophil count Hemoglobin Platelets	CRP Ferritin Serum Pregnancy Test[a]
		Coagulation
		Prothrombin time (PT) Activated partial thromboplastin time (aPTT) International normalized ratio (INR) fibrinogen

Table 3: Local Laboratory Tests (Continued)

Chemistry Panel	CBC with Differential	Other
	Viral Serology	**Disease-specific tumor markers (if applicable)** [b]
	HIV-1, HIV-2 HBsAg & HBcAb (if positive, a PCR test should be performed) HCV antibody/PCR (if the HCV antibody is positive, a screening HCV RNA by any RT-PCR or DNA assay should be performed)	Carcinoembryonic antigen (CEA) for colorectal, lung, stomach, thyroid, pancreatic, breast, and ovarian cancers Carbohydrate antigen 19-9 (CA 19-9) for pancreatic, gallbladder, bile duct, and gastric cancers Carbohydrate antigen 125 (CA- 125) for ovarian cancer Cancer antigen 15-3 (CA15-3) or Cancer antigen 27-29 (CA 27-29) for Breast cancer Prostate Specific Antigen (PSA) for prostate cancer Alpha Fetoprotein (AFP) for liver cancer, and cancers ofthe ovary or testicles

Abbreviations: AFP, alpha fetoprotein; ALC, absolute lymphocyte count; ALT, alanine aminotransferase; ANC, absolute neutrophil count; aPTT, activated partial thromboplastin time; AST, aspartate aminotransferase; BUN, blood urea nitrogen; CA, carbohydrate antigen; CBC, complete blood count; CEA, Carcinoembryonic antigen; CRP, C-reactive protein; DNA, deoxyribonucleic acid; HIV, human immunodeficiency virus; HBcAb, hepatitis B core antibody; HBsAg, hepatitis B surface antigen; HCV: hepatitis C virus; INR, international normalized ratio; LDH, lactate dehydrogenase; PCR, polymerase chain reaction; PSA, prostate specific antigen; PT, prothrombin time; RT-PCR, real-time reverse transcription-polymerase chain reaction assay; WBC, white blood cell

[a] Pregnancy test: for subjects who are female of childbearing potential (see Section9.4.1for childbearing potential definition).

[b] Disease-specific tumor markers will be evaluated only in relevant histologies. Any of these tumor-marker assessment may be performed for histologies not specified as appropriate.

Blood, serum, and other tissues samples will be collected and shipped to the central laboratory and/or other specialty laboratory for safety assessments, translational assessments, and other exploratory assessments and testing. See Section7.11for more details related to central laboratory assessments and testing.

7.9.2.1. Creatinine Clearance

Creatinine clearance is estimated by Cockcroft Gault. If the Investigator determines that Cockcroft Gault is not an appropriate assessment of subject renal function, other methods may be used (eg, 24-hour urine creatinine clearance test) with the approval of the Sponsor Medical Monitor.

7.9.2.2. Pregnancy Testing

Serum pregnancy tests are to be performed for subjects who are female of childbearing potential. Refer to Section9.4.1for definition of female of childbearing potential.

7.9.3. Physical Examination with Neurological Examination

A full physical examination will be performed at the Screening Visit and will include assessment of the following body systems: head, eyes, ears, nose, throat; lymph nodes; skin; extremities; respiratory system; gastrointestinal system; musculoskeletal system; cardiovascular system; and CNS.

Physical and neurological examination abnormalities or changes in clinically significant abnormalities will be assessed by the Investigator to determine whether findings that are deemed clinically significant should be reported as medical history (if prior to enrollment), or AE (if after enrollment). Refer to Section7.3 for more information on medical history reporting and Section9.1 for more information on AE reporting. For ICANS assessments, refer to Section9.7.1.1.

Post-screening physical examinations with neurological examination, including inpatient hospitalization for study treatment, will be performed per institution procedures and/or guidelines.

Weight and height will also be collected at the timepoint specified in the SOA, refer to Appendix B.

7.9.4. Vital Signs

Vital signs assessments will include resting blood pressure (systolic and diastolic), heart rate, oxygen saturation, and temperature. During inpatient hospitalization, vital signs will be measured before, during (if feasible), and within approximately 15 minutes after ABC-01 treatment and then routinely during hospitalization per institution standard procedure and/or guidelines.

The Investigator will assess vital signs abnormalities or changes in clinically significant abnormalities to determine whether findings that are deemed clinically significant should be reported as medical history (if prior to enrollment) or an AE (if after enrollment). Refer to Section7.3 for more information on medical history reporting and Section9.1 for more information on AE reporting.

7.9.5. Immune Effector Cell-Associated Encephalopathy Score

The immune effector cell-associated encephalopathy (ICE) score will be used to assess ICANS. Refer to Table 13 through Table 15 in Appendix C for more information related to grading and management of ICANS.

Baseline ICE score should be assessed prior to ABC-01 treatment and maintained in the source documents.

7.9.6. Eastern Cooperative Oncology Group

Performance status as measured by the ECOG scale will be performed.

7.9.7. Brain Magnetic Resonance Imaging

A brain MRI will be performed during screening to rule out CNS metastasis for subjects who are considered at high-risk for CNS metastasis (eg, NSCLC, breast cancer, colon, or other histology considered high-risk of CNS metastasis by the Investigator or at the request of the Medical

Monitor, who have a history of CNS metastasis, or if there is clinical suspicion of CNS involvement. A brain MRI performed after the completion of the subject's last systemic treatment and within the 28 days prior to signing screening informed consent may be used as the brain MRI to confirm enrollment eligibility. Refer to Section5.3 for eligibility criteria related to CNS metastasis.

A brain MRI may be performed as part of evaluation of new-onset ICANS that occur after ABC-01 treatment.

The MRI will be performed with contrast but may be performed without contrast in case of contraindication. A CT of the brain may be performed if MRI is contraindicated.

7.9.8. Cardiac Function

7.9.8.1. Electrocardiograms

A 12-lead electrocardiogram (ECG) will be performed during screening to confirm eligibility for enrollment. ECG assessments will include durations of PR, QRS, QT, and RR, as well as corrected QT (QTc) interval calculated using Fridericia's formula (QTcF) (Fridericia, 2003).

Abnormalities will be assessed by the Investigator to determine whether findings that are deemed clinically significant should be reported as medical history.

7.9.8.2. Echocardiogram

An ECHO will be performed during screening to confirm eligibility for enrollment. An ECHO performed after the last chemotherapy regimen and within the 28 days prior to signing the screening ICF may be used to assess enrollment eligibility and does not need to be repeated at the Screening Visit.

If the Investigator determines that ECHO is not an appropriate assessment of ejection fraction, other methods may be used (eg, MRI, radionuclide angiography) with the approval of the Sponsor Medical Monitor.

Abnormalities will be assessed by the Investigator to determine whether findings that are deemed clinically significant should be reported as medical history.

7.9.9. Lumbar Puncture

Lumbar puncture (LP) may be performed after ABC-01 treatment to assess new onset of clinically significant ICANS, if clinically feasible. LP is required for any Grade 2 or higher ICANS event. Collected CSF should be sent for standard panel including at least CSF color, clarity, protein, glucose, total number of cells and CSF differential cell count. If infection is suspected, CSF Gram stain, routine culture and molecular tests that detect the genetic material of any microbes present, eg, HSV-1, HSV-2, enterovirus, varicella zoster virus, cytomegalovirus and HHV-6 should be conducted. The Sponsor may request cerebral spinal fluid (CSF) samples to be shipped to the central laboratory and/or specialty laboratory for assessments of key analytes (eg, cytokines, chemokines) and presence of eTCR. The opening pressure with each lumbar puncture will be recorded.

7.9.10. Collection of Additional Body Fluids

Collection of body fluids, such as fluid ascites or pleural effusions, may be performed after ABC-01 treatment to assess new onset of clinically significant AEs. The Sponsor may request samples to be shipped to a central laboratory for assessments of key analytes (eg, cytokines, chemokines) and presence of eTCR+ T cells.

7.10. Disease Assessments

Disease will be assessed by imaging and disease-specific tumor marker measurement in serum as detailed below.

7.10.1. Imaging Assessment

Planned timepoints for all efficacy assessments are provided in the SOA (Appendix B).

CT scans with contrast (or MRI scans) will be done at the Screening Visit to confirm eligibility for enrollment (at least one measurable lesion per Section5.3.1). A recent CT or MRI scan performed per standard of care within 28 days prior to signing the screening ICF may be used to confirm enrollment eligibility and does not need to be repeated if, in the opinion of the Investigator, the clinical status of the subject is unchanged.

- Baseline CT or MRI scan may be performed up to the 15 days prior to initiating lymphodepleting chemotherapy on Day -7 but should be performed as close to Day -7 as feasible.
- If bridging therapy was administered (see Section6.1.3), baseline disease assessment will be done after the completion of bridging therapy and prior to initiating lymphodepleting chemotherapy on Day -7.
- Disease will be assessed by the Investigator per RECIST guidelines v1.1. using the most recent CT or MRI scans, until disease progression or initiation of subsequent anticancer treatment.
- After the Month 24 visit, disease will be assessed by SOC. Refer to Table 11 and Table 12 for the schedule of post-treatment disease assessment timepoints. CT scans with contrast (or MRI scans) of the neck (if applicable), chest, abdomen, pelvis, and other sites of disease will be performed.

The same imaging modality should be used throughout the study if possible. Changes in imaging modality should be discussed with the Sponsor Medical Monitor (eg, if subject develops a contraindication to the IV contrast for CT). The same technique should be used to characterize each identified lesion.

Lesions identified on chest X-ray should be imaged by CT or MRI scan. Ultrasound and fluorodeoxyglucose-positron emission tomography (FDG-PET) alone cannot be used to assess overall tumor response or disease progression. An FDG-PET and a CT scan with contrast (unless contraindicated) or MRI scan will be performed at baseline (prior to initiating lymphodepleting chemotherapy). The FDG-PET may be performed after ABC-01 treatment as determined to be clinically necessary by the Investigator.

Measurability of lesions on CT scan is based on the assumption that CT slide thickness is 5 mm or less. If the slice thickness is greater than 5 mm, the minimum size for a measurable lesion should be at least 10 mm or twice the slice thickness, whichever is greater.

Imaging data (scans) will be submitted centrally and stored by a central imaging vendor designated by Sponsor for future independent central review. The Investigator's assessment will be used for the data analysis and for treatment decision making. For subjects who discontinue the study due to disease progression, any additional imaging data, subsequent to the image used for disease progression assessment, may be submitted to the central imaging vendor.

7.10.2. Disease-specific Tumor Markers

Tumor markers will be measured in serum at specific timepoints to coincide with imaging assessments as detailed in SOA (Appendix B).

According to the type of tumor, the following tumor marker assessment will be performed by local laboratory in relevant histologies.

Tumor markers alone will not be used for response evaluation.

- Carcinoembryonic antigen for colorectal cancer, lung, stomach, thyroid, pancreas, breast, and ovary
- Carbohydrate antigen 19-9 for pancreatic, gallbladder, bile duct, and gastric cancers
- Carbohydrate antigen 125 for ovarian cancer
- Cancer antigen 15-3 or Cancer antigen 27-29 for Breast cancer
- Prostate Surface Antigen for prostate cancer
- Alpha Fetoprotein for liver cancer, and cancers ofthe ovary or testicles.
- Any of the above marker(s) may be assessed for disease histology not mentioned above, if appropriate.

7.11. Central Laboratory Assessments

7.11.1. Sample Collection for Central Assessment

The samples below will be collected and shipped to the central lab and/or specialty lab for central assessment.

Sponsor may request discontinuation of sample collection. Continue sample collection for all listed time points until otherwise instructed by Sponsor. Sites should refer to the Laboratory Manual for details about sample collection requirements.

- Serum/plasma:
  - Immunogenicity studies (eg, anti-ABC-01 antibodies)
  - Key analytes (eg, cytokines, chemokines)
  - Pharmacodynamics (refer to Section7.11.1.2for more information on pharmacodynamics assessment)

  - Exploratory assessments (see Section7.11.2)
- Whole blood:
  - Peripheral blood mononuclear cells (PBMCs)
    - PK (refer to Section7.11.1.1for more information on PK assessment)
    - TCR sequencing
  - Complete blood count (CBC) with differential
  - Cell-free DNA (cfDNA)
  - Exploratory assessments (see Section7.11.2)
- Tumor biopsy:
  - Archived tumor tissue: see Section7.11.4for details.
  - Fresh tumor biopsy: see Section7.11.5for details.
- CSF:
  - Sponsor may request CSF samples collected as part ofthe evaluation of new onset ICANS after ABC-01 treatment to be sent to the central and/or specialty laboratory for exploratory and/or additional safety analysis.
  - Refer to Section7.9.9for more information regarding the lumbar puncture assessment for this study.
- Body fluids
  - Sponsor may request body fluid (eg, fluid from ascites, pleural effusions, etc.) samples collected as part ofthe evaluation ofnew or worsening AEs after ABC-01 treatment to be sent to a central laboratory for exploratory and/or additional safety analysis.

7.11.1.1. Pharmacokinetic Assessments

Whole blood samples will be collected and processed to isolate PBMCs for the measurement of expansion and persistence of ABC-01 in circulation.

Samples will be collected prior to ABC-01 infusion (baseline) and at multiple timepoints after dosing of ABC-01. Instructions for the collection and handling of biological samples will be provided by the Sponsor.

Samples will be used to evaluate the PK of ABC-01 by dPCR and/or flow cytometry. Samples may also be utilized for additional quantification and characterization ofT cell subsets and for additional quantification and characterization ofT cells in periphery utilizing next-generation sequencing analyses.

7.11.1.2. Pharmacodynamic Assessments

Serum and plasma samples for pharmacodynamics assessments will be collected and will include monitoring of key analytes in serum (eg, cytokines, chemokines) over time. Key analytes include but may not be limited to the below:

- Homeostatic/proliferative cytokines: interleukins (IL-2, IL-7, IL-15)
- Inflammatory/immune-modulation cytokines: interferon gamma, IL-6, IL-10, IL-17A, IL-1β, IL-12, granulocyte macrophage-colony stimulating factor, and tumor necrosis factor-alpha
- Correlates of acute phase responses: CRP and ferritin
- Chemokines: IL-8, C-X-C motif chemokine ligand-10, macrophage inflammatory protein 1-alpha, and monocyte chemotactic protein-1

7.11.1.3. Genetics

Peripheral blood mononuclear cells and/or tumor material will be collected from subjects. DNA and/or RNA will be isolated from PBMCs and/or tumor samples and subsequently subject to sequencing.

7.11.1.4. Biomarkers

Biomarkers associated with treatment with ABC-01 are not known and will not be evaluated in this study.

7.11.2. Exploratory Assessments

Blood, serum, and other tissues and samples will be collected and shipped to the central laboratory and/or other specialty laboratory for exploratory assessments and testing as described below:

- PBMCs will be collected for cellular immunogenicity studies, and for exploratory analyses, including characterization of lymphocyte subsets.
- PBMCs and samples of ABC-01 will be banked and analyzed by immunophenotyping, digital polymerase chain reaction, gene expression profiling, and/or subject to functional tests in vitro.
- Whole blood will be collected for cfDNA analysis.
- Serum will be collected for immunogenicity studies, and for evaluation of circulating cytokines and chemokines; see Section7.11.6.

7.11.3. HLA-A*02:01 Test

For subjects without prior documented HLA-A*02:01 positivity using one of the assays described in Appendix F, a blood sample for central HLA testing will be collected at the Prescreening Visit. Subjects who have documented HLA-A*02:01 positivity from an assay other than that described in Appendix Fmay proceed to Screening, however a sample must be provided for central HLA testing during the Screening Period. For subjects who are not required to complete a Prescreening Visit, refer to Section8.1.

7.11.4. Tumor Tissue

Archival or fresh tumor tissue for *TP53* XYZtesting and archival tumor tissue for translational analysis have independent requirements and may be shipped to different laboratories.

- For subjects without prior documented *TP53* XYZpositivity using one of the assays described in Appendix F, a tumor tissue sample will be shipped to a Sponsor -designated central laboratory at the Prescreening Visit. Refer to Section 8.1.1, for requirements regarding *TP53* XYZ testing.
- Archival tumor tissue (FFPE block or approximately 30 unstained slides including 1 H&E slide when available) for translational analysis will be shipped to the central laboratory and/or specialty laboratory at least 7 days prior to starting lymphodepletion chemotherapy for immunohistochemistry and genomic/transcriptomic studies. If archival tumor tissue is not available, a biopsy will be procured and shipped to the central laboratory and/or specialty laboratory if determined to be clinically feasible without conferring undue risk upon the subject.

7.11.5. Fresh Tumor Biopsy

All biopsy tissue will be sent to a central laboratory.

Fresh tumor biopsies will be taken as described below only if determined to be clinically feasible by the Investigator and with the consent of the subject. Institutional guidelines should be followed regarding additional consents required at the time of biopsy. Biopsies are procedures routinely performed and are considered to carry manageable risks, with an expected morbidity or mortality rate of < 2%.

Tumor tissue (pretreatment and post-treatment tumor biopsies collected from the study) will be used for translational analysis, which includes evaluation of ABC-01 migration to tumor site, expression of neoantigens by tumor cells, frequency of neoantigen-specific eTCR cells, composition of the tumor microenvironment, and other exploratory analyses.

An attempt should be made to biopsy lesions that will not impact disease response assessment, and the same lesions should be biopsied at all timepoints if medically feasible.

Biopsies will be required at the following timepoints.

- Baseline: A pretreatment (baseline) biopsy will be collected for all subjects and sent to a central laboratory. The archival biopsy (described in Section7.11.1.4) may be used as the baseline material if the biopsy was completed within 6 months prior to enrollment. If the archival biopsy is older than 6 months from enrollment, there is no archival tissue available, or ifthe archival tissue is ofinsufficient volume or quality to fulfill central laboratory requirements, a fresh biopsy will be collected at least 7 days prior to initiating lymphodepleting chemotherapy, as feasible. This sample will serve as a comparator for biopsies collected post-treatment.
- Day 28: A fresh tumor biopsy will be collected on Day 28. If Day 28 collection is not feasible, the fresh biopsy may be collected at Week 6; if not feasible at Week 6, then it may be collected at Week 12.
- Month 12 (if occurs before Disease Progression)
- Disease progression

An optional fresh tumor biopsy may be collected at Day 7 (-2/+7 days) for subjects consenting to the additional procedure.

7.11.6. Immunogenicity Assessments

Antibodies to ABC-01 will be evaluated in serum samples collected from study subjects according to the SOA (Appendix B). These samples will be tested by the Sponsor or designee.

Serum samples will be screened for antibodies binding to the TCR expressed on ABC-01 drug products. Other analyses measuring T cell immunogenicity may be performed to further characterize the immunogenicity of ABC-01.

The detection and characterization of antibodies to ABC-01 will be performed using a qualified assay method by or under the supervision of the Sponsor.

7.11.7. Sample Storage

Blood, tissue, and other samples, including derivatives from these samples, may be stored for up to 15 years from the date of the last subject dosed for the study. Subjects have the right to have their sample material destroyed at any time by informing the Investigator. The Investigator will inform the Sponsor of requests by any subjects for sample material destruction or ifany subject withdraws their consent for sample storage. For subjects who withdraw consent for sample storage, samples collected up to the date of withdrawn consent for sample storage will remain in storage unless requested to be destroyed by the subject. Samples collected after the date of withdrawn consent will not be stored for future analysis.

Remaining samples may be stored for exploratory analysis including analysis for genetic, transcriptional, epigenetic, molecular or protein markers, as described in Section7.11.2.

8. STUDY PERIODS AND VISITS

Investigative sites will maintain a log ofall screened subjects who were reviewed and evaluated for study participation. Information collected on the screening log should include limited information such as the date of screening, date when the subject was enrolled or the reason why the subject failed screening.

8.1. Prescreening Period

The Investigator is expected to perform a review of a subject's medical history to ensure that known eligibility criteria are met, to the best oftheir knowledge, including documentation supporting that a subject's tumor is positive for the *TP53* XYZmutation and HLA-A*02:01.

Subjects will be required to enter a Prescreening Period, which consists ofa Prescreening Visit, if the subject is not known to be positive for *TP53* XYZ mutation and/or HLA-A*02:01 (ie, prior test was not done, or the result is not available) determined using one of the CLIA -accredited laboratory-based assays describedAppendix F.

The Prescreening Period starts when the subject signs the Prescreening ICF to participate in the prescreening portion of the study and ends on the date the subject enters the Screening Period (see Section8.2), on the date that the subject is determined to not be eligible for screening (eg, tumor is negative for *TP53* XYZ), or on the date that the subject is withdrawn for any reason (see Section5.5for more information on study withdrawal reasons).

8.1.1. Prescreening Visit

Subjects required to complete the Prescreening visit will sign a Prescreening ICF. Refer to Section11.3for more information on informed consent expectations.

- For subjects without prior documented *TP53* XYZ positivity using one of the assays described inAppendix F, an archival or fresh tumor biopsy sample will be obtained and sent to the specialty laboratory for testing using the F1CDx assay.
- For subjects without prior documented HLA-A*02:01 positivity using one of the assays described inAppendix F, a blood sample will be collected and shipped to the specialty laboratory for HLA typing.

Prescreening testing results must be obtained and confirmed to meet study eligibility requirements regarding mutation and HLA status prior to a subject continuing to Screening.

Subjects who have documented HLA-A*02:01 positivity from an assay other than that described inAppendix Fmay proceed to Screening, however a sample must be provided for central HLA testing during the Screening Period.

High level chart review by the investigator including key elements of disease treatment history and medical history including but not limited to prior lines of therapy for the solid tumor, history of any cardiovascular or auto-immune disease, and history of a second malignancy will be performed and may be discussed with the Sponsor Medical Monitor.
Refer to Section7.11.4for information about the procedures associated with the *TP53* XYZ mutation testing. Refer to Section7.11.3for information about the procedures associated with the HLA-A*02:01 testing.

8.1.1.1. Prescreening Subject ID Number Assignment

All subjects who provide written Prescreening informed consent and who performed at least 1 additional Prescreening Visit procedure will be assigned a unique prescreening ID number to be used during prescreening only. The prescreen subject ID number must be used to identify the biopsy sample(s) and/or blood sample collected during prescreening to be used for confirmatory testing.

Subjects who complete the Prescreening Visit and are eligible to be screened for enrollment will receive a unique Subject ID number to be used to identify the subject and subject samples throughout their participation in the study.

8.1.1.2. Prescreening Visit Procedure

The assessments/procedures below will be completed prior to initiating any prescreening visit procedures:

- Prescreening informed consent

After written Prescreening informed consent the assessments/procedures below will be completed.

- Demographics
- AE reporting: refer to Section9.1for AE reporting requirements during the Prescreening period.
- Sample collection for central assessment:
  - Archival tumor tissue or fresh tumor biopsy submission to central lab for *TP53* XYZmutation confirmation. Refer to Section 7.11.4 for information regarding *T P53* XYZ mutation testing.
  - Whole blood: HLA-A*02:01. Refer to Section7.11.3for information regarding HLA-A*02:01 testing

8.2. Screening Period

The Screening Period begins on the date that the subject provided written screening informed consent to participate in the study and ends on the date of leukapheresis, on the date that the subject is determined to not meet eligibility for enrollment (ie, screen failure), or on the date that the subject is withdrawn for any reason (see Section5.5for more information on study withdrawal reasons).

Once the eligibility is confirmed by the Investigator and reviewed by the Sponsor Medical Monitor as described in the Sponsor Medical Monitoring Plan (MMP), the subject can be enrolled and proceed to leukapheresis.

If, at any time prior to enrollment, the subject fails to meet the eligibility criteria, the subject will be considered a screen failure.

8.2.1. Screening Visit

A Screening Visit is conducted where procedures are performed to assess eligibility for study enrollment. Refer to Section11.3 for more information on informed consent expectations.

8.2.1.1. Study Subject ID Number Assignment

All subjects who provide written Screening informed consent and who performed at least 1 additional Screening Visit procedure will be assigned a unique Subject ID number by the Sponsor via EDC.

The same subject ID number that was assigned at screening will be used to identify the subject during the Screening Period and throughout the subject's participation in the study.
Ifa subject completed the Prescreening Visit (see Section8.1.1) and received a Prescreening Subject ID Number (see Section8.1.1.1), the Prescreening Subject ID Number will not be used once the unique Subject ID number is assigned by the Sponsor via EDC at screening.

8.2.1.2. Screening Visit Procedures

The assessments/procedures below will be completed prior to initiating any screening visit procedures:

- Screening informed consent

After written screening informed consent, the assessments/procedures below will be completed:

- Demographics
- Disease history
- Disease treatment history
- Medical history
- Prior and concomitant medications
- Vital signs
- Height
- Weight
- ECG
- ECHO
  - An ECHO performed after the last systemic treatment regimen and within the 28 days prior to signing the screening ICF may be used to assess enrollment eligibility and does not need to be repeated at the Screening Visit.
  - If the Investigator determines that an ECHO is not an appropriate assessment of ejection fraction, other methods may be used (eg, MRI, radionuclide angiography) with the approval of the Sponsor Medical Monitor.
- Physical examination with neurological examination
- ECOG performance status

- Brain MRI (for subjects who are considered at high-risk for CNS metastasis [eg, NSCLC, breast cancer, colon, or other histology considered high-risk of CNS metastasis by the Investigator], who have a high risk of CNS metastasis, or ifthere is clinical suspicion of CNS involvement)
  - A brain MRI performed after the completion of the subject's last systemic treatment and within the 28 days prior to signing the screening ICF may be used to confirm enrollment eligibility.
- Local laboratory assessments:
  - Chemistry panel
  - Coagulation panel including fibrinogen
  - Disease-specific biomarkers (see Section7.9.2and Section7.10.2)
  - CrCl calculation by Cockcroft Gault (see Section7.9.2.1for requirements and exceptions)
  - CBC with differential
  - Serology: HIV (1 and 2), Hepatitis B (surface antigen and core antibody, if any positive, a PCR test should be performed), Hepatitis C (antibody/PCR, if the HCV antibody is positive, a screening HCV RNA by any RT-PCR or DNA assay should be performed).
  - Serum pregnancy test for subjects who are female of childbearing potential
- Sample collection for central assessment:
  - Blood sample for confirmatory HLA typing, if applicable (see Section7.11.3)
  - Baseline archival tumor tissue and fresh tumor biopsy for immunohistochemistry and genomic/transcriptomic studies.
    - Archival tumor tissue
    - Fresh biopsy may be collected any time during screening up to at least 7 days prior to initiating lymphodepleting chemotherapy. If the archival tumor tissue was collected within 6 months prior to enrollment, the baseline fresh biopsy sample is not required.
    - Refer to Section7.11.4and Section7.11.5for more information regarding tumor tissue requirements.
- CT scan with contrast (unless contraindicated) or MRI scan to confirm eligibility (see Section5.3.1). A recent CT or MRI scan performed per standard of care within 28 days prior to signing the screening ICF may be used to confirm enrollment eligibility and does not need to be repeated if, in the opinion of the Investigator, the clinical status of the subject is unchanged.
- AE reporting: refer to Section9.1for AE reporting requirements during the screening period.

8.2.2. Rescreening

If one or more screening assessments did not meet one or more of the eligibility criteria (see Section5.3), the assessment and/or procedure may be repeated as clinically justified and permitted by the Sponsor Medical Monitor.

If there is a substantial delay (eg, > 28 days) between the Screening Visit procedures and the date of enrollment/leukapheresis, the Investigator will discuss with the Sponsor Medical Monitor the screening procedures that will need to be repeated to ensure that the subject remains eligible for enrollment. The Investigator will also discuss with the IRB/IEC if reconsent is necessary.

Ifa subject was previously confirmed to be a screen failure by the Investigator and is considered a viable candidate for rescreening by the Investigator, the subject may repeat the Screening Visit assessments/procedures as detailed in the SOA inAppendix B, including screening informed consent, as permitted by the Sponsor Medical Monitor.

Subjects who are rescreened will retain the same subject ID number assigned at the time of screening.

8.3. Enrollment/Leukapheresis Period

Subjects are enrolled once leukapheresis commences.

Investigators must confirm eligibility by reviewing the Screening assessments against the full eligibility criteria detailed in Section5.3.2prior to the start of leukapheresis. The Leukapheresis visit should occur within approximately 5 business days of eligibility confirmation.
See Section8.3.1.1for details on criteria for initiating leukapheresis. If the specified exclusion criteria are met, leukapheresis must be delayed until the event resolves. If leukapheresis occurs more than 5 days after eligibility confirmation, baseline CBC with differential and chemistry panel, coagulation panel and fibrinogen must be repeated.

8.3.1. Enrollment/Leukapheresis Visit Procedures

The Enrollment/Leukapheresis visit will be completed after eligibility confirmation by the Investigator and reviewed by the Sponsor Medical Monitor. A subject is enrolled once leukapheresis is initiated. Procedures may be performed within 24 hours prior to initiating leukapheresis.

The following assessments/procedures will be done prior to leukapheresis:

- AE assessment
- Concomitant medications assessment
- Vital signs
- Weight
- Physical examination with neurological examination
- ECOG performance status
- Local laboratory assessments

  - Chemistry panel
  - Coagulation panel including fibrinogen
  - CrCl calculation by Cockcroft Gault (see Section7.9.2.1for requirements and exceptions)
  - CBC with differential
  - Ferritin
  - CRP
  - Serum pregnancy test for subjects who are female of childbearing potential
- Sample collection for central assessment:
  - Whole blood: PBMCs, CBC with differential
  - Serum/plasma: key analytes (eg, cytokines, chemokines), immunogenicity studies
  - Baseline fresh tumor biopsy (only if not collected at screening)
    - If the archival tissue biopsy at screening was collected within 6 months prior to enrollment, the baseline fresh biopsy sample is not required.
    - If fresh biopsy is required, it may be collected any time during screening up to at least 7 days prior to initiating lymphodepleting chemotherapy.
    - Refer to Section7.11.4and Section7.11.5for more information regarding tumor tissue requirements.

Prior to leukapheresis initiation the Investigator will review the enrollment visit assessments to confirm that the subject continues to meet enrollment eligibility criteria (see Section5.3). Additionally, the Investigator will review the criteria listed in Section8.3.1.1prior to initiating leukapheresis. If the subject continues to meet enrollment eligibility criteria (see Section5.3), and does not meet any of the criteria in Section8.3.1.1, the procedure below will be completed:

- Leukapheresis per institutional guidelines (refer to Section6.1.1)

Refer to the IPM for information and expectations for leukapheresis, procedures for preparation, and packaging of leukapheresed material for shipment to the manufacturing facility.

8.3.1.1. Criteria for Initiating Leukapheresis

Leukapheresis will not be initiated if any of the following conditions are met:

- The subject no longer meets eligibility criteria (see Section5.3) based on enrollment visit assessments.
- There is evidence of a clinically significant infection.
- The WBC count is ≥ 20,000 cells/µL.
- The subject is receiving corticosteroid therapy at a pharmacological dose (>10 mg/day of prednisone or equivalent doses of other corticosteroids) and/or other immunosuppressive drugs are taken <7 days prior to leukapheresis. Refer to Section5.4for more information on prohibited medications.

If any of these criteria are met and leukapheresis is not initiated, the Investigator will consult with the Sponsor Medical Monitor regarding next steps (eg, leukapheresis delay, repeat assessment).

If any screening assessments or procedures are repeated between screening and the start of conditioning chemotherapy and results are outside the eligibility criteria, the Investigator should discuss the results with the Sponsor Medical Monitor before proceeding with lymphodepleting chemotherapy.

8.4. Lymphodepletion Period

During the Lymphodepletion Period, a nonmyeloablative lymphodepleting regimen will be administered to induce lymphocyte depletion and create an optimal environment for expansion of ABC-01 in vivo. Lymphodepleting chemotherapy may be administered inpatient or outpatient at the discretion of the Investigator.

Subjects must present progressive disease, stable disease, or lack of clinical benefit from previous treatment based on Investigator assessment of subject's clinical condition and upon agreement with the Sponsor Medical Monitor.

See Section8.4.2for criteria for initiating lymphodepleting chemotherapy.

8.4.1. Lymphodepleting Chemotherapy Visit Procedures (Day -7 to Day -3)

The assessments/procedures below will be done on each day (except as specified below) prior to the administration of lymphodepleting chemotherapy. On Day -7, the procedures may be performed the day prior (24 hours) to initiating lymphodepleting chemotherapy:

- AE assessment
- Concomitant medications assessment
- Vital signs
- Weight
- ECOG on Day -7 only
- Physical examination with neurological examination (Day -7 only)
- Local laboratory assessments:
  - CBC with differential
  - Chemistry panel
  - Coagulation panel including fibrinogen on Day -7 only
  - CrCl calculation by Cockcroft Gault (see Section7.9.2.1) for requirements and exceptions on Day -7 only
  - Ferritin on Day -7 only
  - CRP on Day -7 only
  - Disease-specific biomarker on Day -7 only

  - Serum pregnancy test for female subjects of childbearing potential on Day -7 only

- Sample collection for central assessment prior to administering lymphodepleting chemotherapy on Day -7 only:
  - Whole blood: PBMCs, CBC with differential, cfDNA
  - Serum/plasma: key analytes (eg, cytokines, chemokines)
  - Baseline fresh tumor biopsy up to at least 7 days prior to initiating lymphodepleting chemotherapy (only if required and not collected at Screening or Enrollment/Leukapheresis visits)

- CT scan with contrast (unless contraindicated) or MRI scan (neck [if applicable], chest, abdomen, pelvis, and all other sites of disease) to establish baseline disease status for post-treatment disease response assessment must be done prior to administering lymphodepleting chemotherapy. Baseline CT or MRI scan may be performed up to the 15 days prior to initiating lymphodepleting chemotherapy on Day -7.

Prior to lymphodepleting chemotherapy initiation, the Investigator will review the lymphodepleting chemotherapy visit assessments (from each visit in the Lymphodepleting Chemotherapy Period) to confirm that the subject continues to meet enrollment eligibility criteria (see Section5.3). Additionally, on each visit in the Lymphodepleting Chemotherapy Period, the Investigator will review the criteria listed in Section8.4.2prior to initiating the lymphodepleting chemotherapy regimen.

Once the subject is confirmed to be eligible to receive lymphodepleting chemotherapy, the procedure below will be done:

- Lymphodepleting chemotherapy administration (as described in Section6.1.2).

8.4.2. Lymphodepleting Chemotherapy Criteria

The lymphodepleting chemotherapy regimen should not be initiated without Sponsor confirmation of ABC-01 availability for the subject. Further, the lymphodepleting chemotherapy regimen will not be initiated if subject no longer meets any of the eligibility criteria for enrollment, as detailed in Section5.3.

Lymphodepleting chemotherapy will not be initiated if any of the following conditions are met:

- Subject has an ongoing infection or inflammation, even ifasymptomatic. If any of the conditions below are met, a workup should be done to rule out infection or inflammatory state, which may increase the risk for high-grade and fatal toxicities.
  - Temperature > 38 °C within the 72 hours prior to lymphodepleting chemotherapy administration.
  - WBC count or WBC differential result that is concerning for infectious process at any time between enrollment and lymphodepleting chemotherapy (eg, WBC > 20,000, rapidly increasing WBC, or differential with a high percentage of segments/bands).

- Subject has evidence of infection/inflammation revealed during the physical examination and/or AE assessment.
- Subject received any systemic antimicrobials for the treatment of a known or suspected infection within the 48 hours prior to lymphodepleting chemotherapy.
  - Use of antimicrobials as prophylaxis is allowed. Treatment course of any antimicrobials given for a known or suspected antecedent infection should be complete as per infectious disease consult (if applicable) recommendation before stopping or switching to prophylactic antimicrobials.
- Subject has a confirmed infectious process for which antimicrobials are not available (eg, viral pneumonia). The infection must be clinically resolved as confirmed by the Investigator in consultation with an infectious disease service (if applicable) prior to lymphodepleting chemotherapy administration.
- There is clinical suspicion for an infection based on blood, urine, or other body fluid cultures (eg, growth, other infectious workup performed such as bacterial, PCR, stool studies, imaging studies). If there was growth, the most recent cultures must show no growth for at least 48 hours prior to lymphodepleting chemotherapy. If cultures are unlikely to be positive within 48 hours (eg, fungal infection), adequate time must be allowed for the cultures to become positive.

If any of these criteria are met, the Investigator will consult with the Sponsor Medical Monitor regarding next steps (eg, lymphodepleting chemotherapy delay, repeat assessment).

8.4.3. Lymphodepleting Chemotherapy Administration

Once the Investigator confirms that none of the conditions detailed in Section8.4.2are met, lymphodepleting chemotherapy administration may proceed as described below:

Day -7 and Day -6:

1. Administer IV hydration per institutional guidelines.
2. Administer DrugC, 30 mg/kg IV for approximately 60 minutes. For obese subjects (body mass index > 35), dose is calculated using adjusted body weight. DrugC dose may be adjusted for renal insufficiency per institutional guidelines.
3. Administer mesna (sodium 2-mercaptoethanesulfate) per investigator discretion or institutional guidelines, unless contraindicated. Mesna is a detoxifying agent used to inhibit hemorrhagic cystitis induced by chemotherapy.
4. Administer DrugF, 25 mg/m2 for approximately 30 minutes. For obese subjects (body mass index > 35), dose is calculated using adjusted body weight. DrugF dose may be adjusted for renal insufficiency per institutional guidelines.

Day-5, Day -4, Day -3:

1. Administer IV hydration per institutional guidelines.

2. Administer DrugF, 25 mg/m2 for approximately 30 minutes. For obese subjects (body mass index > 35), dose is calculated using adjusted body weight. DrugF dose may be adjusted for renal insufficiency per institutional guidelines.

Any changes in the dose or other aspects of the lymphodepleting chemotherapy regimen must be discussed with the Sponsor Medical Monitor prior to administration. (eg, in case of shortage of DrugF, other equivalent chemotherapy agents [eg, bendamustine] could be utilized after discussion and agreement with the Sponsor Medical Monitor).

Subjects should be well hydrated but closely monitored to avoid fluid overload. Subjects should be instructed to drink plenty of liquids during and for 24 hours following lymphodepleting chemotherapy administration.

Refer to the SOA inAppendix Bfor details on the assessments/procedures to be performed during the Lymphodepletion Period.

Lymphodepleting chemotherapy must be repeated if ABC-01 administration is delayed for more than 14 days after termination of lymphodepletion, unless agreed between Investigator and Sponsor Medical Monitor. In all cases of ABC-01 delay in administration, contact the Sponsor Medical Monitor for guidance.

8.5. Treatment Period

Subjects will be admitted to the hospital on Day 0, following 2 days of rest (Day -2 to Day -1) after the final administration of the lymphodepleting chemotherapy regimen. Subjects will remain hospitalized for a minimum of 8 days (Day 0 to Day 7). The Treatment Period will begin on Day 0 when subjects receive an infusion of ABC-01 and ends on Day 13.

Subjects must present progressive disease, stable disease, or lack of clinical benefit from previous treatment based on Investigator assessment of subject's clinical condition and upon agreement with the Sponsor Medical Monitor.

Exclusion criteria for ABC-01 administration are presented in Section8.5.1.1. Assessments/procedures for the Treatment Period are detailed in the SOA inAppendix B.

8.5.1. Treatment Visit Procedures (Day 0)

The assessments/procedures below will be done prior to the administration of ABC-01:

- AE assessment
- Concomitant medications assessment
- Vital signs (per institutional guidelines, at minimum prior to ABC-01 infusion, during ABC-01 infusion (if feasible), and within approximately 15 minutes post-infusion)
- Weight
- ECOG performance status
- Physical examination with neurological examination
- Baseline ICE score should be assessed prior to ABC-01 treatment and maintained in the source documents

- Local laboratory assessments (may be performed within the 24 hours prior to ABC-01 administration):
  - Chemistry panel
  - Coagulation panel including fibrinogen
  - CrCl calculation by Cockcroft Gault (see Section7.9.2.1for requirements and exceptions)
  - CBC with differential
  - Ferritin
  - CRP
  - Serum pregnancy test for subjects who are female of childbearing potential
- Sample collection for central assessment
  - Whole blood: PBMCs, CBC with differential, cfDNA
  - Serum/plasma: key analytes (eg, cytokines, chemokines)

Prior to ABC-01 administration the Investigator will review the Treatment Visit (Day 0) assessments to confirm that the subject continues to meet enrollment eligibility criteria (see Section5.3and Section8.5.1.1for exceptions to certain enrollment eligibility criteria). Additionally, the Investigator will review the criteria listed in Section8.5.1.1prior to initiating ABC-01 administration to confirm that the subject is eligible to receive ABC-01.

Once the Investigator confirms the subject to be eligible for ABC-01 administration, the procedure below will be performed:

ABC-01 administration (as described in Section8.5.1.2)

rIL-2 administration (as described in Section8.5.1.3)

8.5.1.1. ABC-01 Administration Criteria

ABC-01 administration may not be initiated if any of the following conditions are met:

- Subject has an ongoing infection or inflammation, even ifasymptomatic. If any of the conditions below are met, a workup should be done to rule out infection or inflammatory state which may increase the risk for high-grade and fatal toxicities.
  - Temperature > 38 °C within the 72 hours prior to ABC-01 administration
  - WBC count or WBC differential result that is concerning for infectious process at any time between lymphodepleting chemotherapy and ABC-01 administration (eg, WBC > 20,000, rapidly increasing WBC, or differential with a high percentage of segments/bands).
- Subject has evidence of infection/inflammation revealed during the physical examination and/or AE assessment.
- Subject received any systemic antimicrobials for the treatment of a known or suspected infection within the 48 hours prior to ABC-01 treatment.

- Use of antimicrobials as prophylaxis is allowed. Treatment course of any antimicrobials given for a known or suspected antecedent infection should be complete as per infectious disease consult (if applicable) recommendation before stopping or switching to prophylactic antimicrobials.

- Subject has a confirmed infectious process for which antimicrobials are not available (eg, viral pneumonia). The infection must be clinically resolved as confirmed by the Investigator in consultation with an infectious disease service (if applicable) prior to ABC-01 treatment.
- There is clinical suspicion for an infection based on blood, urine, or other body fluid cultures (eg, growth, other infectious workup performed such as bacterial, PCR, stool studies, imaging studies). If there was growth, the most recent cultures must show no growth for at least 48 hours prior to ABC-01 administration. If cultures are unlikely to be positive within 48 hours (eg, fungal infection), adequate time must be allowed for the cultures to become positive.
- > Grade 3 non-hematologic AEs occurring after lymphodepletion, where nausea, vomiting, diarrhea, and constipation are excluded.
- ECOG performance status >1, unless discussed and agreed between Investigator and Sponsor Medical Monitor.

If any of these criteria are met, the Investigator will consult with the Sponsor Medical Monitor regarding next steps (eg, ABC-01 administration delay, repeat assessment).

8.5.1.2. ABC-01 Administration

Approximately 1 hour prior to ABC-01 administration, the medications below should be administered:

3. Acetaminophen, 500 to 1,000 mg orally (PO) or equivalent
4. Diphenhydramine, 12.5 to 25 mg IV or 25 mg PO or equivalent

ABC-01 administration is expected to start as soon as possible and approximately within 30 minutes of thaw completion time. ABC-01 will be infused IV through central venous access (eg, port or peripherally inserted central catheter) without leukodepleting filters. Catheter care will be followed per institutional guidelines. IV administration may be done via gravity drip or via IV pump set to infuse each bag over approximately 15 to 20 minutes. The entire infusion duration will depend on the number of bags and is expected to be completed in less than an hour.

Vital signs will be measured before, during (if feasible), and within 15 minutes post-infusion and then routinely during hospitalization per institution standard procedure and/or guidelines. Other infusion procedures will follow institutional guidelines for the infusion of cell products.

Instructions for the preparation of ABC-01 for infusion (thawing ABC-01, inspection of product) and administration of ABC-01 (eg, fusing the line, spiking the bag) are provided in the IPM.

The dose levels of ABC-01 to be evaluated are presented in Table 2.

If ABC-01 administration is delayed > 14 days, lymphodepletion must be repeated unless agreed between Investigator and Sponsor Medical Monitor. In all cases of ABC-01 delay in administration, contact the Sponsor Medical Monitor for guidance.

8.5.1.3. Recombinant IL-2 Administration

Recombinant IL-2 will be administered to support T cell expansion and persistence.

Recombinant IL-2 will be administered SC starting on Day 0 at a dose of 300,000 IU/kg approximately 4 hours following ABC-01 treatment. The rIL-2 may be administered up to 24 hours after ABC-01 treatment due to AE or other clinical reasons. The same rIL-2 dose will be administered once daily for a maximum of 8 doses as tolerated.

Administration of rIL-2 may be discontinued or interrupted as clinically necessary per institutional guidelines. Investigator will inform the Sponsor Medical Monitor if rIL-2 administration is discontinued or interrupted.

Acetaminophen 250 to 500 mg PO every 4 to 6 hours may be taken for the duration of rIL-2 administration to prevent pyretic reactions.

Refer to the product label for guidance on packaging, storing, preparing, and administering rIL-2. Refer to the product label and/or institutional guidelines for the management of toxicities related to rIL-2.

8.5.2. Inpatient Monitoring After ABC-01 Treatment (Day 1 through Hospital Discharge)

Subjects will be monitored (during and after treatment with ABC-01 until hospital discharge [at least 7 days after ABC-01 administration]) for signs and symptoms of CRS, ICANS, or other treatment-related AEs. Assessments/procedures are detailed in the SOA (Appendix B).

The assessments/procedures below are expected to be done as part of subject monitoring during and after ABC-01 treatment until hospital discharge (at least 7 days after ABC-01 treatment) to monitor for signs and symptoms of CRS, ICANS or other treatment-related AEs.

- AE assessment
- Concomitant medications assessment
- Vital signs: routine monitoring per institutional guidelines
- Physical examination with neurological examination: routine monitoring per institutional guidelines
- Local laboratory assessments: Daily until hospital discharge (unless specified otherwise below)
  - Chemistry panel
  - CBC with differential
  - Coagulation panel including fibrinogen (Day 1, 3, 5, and 7)
  - CRP (Day 1, 3, 5, and 7)
  - Ferritin (Day 1, 3, 5 and 7)

- Sample collection for central assessment: On Day 1, Day 3, Day 5, and Day 7; and 10 days after ABC-01 treatment if subject has not been discharged from the hospital (unless specified otherwise below)
  - Whole blood: PBMCs, CBC with differential
    - Whole blood collection timepoints during inpatient monitoring that fall on a Friday, weekend, or observed holiday (refer to the Central Lab Manual) may be missed as permitted by the Sponsor.
  - Whole blood: cfDNA (on Day 7 only)
  - Serum/plasma: key analytes
  - Optional fresh biopsy (on Day 7 [-2/+7 days] only)
- rIL-2 administration: daily through Day 7 (see Section6.3)

The following assessments/procedures will be performed in the event the subject experiences CRS or ICANS:

- For Grade 3 or higher CRS or ICANS, the samples below will be collected at onset or first reoccurrence (if not already collected on the same date as onset) and every 48 hours after onset and until Grade < 3 or at hospital discharge, whichever occurs first:
  - Sample collection for central assessment:
    - Serum/plasma: key analytes (eg, cytokines, chemokines)
    - Whole blood: PBMCs and other exploratory analyses, CBC with differential
      - Whole blood collection timepoints during inpatient monitoring that fall on a Friday, weekend, or observed holiday (refer to the Central Lab Manual) may be missed as permitted by the Sponsor.
- For ICANS (Grade 2 or higher), a lumbar puncture is recommended at onset (see Section7.9.9). A sample may be sent to the central and/or specialty laboratory to assess cellularity (PK, Section7.11.1.1) and presence of key analytes (pharmacodynamics, see Section7.11.1.2).

Subjects should remain in the hospital for at least 7 days following treatment with ABC-01 and should not be discharged from the hospital until all ABC-01-related non-hematological toxicities have resolved to Grade 1 or to baseline. Subjects may be discharged at the Investigator's discretion with noncritical and clinically stable or improving toxicities (eg, renal insufficiency) which may be > Grade 1 at the time of discharge. These cases should be discussed with the Sponsor Medical Monitor.

At the time of discharge, subjects should be instructed to remain within proximity of the study center for at least 4 weeks after ABC-01 treatment. Subjects and their caregivers should be educated on potential CRS and ICANS symptoms (eg, fever, dyspnea, confusion, aphasia, dysphagia, somnolence, encephalopathy, ataxia, or tremor) and instructed on how to immediately contact the Investigator and/or seek immediate medical attention at the onset of any of these symptoms.

8.5.2.1. Rehospitalization Visits due to ABC-01 Adverse Events

If a subject is rehospitalized due to ABC-01-related AEs, the samples described below will be assessed locally and collected and sent to the central and/or specialty laboratories:

- Local laboratory assessments (daily):
  - Chemistry panel
  - CBC with differential
- Sample collection for central assessment (day of readmission, weekly, and day of discharge):
  - Serum/plasma: key analytes (eg, cytokines, chemokines)
  - Whole blood: PMBCs, CBC with differential
    - Whole blood collection timepoints during inpatient monitoring that fall on a Friday, weekend, or observed holiday (refer to the Central Lab Manual) may be missed as permitted by the Sponsor.

For samples to be collected at the onset of Grade 3 or higher CRS and/or Grade 2 or higher ICANS, refer to Section8.5.2.

Ifa timepoint for sample collection due to rehospitalization coincides with a scheduled visit, the sample collected for the scheduled visit will be used (eg, if the subject was readmitted on Day 14, one serum/plasma sample will be collected to meet the Day 14 visit requirement and the rehospitalization requirement).

8.6. Follow-up Period

8.6.1. Post-treatment Follow-up Period

Post-treatment follow-up period will begin on Day 14 and continue to Day 84. The assessments/procedures for the Post-treatment Follow-up Period are specified in the SOA in Appendix B.
Subjects will complete the visits below at the timepoints specified in the SOA inAppendix B.

Visit days are calculated using the date of ABC-01 treatment as Day 0. Due to the required hospitalization, see Section8.5.2, post-treatment assessment visits may coincide with the visits for inpatient monitoring post ABC-01 treatment. When treatment visits and post-treatment visits overlap, the procedures associated to post-treatment visits will be performed (eg, if subject is the hospital at Day 14), the Day 14 post-treatment assessment visit procedures will be done instead of the inpatient monitoring procedures presented in Section8.5.2.

8.6.1.1. Post-treatment Visits

Subjects will complete the visits below at the timepoints specified inAppendix B.

- Day 14 (±2 days)
- Day 28 (±3 days)
- Week 6 (±7 days)

- Week 12 (±7 days)

The assessments/procedures below will be done during the visits in the Post-treatment Assessment period at the timepoint specified in Appendix B.

- AE assessment
- Concomitant medications assessment
- Vital signs
- Physical examination with neurological examination
- Local laboratory assessments
  - Chemistry panel
  - Coagulation panel including fibrinogen (Days 14 and Day 28 only)
  - CBC with differential
  - CRP and ferritin (Days 14 and Day 28 only)
  - Disease-specific tumor markers (Week 6 and Week 12 only)
- Sample collection for central assessment:
  - Whole blood:
    - PBMCs
    - CBC with differential
    - cfDNA (Day 28, Week 12 only)
  - Serum/plasma:
    - Key analytes (eg, cytokines, chemokines)
    - Immunogenicity studies (Day 28 and Week 12 only)
  - Fresh tumor biopsy: Day 28 (or Week 6 or Week 12 if Day 28 is not clinically feasible)
- Disease/overall response assessment (Week 6 and Week 12 only):
  - CT or MRI scan (neck [if applicable], chest, abdomen, pelvis, and all other sites of disease)
  - Disease response assessment by RECIST v1.1

8.6.2. Extended Follow-up Period

The Extended Follow-up Period will begin after Day 84 with the first study visit on Week 18 and continue to Month 24.

The assessments/procedures for the Extended Follow-up Period are specified in the SOA in Appendix B.

8.6.2.1. Extended Follow-up Visits

Subjects will complete the visits below at the timepoint specified inAppendix Bafter completing the post-treatment follow-up period. Subjects are considered in Extended Follow-up Period after the completion of the Week 12 visit or if the subject early exited the study prior to completing the Month 24 visit (see Section8.6.5). Visit days are calculated using the date of ABC-01 treatment as Day 0.

- Week 18 (±7 days)
- Month 6 (± 14 days)
- Month 9 (± 14 days)
- Month 12 (± 14 days)
- Month 15 (± 14 days)
- Month 18 (± 14 days)
- Month 24 (±14 days)

The assessments/procedures below will be done in the Extended Follow-up Period at the timepoint specified in theAppendix B.

- AE assessment through Month 6
- Targeted AE assessment after Month 6 through Month 24
- Concomitant medications assessment through Month 6
- Targeted concomitant medications assessment after Month 6 through Month 24
- Vital signs
- Physical examination with neurological examination
- Local laboratory assessments
  - Chemistry panel
  - CBC with differential
  - Disease-specific tumor markers
- Sample collection for central assessment:
  - Whole blood:
    - PBMCs
    - CBC with differential
    - cfDNA (Month 6, Month 12, Month 24 only)
  - Serum/plasma:
    - Key analytes (eg, cytokines, chemokines)
    - Immunogenicity studies (Month 9 and Month 18 only)

- Fresh tumor biopsy: Month 12 (if determined to be clinically feasible by the Investigator)

- Disease/overall response assessment:
  - CT or MRI scan (neck [if applicable], chest, abdomen, pelvis, and all other sites of disease)
  - Disease response assessment by RECIST v1.1
- Subsequent anticancer therapy if the subject experienced disease progression after ABC-01 treatment
- Survival status: Sponsor may ask for survival status outside of the scheduled visits (eg, for survival sweep), which may be confirmed by telephone or other type of contact with the subject or subject's caregiver.

8.6.3. Long-Term Follow-up Period

The Long-Term Follow-up Period will begin after Month 24, with the first study visit at Month 30 and continue to Month 180.

Subjects will complete the visits below at the timepoint specified in the SOA in Appendix B after completing the Extended Follow-up Period. Subjects are considered in the Long-Term Follow-up Period after the completion of the Month 24 visit or if the subject early exited the study after completing the Month 24 visit (see Section 8.6.5). Visit days are calculated using the date of ABC-01 treatment as Day 0.

The Sponsor may ask for survival status outside of the scheduled visits (eg, for survival sweep), which may be confirmed by telephone or other type of contact with the subject or subject's caregiver.

The assessments/procedures for the Long-Term Follow-up Period are specified in the SOA in Appendix B.

Month 180 is the final visit in the Long-Term Follow-up period and is also the EOS visit.

8.6.4. Long-Term Follow-up Visits

Subjects will complete the visits below at the timepoint specified in Appendix B after completing the Extended Follow-up Period.

- Month 30 (±30 days)
- Month 36 (±30 days)
- Month 42 (±30 days)
- Month 48 (±30 days)
- Month 54 (±30 days)
- Month 60 (±30 days)
- Month 72 / Year 6 (±30 days)

- Month 84 / Year 7 (±30 days)
- Month 96 / Year 8 (±30 days)
- Month 108 / Year 9 (±30 days)
- Month 120 / Year 10 (±30 days)
- Month 132 / Year 11 (±30 days)
- Month 144 / Year 12 (±30 days)
- Month 156 / Year 13 (±30 days)
- Month 168 / Year 14 (±30 days)
- Month 180 / Year 15 (±30 days)

The assessments/procedures below will be done during the Long-Term Follow-up Period at the timepoint specified in the SOA (Appendix B).

- Sample collection for central assessment:
  - Whole blood: PBMCs, CBC with differential
- SOC disease/overall response assessment (imaging and disease-specific tumor markers)
- Subsequent anticancer therapy if the subject experienced disease progression after ABC-01 treatment
- Survival status: Sponsor may ask for survival status outside of the scheduled visits (eg, for survival sweep) which may be confirmed by telephone or other type of contact with the subject or subject's caregiver.

Month 180 is the final visit in the Survival Follow-up period and is also the EOS visit (see Section4.2).

8.6.5. Follow-up Visits for Early Exit Subjects

Subjects who were treated with ABC-01 and early exit from the study prior to Week 12 will be asked to complete the Week 12 visit procedures and will continue follow-up as described below.

All subjects who were treated and early exited the study (eg, due to disease progression or other reason specified in Section5.5) will complete the assessments/procedures specified below at the scheduled timepoints for Extended Follow-up Period if early exited prior to Month 24, or at the scheduled timepoints for the Long-Term Follow-up Period if early exited after Month 24. Subjects who undergo leukapheresis and/or lymphodepleting chemotherapy but do not receive ABC-01 will be required to complete safety follow up for at least 30 days or until the resolution of any ongoing procedure or lymphodepleting chemotherapy -related AE(s) and will then come off study.

Subjects who early exit from the study due to partial withdrawn consent have the right to withdraw consent from completing a Week 12 visit, as described above, and/or any of the procedures described below at any time. Specific procedures applicable to the withdrawn consent

will be discussed between the subject and the Investigator and documented in the study source documents.

See Section8.6.7for study procedures required at time of disease progression.

- AE assessment until 30 days after the last study procedure was completed or initiation of anticancer therapy, whichever is earliest.
  - AE reporting requirements will follow the requirements applicable to the study period that early exit occurred. For example, if early exit occurred on Month 6 or earlier all AEs are reported, if between Month 6 and Month 24, only targeted AEs are reported. Refer to Section9.1.
- Concomitant medications assessment until 30 days after the last study procedure was completed or initiation of anticancer therapy, whichever is earliest.
  - Concomitant medications reporting requirements will follow the requirements applicable to the study period that the visit occurred. For example, if the early exit occurred on Month 6 or earlier all concomitant medications are reported, if between Month 6 and Month 24, only targeted concomitant medications are reported.
- Subsequent anticancer therapy
- SOC disease assessment including disease-specific markers until initiation of subsequent anticancer therapy
- Secondary malignancy assessment and reporting for subjects who were dosed with ABC-01
- Sample collection for central assessment for subjects who were dosed with ABC-01 to assess clonal expansion
  - Whole blood: PBMCs, CBC with differential
- Survival status
  - Sponsor may ask for survival status outside of the scheduled visits (eg, for survival sweep), which may be confirmed by telephone or other type of contact with the subject or subject's caregiver.

8.6.6. Unscheduled Visits

Unscheduled visits may be completed in addition to the required study visits to assess AEs, disease progression, repeat study procedures, or other reason at the discretion of the Investigator in consultation with the Sponsor Medical Monitor.

Unscheduled visits and the procedures completed as part of the unscheduled visit will be reported in the eCRF.

8.6.7. Disease Progression Visit

For subjects who experience disease progression after ABC-01 treatment, the assessments/procedures presented in the SOA inAppendix B(and outlined below) will be

completed as discussed and agreed by the Sponsor medical monitor as an unscheduled Disease Progression Visit. The Disease Progression visit may be conducted either separately from or in place of a scheduled study visit and should be completed as soon as possible after PD is determined and before new anticancer therapy is initiated, if possible.

- AE assessment
  - Adverse event reporting requirements will follow the requirements applicable to the study period that the visit occurred. For example, if the disease progression visit occurred on Month 6 or earlier all AEs are reported, if between Month 6 and Month 24, only targeted AEs are reported. Refer to Section9.1.
- Concomitant medications assessment
  - Concomitant medications reporting requirements will follow the requirements applicable to the study period that the visit occurred. For example, if the disease progression visit occurred on Month 6 or earlier all concomitant medications are reported, if between Month 6 and Month 24, only targeted concomitant medications are reported.
- Vital signs
- Physical examination with neurological examination
- Local laboratory assessments
  - Chemistry panel
  - CBC with differential
- Sample collection for central assessment:
  - Whole blood: PBMCs, CBC with differential, cfDNA
  - Serum/plasma: key analytes (eg, cytokines, chemokines), immunogenicity studies
- Disease/overall response assessment:
  - CT or MRI scan (neck [if applicable], chest, abdomen, pelvis, and all other sites of disease) (unless performed within the previous 6 weeks)
  - Disease response assessment by RECIST v1.1 (unless performed within the previous 6 weeks)
  - Disease-specific tumor markers
- Fresh tumor biopsy: tumor biopsy for exploratory analyses

After completion of the Disease Progression Visit, subjects will proceed to Follow-up Visits for Early Exit Subjects completed as outlined in Section8.6.5.

8.6.8. Post-study Care

The Sponsor will not provide additional care for study subjects after completion of the study.

9. SAFETY MONITORING

9.1. Adverse Events and Reporting

An adverse event is any untoward medical occurrence experienced by a clinical trial subject from the time of study entry (initial screening informed consent), which may or may not have a relationship to the study treatment and/or study procedures. The types of reportable AEs for this study and reporting procedure and requirements are listed in Table 4. Investigators are responsible for assessing and determining if such medical occurrences are reportable as AEs.

Table 4: Adverse Events and Report

Adverse Event Type	Reporting Procedure / Requirement
Adverse Events: AEs may be a new medical event, sign, or symptom; or worsening of a pre-existing medical condition (eg, increased severity, increased frequency, prolonged duration, or worse outcome). New or worsening clinical sequalae, indicated by abnormal laboratory values and/or vital signs, may be reportable as AEs. Clinically significant abnormal laboratory and/or vital sign values may be reported as AEs if the criteria below are met: • Is clinically significant or represents a clinically significant change from baseline as determined by the Investigator • Associated with clinical symptom • Required medical intervention or change in concomitant therapy (eg, potassium supplementation for hypokalemia) AEs related to disease progression should be reported as the worsening signs and symptoms of the malignancy and not reported as the primary tumor histology. Events that should not be considered reportable as an adverse event: • A pre-existing medical condition that has not worsened during the study. • Pre-existing conditions that require elective cosmetic surgery or medical procedure. • Disease progression as a medical event. Worsening of sign and symptoms of the primary study malignancy should be assessed as AEs. • Abnormal laboratory values or vital signs, or changes from baseline value, without clinical significance	Data for all AEs are recorded in the eCRFs. Refer to Table 6 for AE reporting requirements per study period. AEs should be reported as a diagnosis rather than individual signs and symptoms whenever a diagnosis is available except for signs and symptoms of CRS (refer to the most recent version of the IB for more information related to CRS and ICANS). CRS is reported as an AE, and its associated signs and symptoms are also reported and assessed as individual AEs. ICANS is reported as an AE, and its associated signs and symptoms are also reported as assessed as individual AEs. Events that do not meet AE criteria are not reported in the eCRFs as AEs.

Table 4: Adverse Events and Report (Continued)

Adverse Event Type	Reporting Procedure / Requirement
Targeted Adverse Events: An AE that meets the criteria below: • CRS • Neurologic AE • Persistent Hematologic AE • Infection • Autoimmune disorder • Secondary malignancy (development ofa new malignancy after the treatment of ABC-01)	Data for all targeted AEs are recorded in the eCRFs. Refer toTable 5for AE reporting requirements per study period.
Serious Adverse Event: A SAE is an AE that results in any of the outcomes below: • Fatal • Life-threatening (ie, places the subject at an immediate risk of death as it occurred and not hypothetically if the event were more severe) • Inpatient hospitalization (requires admission to a healthcare facility) or prolongation of hospitalization or required an escalation of care during hospitalization (eg, required movement from routine care in the hospital to the intensive care unit) • Persistent or significant disability/incapacity (eg, substantial disruption in ability to conduct normal life functions) • Congenital anomaly/birth defect • Important medical event Death due to disease progression should be reported as the primary tumor histology. The occurrences below should not be considered reportable as a SAE: • Hospitalization for palliative or hospice care • Protocol-required hospitalization (eg, for NT-175 treatment and monitoring, efficacy measurement for the study) • Planned hospitalization for a pre-existing condition. • Hospitalization for disease progression	All SAEs are reported to the Sponsor within 24 hours of first awareness of the SAE. Refer to Table 5for AE reporting requirements per study period. SAEs are reported on a completed SAE form and sent via email to safety.neogene@apcerls.comor via fax to +1 844-245-7212 if electronic reporting is not available. For CRS reported as an SAE, CRS will be reported as the diagnosis and the individual signs and symptoms associated to CRS will be reported as individual AEs and indicated as SAEs in the EDC system, if SAE criteria is met for the specific AE. For CRS reported as an SAE, the individual signs and symptoms of CRS that meet SAE criteria will not be reported as individual SAEs (ie, separate from the associated CRS). Data for all SAEs and Targeted SAEs are recorded in the eCRFs. SAEs will require additional information that may be captured outside of the eCRFs. SAE data in the SAE report and eCRFs must reconcile (eg, updated SAE outcome report to Sponsor must be updated to match in the eCRF). Events that do not meet AE criteria are not reported in the eCRFs as AEs.

Table 4: Adverse Events and Report (Continued)

Adverse Event Type	Reporting Procedure / Requirement
Targeted SAEs: A SAE that meets the criteria below: • CRS • Neurologic AE • Persistent Hematologic AE • Infection • Autoimmune disorder • Secondary malignancy (development ofa new malignancy after the treatment of ABC-01).	All targeted SAEs are reported to the Sponsor within 24 hours of first awareness of the SAE. Refer toTable 5for AE reporting requirements per study period. Targeted SAEs are reported on a completed SAE form and sent via email to safety.neogene@apcerls.comor via fax to +1 844-245-7212 if electronic reporting is not available. For CRS reported as a SAE, the individual signs and symptoms associated to the CRS will remain reported as AEs but are not reported as individual SAEs, unless the AE meets SAE criteria. For ICANS reported as a SAE, the individual signs and symptoms associated to ICANS will remain reported as AEs but are not reported as individual SAE. Data for all SAEs and Targeted SAEs are recorded in the eCRFs. SAEs will require additional information that may be captured outside of the eCRFs. SAE data in the SAE report and eCRFs must reconcile (eg, updated SAE outcome report to Sponsor must be updated to match in the eCRF).

Abbreviations: AE, adverse event; CRS, cytokine release syndrome; eCRF, electronic Case Report Form; EDC, electronic data capture; IB, Investigator's Brochure; SAE, serious adverse event

9.2. Adverse Event Assessment

Investigators are responsible for assessing all AEs for severity and causality. CRS will be assessed for severity and causality independent of the associated AEs that are individual signs and symptoms of CRS events.

9.2.1. Adverse Event Severity Grading

The severity of AEs is graded according to NCI CTCAE v5.0. A copy of the NCI CTCAE grading scale can be downloaded from the Cancer Therapy Evaluation Program home page (http://ctep.cancer.gov).

9.2.1.1. Cytokine Release Syndrome and ICANS Severity Grading

Cytokine release syndrome as a syndrome requires additional severity grading according to the American Society for Transplantation and Cellular Therapy (ASTCT) CRS Consensus Grading (Lee, 2019).

The individual signs and symptoms associated with CRS are graded according to the NCI CTCAE (v5.0).

Neurological symptoms associated with the categorization of CRS are graded according to the NCI CTCAE.

Neurotoxicity associated with ABC-01 requires additional severity grading according to the ASTCT ICANS Consensus Grading (Lee, 2019). The ASTCT ICANS Consensus Grading includes the ICE assessment.
See Table 13 and Table 14 in Appendix C for more information on the ASTCT CRS Consensus Grading and ASTCT ICANS Consensus Grading.

9.2.2. Adverse Event Causality

Investigators are responsible for assessing the relationship of each AEs to one or more of the study agents or procedures below. An AE is considered related if, per Investigator clinical judgment, there is reasonable possibility that the event may be caused by a study agent and/or study procedure.

- IP (ABC-01)
- Lymphodepleting chemotherapy
  - DrugC
  - DrugF
  - Mesna
- rIL-2
- Leukapheresis
- Any other protocol-required procedure

AEs that meet the SAE reporting criteria may require alternative causality in addition to the above (eg, disease progression, concurrent disease, concomitant medications). See Table 4 for SAE criteria.

9.3. Adverse Event Reporting Period

For subjects who are enrolled and treated with ABC-01, AE reporting starts from the time of initial screening informed consent and ends at Month 24, disease progression, or early study exit (see Section 5.5 for early study exit criteria), whichever occurs first.

The requirement for type of AE to be reported differ by study period, see Table 5.

Prior to enrollment (initiation of leukapheresis), AEs that meet SAE criteria (see Table 4 for SAE criteria) or AEs that are related to study treatment or procedures (eg, biopsy) are reported.

Starting from enrollment (initiation of leukapheresis) through Month 6 after ABC-01 treatment, all AEs are reported, regardless of AE type.

After Month 6 through Month 24, or disease progression, or early study exit, whichever occurs first after ABC-01 treatment, AEs that meet the targeted AE criteria are reported. Targeted SAEs will require the same reporting timeline as defined in Section9.1.

Adverse events for subjects who will not receive ABC-01 treatment (eg, prescreen fail, screen fail, enrolled but not able to receive treatment) will be reported as defined in Table 5up until 30 days from the last study-specific procedure done or until the initiation of anticancer therapy, whichever occurs first.

Table 5: Adverse Event Reporting Period

<table>
<tr><th>Prescreening and Screening</th><th>Enrollment/ Leukapheresis</th><th>Lymphodepleting Chemotherapy</th><th>Treatment</th><th>Post-treatment Follow-up and Extended Follow-up</th><th>Long-Term Follow-up</th></tr>
<tr><td rowspan="2">AE Type:
SAEs related to study procedures
AEs related to study treatment or procedure (eg, biopsy)
Reporting Period:
Prescreening ICF or screening ICF to enrollment (leukapheresis start)</td><td colspan="4">AE Type:
All AEs/SAEs
Reporting Period:
Enrollment through Month 6</td><td rowspan="2">AE Type:
Secondary malignancies and ABC-01-related targeted SAES
Reporting Period:
After Month 24 through Month 180</td></tr>
<tr><td colspan="4">AE Type:
Targeted AEs/SAEs
Reporting Period:
After Month 6 through Month 24</td></tr>
</table>

Abbreviations: AE, adverse event; SAE, serious adverse event

9.4. Pregnancy and Lactation Reports

9.4.1. Pregnancy and Birth Control

There is no relevant clinical experience with ABC-01 in pregnant or lactating women, and animal reproductive studies have not been performed. Therefore, ABC-01 should not be administered to pregnant women or women who are breastfeeding. Women of childbearing potential must have a negative pregnancy test prior to enrollment because of the potentially dangerous effects of the lymphodepleting chemotherapy on the fetus.
The study will follow the recommendations from the Clinical Trial Facilitation Group (CTFG, 2014) as described below.

A woman is considered of childbearing potential (eg, fertile) following menarche and until becoming postmenopausal unless permanently sterile. Permanent sterilization methods include hysterectomy, bilateral salpingectomy, and bilateral oophorectomy.

A postmenopausal state is defined as no menses for 12 months without an alternative medical cause. A high follicle-stimulating hormone level in the postmenopausal range may be used to confirm postmenopausal data in women not using hormonal contraception or hormonal replacement therapy. However, in the absence of 12 months ofamenorrhea, a single follicle stimulating hormone measurement is sufficient.

A man is considered fertile after puberty unless permanently sterile by bilateral orchidectomy.

Acceptable methods of birth control are described in Appendix E.

9.4.2. Pregnancy and Lactation Reporting

Pregnancy and lactation reporting is required from time of study enrollment (initiation of leukapheresis) through 12 months after lymphodepleting chemotherapy completion or administration of ABC-01, whichever is longer.

Pregnancy and lactation events are reported to the Sponsor within 24 hours of first awareness. Pregnancies and lactation events are reported on a completed Pregnancy Report Form and sent via email tosafety.neogene@apcerls.comor via fax to +1 844-245-7212.

The pregnancy should be followed until its conclusion, even if the subject had completed and/or early exited the study, with the outcome reported to above mentioned email or fax. Any SAE occurring as an adverse pregnancy outcome after the subject completed the study will be reported to the Sponsor per the SAE reporting requirements (see Section9.1).

9.5. Sponsor Reporting Requirements

All SAEs will be reported to health authorities per local reporting guidelines.

The Sponsor may expedite reporting of serious adverse drug reactions or suspected unexpected serious adverse reactions (SUSARs) in accordance with relevant local and/or country legislation or regulation. The Sponsor will notify worldwide regulatory agencies and the relevant IRB/IEC in concerned member states of applicable SUSARs per current regulations.

The Sponsor will assess expectedness for SAEs using reference safety information specified in the IB.

All Investigators will receive a safety letter regarding SUSAR reports associated with any study drug. The Investigator will be responsible for notifying the local IRB/IEC of SUSAR reports as required by local regulatory agencies and per institutional policy.

9.6. Dose Limiting Toxicities

Dose-limiting toxicities are AEs related to ABC-01 that meet the criteria inTable 6within the first 28 days (Day 28) after ABC-01 treatment (Day 0).

Dose-limiting toxicities will be reviewed by the SRT before escalating to the next dose level after each planned dose level.

Table 6: Dose Limiting Toxicity Criteria

DLT Criteria	Exceptions to DLT Criteria
ABC-01 related Grade 5 adverse event	• None
ABC-01 related Grade 4 non-hematologic toxicity, regardless of duration[a]	• Grade 4 fever (pyrexia) or anorexia • Grade 4 transaminase, alkaline phosphatase, bilirubin, or other liver enzyme or function test elevation that improves to ≤ Grade 3 within 72 hours of initial onset and ≤ Grade 2 within 14 days of initial onset • Grade 4 infections that improve to ≤ Grade 3 within 7 days of initial onset
ABC-01 related Grade 3 non-hematologic toxicity that do not resolve to ≤ Grade 2 within 96 hours of onset[a]	• Grade 3 fever (pyrexia) or anorexia • Grade 3 nausea • Grade 3 transaminase, alkaline phosphatase, bilirubin, or other liver enzyme or function test elevation that improves to ≤ Grade 2 within 14 days of initial onset • Grade 3 infection that improves to ≤ Grade 2 within 14 days of initial onset • Grade 3 tremor, malaise, chills, headache, akathisia, restlessness, and insomnia of any duration • Grade 3 cardiac and/or pulmonary symptoms not causing irreversible organ damage that can be managed with appropriate non-invasive supportive care (eg, cough atelectasis, dyspnea) • Grade 3 skin rash, urticaria, or allergic reaction
Grade 4 infusion related reactions Seizure of any grade attributable to the TCR engineered T cell therapy Grade 3 ICANS not resolving to Grade 2 or lower within 72 hours Grade 4 CRS not resolving to Grade 3 or lower within 72 hours, or Grade 3 CRS not resolving to Grade 2 or lower within 14 days with the management of systemic corticosteroids and other supportive treatment.	• None

Abbreviations: CRS, cytokine release syndrome; DLT, dose limiting toxicity; ICANS, immune effector cell-associated neurotoxicity syndrome; TCR, T cell receptor
[a] Individual AEs that are symptoms of CRS should not be reported as a DLT.

9.7. Toxicity Management

9.7.1. Potential Toxicities Associated with ABC-01

To date, the following important identified risks have been associated to adoptive T-cell therapies.

- CRS
- ICANS

While CRS and ICANS have been observed with adoptive T cell therapies, these reactions have occurred much more frequently and with greater severity with CAR T cell therapies than with TCR-based or TIL-based therapies to date. Nevertheless, if observed, the goal of the CRS and ICANS management is to prevent life-threatening conditions while preserving the benefits of potential antitumor effects.

Refer to the IB for additional information about these risks and other important potential risks associated with ABC-01 and possible complications associated to malignancy and cancer treatment.

9.7.1.1. Cytokine Release Syndrome and ICANS Management

During rIL-2 administration (refer to Section8.5.1.3), it is possible that symptoms of rIL-2 toxicities, CRS, and ICANS will overlap. During rIL-2 administration, such symptoms should be managed as a rIL-2-associated toxicity (see Section9.7.1.2) unless there is a clear suspicion that the symptoms are CRS or ICANS. CRS should be managed according to its ASTCT severity grade (Lee, 2019). ICANS should be managed according to its ASTCT severity grade (Lee, 2019). Refer to Section9.2.1for more information on AE grading.

All cases ofCRS and ICANS should be communicated to the Sponsor Medical Monitor and management should be discussed.

Refer to the current version of the IB for CRS, neurological toxicity events, and other potential risks associated to ABC-01. Refer toTable 17inAppendix Cfor toxicity management guidelines for CRS and ICANS associated to ABC-01.

9.7.1.2. Recombinant IL-2-associated Toxicity

Recombinant IL-2 toxicities will be managed according to institutional guidelines. Refer to Appendix Dfor more details regarding rIL-2 toxicity management.

9.8. Safety Review Team and Criteria to Pause Enrollment

9.8.1. Safety Review Team

An SRT will be appointed for this study comprised ofprinciple investigators and selected Sponsor representatives.

- In Dose Escalation, the SRT will meet after each cohort of3 subjects are treated and followed for 28 days, or at a decision point based on the BOIN design, to assess

DLTs and make recommendations on the conduct of the study based on the totality of the data. At the end of Dose Escalation, the SRT will determine the MTD.

- In the Disease Histology Evaluation portion of the study, the SRT will convene approximately every 4 months and/or every time 6 subjects have been treated and followed for tumor response in a single histology, whichever occurs first. The SRT will determine a RP2D and make recommendations for further evaluation of ABC-01 at the RP2D in disease specific cohorts.
- In Part 2 (Disease Cohort Expansion) the SRT will continue to meet every 4 months to review the accruing safety and anti-tumor activity data.

Throughout the study, the SRT may hold ad-hoc meetings based on emerging data or upon the request of a SRT member. SRT membership and conduct is defined in the SRT charter.

9.8.2. Criteria to Pause Enrollment

The SRT will assess criteria to pause enrollment and treatment. Enrollment and treatment will be paused if the events below occur after ABC-01 treatment:

- When life-threatening SAE(s) (Grade 4) is reported on 2 subjects within 30 days of administration of ABC-01 at Dose Level 1 unless it is clearly due to disease progression.
- Any Grade 5 SAE within 30 days from administration of ABC-01 unless it is clearly due to disease progression.
- Any Grade 5 SAE that is at least possibly related to ABC-01.

The SRT will also assess additional criteria to pause enrollment after MTD has been determined. Enrollment will be paused if there is more than 80% probability, based on the Bayesian predictive distribution with a flat prior, that any of the following will happen in > 30% of treated subjects develop any of the following (that are at least possibly related to ABC-01):

- Grade 4 CRS
- Any Grade 4 AEs associated to infusion reaction / hypersensitivity
- Any Grade 4 cardiac event
- Grade 4 ICANS

The Sponsor will notify the applicable regulatory authority, within the applicable safety reporting timeline, if any of the pausing rules are met. Enrollment may be resumed upon recommendation of the SRT.

10. STATISTICAL ANALYSIS

Analysis and reporting will be done using all data from all subjects at the time the study ends.

The statistical analysis plan will be finalized prior to database lock and will include a more technical and detailed description of the statistical analyses described in this section. This section is a summary of the planned statistical analyses of the most important endpoints including primary and key secondary endpoints.

10.1. Hypothesis

No formal hypothesis testing is planned for this study.

10.2. Population

10.2.1. Analysis Sets

The subject analysis sets are defined inTable 7.

Table 7: Subject Analysis Sets

Subject Analysis Set	Description
All Enrolled Set	All subjects who are enrolled in the study and have initiated leukapheresis.
Safety Analysis Set	All subjects who are treated with ABC-01.
DLT-Evaluable Set	All subjects treated at the assigned dose level (Table 2) who meet either criterion: • Followed for the 28-day DLT evaluation period, or • Experienced a DLT within the DLT evaluation period
Efficacy Analysis Set	All subjects with measurable disease at baseline who received ABC-01 treatment and have at least 1 post-treatment disease assessment.
PK/Pharmacodynamics Analysis Set	All subjects who are treated with ABC-01 and have baseline and post-treatment PK and pharmacodynamics data

Abbreviations: DLT, dose-limiting toxicity; PK, pharmacokinetics

10.2.2. Sample Size Rationale

No formal statistical power calculations to determine sample size were performed for this study.

During Dose Escalation, a model-assisted BOIN design will be used to find the MTD of ABC-01 with 3 subjects enrolled and treated at each dose level. The BOIN design received the fit for purpose designation from U.S. Food and Drug Administration (FDA, 2021) as a tool for drug development to determine MTD (Liu, 2015;Yuan, 2016).

Approximately 24 subjects will be enrolled. The actual number of subjects enrolled in the Dose Escalation cohorts will be determined by the incidence of DLTs observed and will depend on the number of dose levels that are tested, and the number of subjects dosed at each dose level.

With a targeted toxicity rate of 30%, a true toxicity rate of 0.6 ×the target rate or lower is deemed potentially subtherapeutic such that dose escalation is indicated, and a true toxicity rate of 1.4 ×the target rate or higher is deemed overly toxic such that de-escalation is indicated. For non-informative priors, the prespecified optimal escalation and de-escalation boundaries for the observed toxicity rate are 23.6% and 35.9%, respectively. Hence, if the observed toxicity rate at the current dose level is ≤ 23.6%, then the dose is escalated for the next dose cohort. If the observed toxicity rate is > 35.9%, then the dose is de-escalated for the next dose cohort.

Furthermore, if there is a higher than 95% posterior probability that the true toxicity rate associated with the current dose level is above the target toxicity rate and there are at least 3 subjects enrolled in that dose level, then that dose level and higher dose levels will be eliminated.

The steps to implement the BOIN design are as follows:

Step 1: Subjects in Cohort 1 are dosed at Dose Level 1 (see Table 2).

Step 2: To assign a dose to the next cohort of subjects, conduct dose escalation according to the rules displayed in Table 8.

Step 3: Repeat Step 2 until the maximum sample size of 24 subjects is reached or stop the study early if the number of evaluable subjects treated at the current dose are ≥ 9 and the decision according to Table 8 is to stay at the current dose.

A flowchart for study conduct using the BOIN design is provided in Figure 2.

Table 8: Dose Escalation/De-escalation Rule for the BOIN Design

Actions[a]	Number of Evaluable Subjects at the Current Dose						
	3	**4**	**5**	**6**	**7**	**8**	**9**
Escalate if # of DLT ≤	**0**	0	1	**1**	1	1	**2**
De-escalate if # of DLT ≥	**2**	2	2	**3**	3	3	**4**
Eliminate if # of DLT ≥	**3**	3	4	**4**	5	5	**5**

Abbreviations: BOIN, Bayesian Optimal Interval; DLT, dose-limiting toxicity
[a] When none of the actions (ie, escalate, de-escalate, or eliminate) is triggered, stay at the current dose for treating the next cohort of subjects. Note that "number of DLT" is the number of subjects with at least 1 DLT.

The dose escalation/de-escalation rules are as follows:

"Eliminate" means eliminate the current and higher doses from the study to prevent treating any future subjects at these doses because they are overly toxic.

When eliminating a dose, automatically de-escalate the dose to the next lower level. In the situation where the first dose level is eliminated, the protocol may be amended to study a lower dose level.

If none of the actions (ie, escalation, de-escalation, or elimination) are triggered, treat the new subjects at the current dose.

If the current dose is the lowest dose and the rule indicates dose de-escalation, treat the new subjects at the lowest dose unless the number of DLTs reaches the elimination boundary, at which point terminate the study for safety.

If the current dose is the highest dose and the rule indicates dose escalation, treat the new subjects at the highest dose. Table 9 shows the operating characteristics of the trial design based on 1000 simulations of the trial using the shiny app "BOIN" (BOIN v.2.7.6.0) available at http://www.trialdesign.org. The operating characteristics show that the design selects the true MTD, if any, with high probability and allocates more subjects to the dose levels with the DLT rate closest to the target of 30%.

Table 9: Operating Characteristics of the BOIN design

	Dose Level			Number of Subjects	% Early Stopping
	1	2	3		
Scenario 1					
True DLT rate	0.30	0.48	0.67	13.5	12.7
Selection %	68.2	18.4	0.7		
% Subjects treated	62.7	32.2	5.1		
Scenario 2					
True DLT rate	0.13	0.30	0.44	17.6	0.1
Selection %	23.3	54.0	22.6		
% Subjects treated	35.0	43.2	21.9		
Scenario 3					
True DLT rate	0.05	0.13	0.30	17.6	0.0
Selection %	1.4	21.8	76.8		
% Subjects treated	21.8	33.5	44.7		
Scenario 4					
True DLT rate	0.15	0.20	0.30	17.1	0.6
Selection %	12.2	28.8	58.4		
% Subjects treated	29.4	34.6	36.1		

Abbreviations: BOIN, Bayesian Optimal Interval; DLT, dose limiting toxicity
Note: "% Early Stopping" refers to early stopping due to excessive DLT.

To collect additional safety, PK/PD, and preliminary anti-tumor activity data, up to 40 additional patients will be enrolled in Part 1 for specific histologies. These patients will contribute to the determination of RP2D.

With up to a total 64 subjects treated in Part 1, there is at least 95% probability to observe any adverse events with more than 5% incidence rate.

Upon review cumulative data regarding safety and benefit in Part 1, the SRT may suggest opening expansion cohort(s) for specific indication(s) at the RP2D and enroll additional subjects

in each specific histology indication during Part 2. Upon Part 1 and Part 2 completion, each indication may enroll up to a total of 20 subjects so that the marginal error of a 90% confidence interval is controlled at 20% of below.

10.2.3. General Considerations

For safety, efficacy, and PK/PD analysis, baseline is defined as the last available value before treatment of ABC-01.

For statistical analysis, in general, continuous data will be summarized by descriptive statistics, including number of subjects, mean, standard deviation, median, minimum, and maximum. Categorical data will be summarized by the number and percentage of subjects.

10.2.4. Primary Endpoints Analysis

DLT rate at each dose level will be summarized. Other safety information, such as SAEs, TEAEs, and laboratory assessments, will be summarized as supportive information. Safety analyses will be performed on the Safety Analysis Set.

10.2.5. Secondary Endpoints Analysis

Efficacy endpoints such as objective response rate (ORR), best objective response (BOR), duration of response (DOR), clinical benefit rate (CBR), time to response (TTR), progression-free survival (PFS), and overall survival (OS) will be analyzed to evaluate preliminary anti-tumor activities of ABC-01. These endpoints will be defined and analyzed by dose level based on Investigator assessment based on RECIST v1.1. Responders (complete response [CR] or partial response [PR]) are defined as subjects who have achieved subsequent confirmation of the first observation of CR or PR based on RECIST v1.1.

10.2.5.1. Objective Response Rate

The incidence of objective response and exact 2-sided 90% confidence interval will be generated for the Efficacy Analysis Set. Additional analyses may be conducted in subgroups defined by tumor histology.

10.2.5.2. Best Objective Response

The incidence of subjects with CR, PR, SD, progressive disease, and not evaluable as BOR to treatment and exact 2-sided 90% confidence interval about the incidence will be generated for the Efficacy Analysis Set. Additional analyses may be conducted in subgroups defined by tumor histology.

10.2.5.3. Time to Response

Time to response will be evaluated among responders in the Efficacy Analysis Set and is defined as the time from the ABC-01 dose to the first observed response (CR or PR) which is subsequently confirmed per RECIST v1.1. Kaplan-Meier estimates and 2-sided 95% confidence intervals will be generated for TTR.

10.2.5.4. Duration of Response

Duration of response will be evaluated among responders in the Efficacy Analysis Set and is defined as the time from the first observed response (CR or PR), which is subsequently confirmed up through disease progression per RECIST v1.1 or death from any cause. Subjects who are alive and remain in response at the time of analysis will have DOR censored at the last evaluable disease assessment prior to the data cutoff date for the analysis. Death in the absence of documented disease progression will be considered a DOR event. The DOR for subjects who received new anticancer therapy in the absence of documented disease progression will be censored at the last evaluable disease assessment prior to the new therapy.

Kaplan-Meier estimates and 2-sided 95% confidence intervals will be generated for DOR. Duration of response will be derived using disease assessments obtained on study prior to the initiation of new anticancer therapy.

10.2.5.5. Clinical Benefit Rate

The CBR will be evaluated in the Efficacy Analysis Set. Clinical benefit is defined as the best response of CR, PR, or SD. The incidence of subjects with clinical benefit will be summarized, and exact 2-sided 90% confidence interval about the incidence will be generated for the Efficacy Analysis Set. Additional analyses may be conducted in subgroups defined by tumor histology.

10.2.5.6. Progression-free Survival

Progression-free survival will be evaluated in the Efficacy Analysis Set and is defined as the time from the ABC-01 dose to disease progression per RECIST v1.1 or death from any cause. The PFS for subjects who are alive and without documented progression at the time of the analysis will be censored at the last evaluable disease assessment prior to the data cutoff date for the analysis. Deaths in the absence of documented disease progression will be considered a PFS event. The PFS for subjects who receive new anticancer therapy in the absence of documented disease progression will be censored at the last evaluable disease assessment prior to the new therapy. Kaplan-Meier estimates and 2-sided 95% confidence intervals will be generated for PFS. Progression-free survival will be derived using disease assessments obtained on study prior to the initiation of new anticancer therapy.

10.2.5.7. Overall Survival

Overall survival will be evaluated in the Efficacy Analysis Set and is defined as the time from the ABC-01 dose until death from any cause. Overall survival for subjects alive at the time of analysis will be censored at the last date known to be alive prior to the data cutoff date, or the data cutoff date, if the subject is known to have been alive after the data cutoff date.

Kaplan-Meier estimates and 2-sided 95% confidence intervals will be generated for OS. Estimates of the proportion of subjects alive at 6-month intervals will be provided.

10.2.6. Exploratory Endpoints Analysis

Summary statistics will be provided for PK and pharmacodynamic parameters including selected biomarkers based on PK/Pharmacodynamics Analysis Set. Immunogenicity analysis may be summarized.

ABC-01 product characteristics and manufacturing success rate will be summarized for drug products manufactured for the All Enrolled Set.

10.3. Interim Analysis

An SRT will review safety data and make recommendations on further study conduct on an ongoing basis. Refer to Section9.8.1for further details on the SRT.

No formal interim analysis is planned.

10.4. Procedures for Reporting Changes to the Planned Analysis

Details of the statistical analyses and considerations will be described in the Statistical Analysis Plan. Major changes to the planned analysis will be included in a protocol amendment.

11. ETHICAL CONSIDERATIONS

11.1. Good Clinical Practice

The Investigator, Sponsor and its third-party vendors will ensure that the study is conducted in accordance with ICH/GCP (ICH E6[R2] addendum) guidelines for GCP and applicable laws and regulations.

11.2. Ethics Review

This study will be conducted under a US Investigational New Drug (IND) application in accordance with recognized international scientific and ethical standards, including by not limited to the ICH guidelines for GCP, and the original principles embodied in the Declaration of Helsinki. These standards are consistent with the requirements of the US Code of Federal Regulation Title 21, Part 312, as well as local legislation.

11.3. Informed Consent Process

The Investigator is responsible for obtaining written informed consent from each subject prior to their participation in the clinical study and re-consenting subjects as relevant to their continued participation in the study. Informed consent must be obtained with an ICF that has been approved by the IRB/IEC.

The informed consent process followed by the Investigator is expected to align with the principles of ICH/GCP, relevant country regulation, IRB/IEC policy, and institution policy. The informed consent process will be documented as part of the subject's study records.

For this clinical study, subjects will be asked to consent to the below:

- Prescreening consent to perform blood (HLA) and biopsy (mutation) collection to confirm HLA-A*02:01 and positivity for *TP53* XYZ mutation, respectively, including consent for genomic testing and planned sample retention for future analysis. Refer to Section8.1for requirements to enter the Prescreening Period.
- Screening consent to perform screening activities to enroll in the study, including consent for genomic testing, planned sample retention for future analysis, and optional fresh tumor biopsy collections for translational analysis.

Subjects who refuse to consent to the main study will not be allowed to participate in the study.

11.4. Data Privacy

For this study, subjects will be identified by a unique subject ID number (see Section8.1.1.1and Section8.2.1.1for more information related to subject ID generation). Year of birth may be used as an additional form of ID (eg, to label leukapheresed material). The Investigator will ensure that procedures are in place and followed to maintain strict anonymity of each subject's identity and protection from disclosure to unauthorized parties.

Per federal regulations, ICH/GCP guidelines, the parties below may have access to information about a subject's identity and other personal information within source documents and/or medical records for the purposes of verifying study data.

- Sponsor and its third-party vendors (eg, contract research organization, contract clinical trial monitor, auditors)
- IRB/IEC
- Regulatory agencies

The parties listed above will follow procedures to maintain strict anonymity of each subject's identity and protection from disclosure to unauthorized parties. The Investigator is responsible for informing the potential disclosures of identity and personal information to the subject during the informed consent process.

11.4.1. Financial Disclosures

Investigators will provide documentation disclosing their financial interest or arrangements with the Sponsor or proprietary interests in the IP during the course of their involvement in this clinical study.

Financial disclosure documentation will be provided prior to the Investigator's participation in the study. The Investigator is responsible for notifying the Sponsor and providing documentation of any changes in reportable interests during the study and up to 1 year after completion of the study. See Section4.2 for the definition of study completion.

12. OVERSIGHT

12.1. Safety Monitoring, Quality Control, and Assurance

12.1.1. Safety Review Team

See Section9.8.1for details on the SRT.

12.2. Quality Control and Assurance

12.2.1. Monitoring

Monitoring will be performed to ensure that the protocol is principally ensured through the Investigator's supervision at the site and through the sponsor monitoring of the trial. The purpose of trial monitoring is to ensure that the study is conducted in accordance with GCPs and with applicable regulatory requirements. The monitor will conduct this activity in accordance with a Monitoring Plan.

12.2.2. Audits and Inspection

Regulatory authorities may conduct inspections at the site. The Investigator will notify the Sponsor immediately upon notification of an inspection by a regulatory authority. The Investigator will provide access to records, facilities, and personnel for the purpose of the inspection by regulatory authorities. Audits may be conducted by the Sponsor and/or a contractor. The Investigator will provide access to records, facilities, and personnel for the purpose of the audit.

12.2.3. Protocol Deviations

The Investigator is responsible for ensuring that the study is conducted in accordance with the protocol and ICH/GCP regulations and guidelines.

All deviations from the protocol and/or ICH/GCP will be documented as part ofthe study records and reviewed and assessed by the Sponsor. Deviations will be reported by the Investigator to the IRB/IEC in accordance with the IRB/IEC policy.

Deviations that may require additional follow-up include deviations that:

- Significantly impact the completeness, accuracy, and/or reliability of study data
- Significantly affects a subject's rights, safety, or well-being

The Investigator is responsible for immediately informing the Sponsor ifa deviation meets either or both definitions above.

12.2.4. Records

12.2.4.1. Source Documentation

Source documents are original documents, data, and records within which the study data is collected.

Source documentation for this study refers to original documents/records maintained by the Investigator and includes, but is not limited to:

- Subject medical records, hospital records, and/or clinic records
- Results of diagnostic tests (eg, ECG printout)
- Local laboratory, imaging, and/or radiology reports
- Study-related pharmacy records

Source documents will comply with Good Documentation Practices/ICH/GCP principles and applicable regulatory guidance, including ALCOA+C.

12.2.4.2. Electronic Case Report Forms

The Investigator is responsible for ensuring that data recorded in the eCRFs (within the electronic database capture [EDC] system) is complete, reliable, accurate, supported by source, and entered contemporaneously (eg, within 5 business days or another timeframe specified by the Sponsor). The Investigator will provide documentation supporting review and approval of the data entered in the eCRFs as instructed by the Sponsor (eg, via electronic signature per eCRF casebook within the EDC system). Data entries will be completed in the English language. Refer to the eCRF Completion Guideline for eCRF completion instructions.

Each subject who provides written informed consent and was assigned a subject ID number will have an eCRF casebook which will require data entry. Data entry will be done by the Investigator, or a study team member authorized by the Investigator for data entry.

Data entered in the eCRFs will be routinely source verified and/or reviewed (for completeness, accuracy, and consistency) by the Sponsor and/or its third-party vendor (eg, contract research organization, contract clinical trial monitor) within the EDC system. The Investigator will be made aware of data discrepancies identified and queried to either confirm the data as entered or confirm ifa data update is required. Ifa data update is required, the reason for update is provided (eg, data entry error). Ifa data update requires changes to the source data, the appropriate correction must be made to the source document, following Good Documentation Practice and ICH/GCP guidelines.

At study completion, the Sponsor will provide a read-only archive copy of the site eCRFs casebooks. The archive will be stored in accordance with the record retention requirements defined in Section12.2.4.4.

12.2.4.3. Access to Study Information

The Investigator is responsible for ensuring that all documents supporting the clinical study (eg, essential study documents, study source documents, subject ICFs, medical records, laboratory reports, pharmacy records, IP documents, radiology records, key correspondences) is retrievable and made readily available for clinical trial monitoring, or audit, or inspection by the Sponsor and its third-party vendors, regulatory authorities, and the IRB/IEC.

12.2.4.4. Records Retention

The Investigator is responsible for ensuring that all documents supporting the clinical study are maintained throughout the site's participation in the study, including, but not limited to essential study documents, study source documents, subject ICFs, medical records, laboratory reports, pharmacy records, IP documents, radiology records, and key correspondences.

Essential documents should be retained according to applicable local regulatory requirement(s) or as needed by the Sponsor.

The Investigator will provide the Sponsor with the location where study files and records are retained. If documents will be moved to an alternative location (eg, for off-site archival, to support subject transfer to another study site), the Investigator will notify the Sponsor and provide the updated location prior to moving the documents.

12.3. Study Termination or Study Site Closure

The Sponsor and the Investigator reserve the right to terminate the Investigator's participation in the study per the terms of the agreement in the study contract. The Investigator is responsible for notifying the IRB/IEC of study completion or early discontinuation.

The Investigator may initiate study-site closure at any time, provided there is reasonable cause and sufficient notice is given in advance of the intended termination.

Reasons for the early closure of a study site by the Sponsor or Investigator may include but are not limited to:

For study termination:

- Discontinuation of further study intervention development

For site termination:

- Failure of the Investigator to comply with the protocol, the requirements of the IRB/IEC or local health authorities, the Sponsor's procedures, or GCP guidelines
- Inadequate or no recruitment (evaluated after a reasonable amount of time) of subjects by the Investigator
- Total number of subjects included earlier than expected

If the study is prematurely terminated or suspended, the Sponsor shall promptly inform the Investigators, the IECs/IRBs, the regulatory authorities, and any contract research organization(s) used in the study of the reason for termination or suspension, as specified by the applicable regulatory requirements. The Investigator shall promptly inform the subject and should ensure appropriate subject therapy and/or follow-up.

13. PUBLICATION POLICY

A clinical study report will be prepared and provided to the regulatory agency. The Sponsor will ensure that the report complies with ICH/GCP guidelines for clinical study reports. An abbreviated report may be prepared in addition to the final clinical study report.

The Investigator may communicate, orally present, or publish in scientific journals or other scholarly media after all the conditions below are met.

- The results of the study in its entirety have been publicly disclosed by or with the consent of the Sponsor in an abstract, manuscript, or presentation form; or the study has been completed at all study sites and for all study subjects for at least 12 months, including any protocol-specified optional follow up data collection.
- The Investigator submits to the Sponsor any proposed publication or presentation along with the respective scientific journal or presentation forum at least 30 days before the submission of the publication or presentation.
- No such communication, presentation, or publication includes the Sponsor's confidential information.
- The Investigator complies with the Sponsor's request to delete references to its confidential information (other than the study results) in any paper or presentation and agrees to withhold publication or presentation for an additional 60 days to seek intellectual property if deemed necessary.
- Study results will not include any sequence or structure ofthe TCR or details regarding potency enhancement or manufacturing process.

14. FINANCING AND INSURANCE

The Sponsor has insurance that provides compensation for study-related illness of injury pursuant to the information outlined in the injury section ofthe ICF.

Investigators and their study staff may be asked to provide services performed for the protocol (eg, attendance at an Investigator meeting). If required under the applicable statutory and regulatory requirements, the Sponsor will capture and disclose to federal and state agencies any expenses paid or reimbursed for these services, including a clinical study payment, meal, travel expenses or reimbursement, consulting fees, and any other transfer of value.

15. REFERENCES

16. APPENDICES

APPENDIX A. INVESTIGATOR'S AGREEMENT

ABC-01-001 Protocol, Amendment 5, 11 Dec 2024

ABC Pharm LP Pharmaceuticals LP

This protocol has been approved by ABC Pharm LP .

Sponsor Medical Monitor Name (Printed) Signature

Date

INVESTIGATOR STATEMENT

I have read the protocol in its entirety and agree that it contains all the necessary information for me and my staff to conduct the study as described. I will conduct the study as outlined herein and will make a reasonable effort to complete the study within the time designated.

I agree to comply with the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) Harmonised Tripartite Guidelines on GCP and applicable national or regional regulations and guidelines. I will ensure all study personnel under my supervision has access to the protocol and all relevant information provided by ABC Pharm LP . I will discuss this material with them to ensure they are fully informed about the IP and the study.

I agree and will ensure that financial disclosure statements are completed by me (as the principal Investigator), all sub-Investigators, and, if applicable, our spouses, legal partners, or and dependent children for up to one year after the study is completed.

I agree to ensure that the confidential information contained in this document will not be used for any other purpose other than conduct of the clinical investigation without prior written consent from ABC Pharm LP .

Principal Investigator Name (Printed) Signature

Date Site Number

APPENDIX B. SCHEDULES OF ASSESSMENTS

The Schedule of Assessments for the Screening Period through the Treatment Period is presented in in Table 10, for the Post-treatment Period through the Extended Follow-up Period in Table 11, and for the Long-Term Follow-up Period (Month 30 through Month 180), Disease Progression Visit, and Early Exit Follow-up in Table 12.

Table 10: Schedule of Assessments for Study ABC-01-001: Screening Period Through Treatment Period

Period	Pre screening Period[a]	Screening Period	Enrollment / Leukapheresis Period[t]	Lymphodepletion Period					Treatment Period							
Visit	**Pre screening**	**Screening**	**Enrollment / Leukapheresis**	**Lymphodepleting Chemotherapy Day**					**Treatment Day**							
Days		**28 day window**	**~ - 28**	**-7**	**-6**	**-5**	**-4**	**-3**	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Study Procedures																
Informed Consent	X	X														
Eligibility Confirmation			X													
Demographics	X	X[d]														
Medical History	X	X														
Disease History		X														
Disease Treatment History	X	X														
Vital Signs		X	X	X[h]	X	X	X	X	X[i]	X[i]	X[i]	X[i]	X[i]	X[i]	X[i]	X[i]
Height		X														
Weight		X	X	X[h]	X	X	X	X	X							
Physical Examination with Neurological Examination[z]		X	X	X[h]					X[bb]	X	X	X	X	X	X	X
ECOG Performance Status		X	X	X[h]					X							
ECG		X														
ECHO[e]		X														
Brain MRI[f]		X[g]														

Table 10: Schedule of Assessments for Study ABC-01-001: Screening Period Through Treatment Period (Continued)

Period	**Pre screening Period[a]**	**Screening Period**	**Enrollment / Leukapheresis Period[t]**	**Lymphodepletion Period**					**Treatment Period**							
Visit	**Pre screening**	**Screening**	**Enrollment / Leukapheresis**	**Lymphodepleting Chemotherapy Day**					**Treatment Day**							
Days		**28 day window**	**~ - 28**	**-7**	**-6**	**-5**	**-4**	**-3**	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Adverse Events	X[j]	X[j]	X	X[h]	X	X	X	X	X	X	X	X	X	X	X	X
Concomitant Medications and Procedures		X[k]	X	X[h]	X	X	X	X	X	X	X	X	X	X	X	X
Procedures for ABC-01 Manufacture																
Leukapheresis			X													
Disease Assessment Procedures																
CT or MRI (for eligibility)[l]		X														
Disease-specific tumor markers[l]		X		X												
Baseline CT or MRI[m]				X												
Treatment Administration																
Lymphodepleting Chemotherapy Administration				X	X	X	X	X								
ABC-01 Administration									X							
rIL-2 Administration[n]									X	X	X	X	X	X	X	X

Table 10: Schedule of Assessments for Study ABC-01-001: Screening Period Through Treatment Period (Continued)

Period	Pre screening Period[a]	Screening Period	Enrollment / Leukapheresis Period[t]	Lymphodepletion Period					Treatment Period							
Visit	**Pre screening**	**Screening**	**Enrollment / Leukapheresis**	**Lymphodepleting Chemotherapy Day**					**Treatment Day**							
Days		**28 day window**	**~ - 28**	**-7**	**-6**	**-5**	**-4**	**-3**	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Specimen Collection/Analysis - For *TP53* XYZmutation and HLA typing																
Central HLA-A*02:01 typing[b]	X	X[b]														
Central *TP53* XYZ mutation testing[b]	X[c]															
Central Specimen Collection/Analysis - For Translational Research																
Archival Tumor Biopsy Shipment[o]		X														
Fresh Tumor Biopsy (baseline)		X[p]														
Optional Fresh Tumor Biopsy																X[aa]
Blood Sample for PBMCs[r,q]			X	X[h]					X	X[q]		X[q]		X[q]		X[q]
Blood Sample for CBC with Differential[r,q]			X	X[h]					X	X[q]		X[q]		X[q]		X[q]
Blood Sample for cfDNA				X[h]					X							X

Table 10: Schedule of Assessments for Study ABC-01-001: Screening Period Through Treatment Period (Continued)

Period	Pre screening Period[a]	Screening Period	Enrollment / Leukapheresis Period[t]	Lymphodepletion Period					Treatment Period							
Visit	**Pre screening**	**Screening**	**Enrollment / Leukapheresis**	**Lymphodepleting Chemotherapy Day**					**Treatment Day**							
Days		**28 day window**	**~ - 28**	**-7**	**-6**	**-5**	**-4**	**-3**	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Serum Sample for Key Analytes[r] (eg, cytokines, chemokines)			X	X[h]					X	X		X		X		X
Serum Sample for Immunogenicity Studies			X													
CSF and/or other bodily fluids		Only required if requested by the Sponsor for assessment of new onset ICANS or other clinically significant AEs														
Local Safety Laboratory Analysis[w]																
Chemistry Panel[s]		X	X[t]	X[h]	X	X	X	X	X	X	X	X	X	X	X	X
CrCl Calculation[u]		X	X	X[h]					X							
CBC with Differential[s]		X	X[t]	X[h]	X	X	X	X	X	X	X	X	X	X	X	X
Coagulation panel including fibrinogen		X	X[t]	X[h]					X	X		X		X		X
CRP			X	X[h]					X	X		X		X		X
Ferritin			X	X[h]					X	X		X		X		X
Pregnancy Test[v]		X	X	X[h]					X							
Serology[x]		X														
CSF[y]									If feasible, for any Grade 2 or higher ICANS event							

Abbreviations: AE, adverse event; CBC, complete blood count; CNS, central nervous system; CrCl, creatinine clearance; CRP, C-reactive protein; CT, computed tomography; cfDNA, cell free DNA; ECG, electrocardiogram; ECHO, echocardiogram; ECOG, Eastern Cooperative Oncology Group; HLA , human leukocyte

antigen; ICANS, immune effector cell-associated neurotoxicity syndrome; ICF, informed consent form; MRI, magnetic resonance imaging; NSCLC, non-small cell lung cancer; PBMCs, peripheral blood mononuclear cells; rIL-2, recombinant interleukin-2; SAE, serious adverse event; SC, subcutaneous

[a] Required if subjects are not known to be positive for *TP53* XYZ mutation and/or HLA-A*02:01 (ie, prior test was not done using one of the assays described in Appendix F or the result is not available) and must wait for the confirmatory *TP53* XYZ result and/or HLA-A*02:01 result, respectively, prior to signing the screening ICF at the Screening Visit.

[b] Subjects required to complete the Prescreening Visit will sign a Prescreening ICF and will have a central test to confirm positivity for *TP53* XYZ mutation and/or HLA-A*02:01 to confirm enrollment eligibility, see Section 5.3. Subjects who are not required to complete the Prescreening Visit will have blood collected for central HLA testing at the Screening Visit.

[c] Archival tumor biopsy will be shipped to the central laboratory and/or specialty laboratory for confirmatory *TP53* XYZ mutation testing. If archival material is not available or not available in sufficient quantity, a fresh tumor biopsy will be performed and used for confirmatory *TP53* XYZ mutation testing.

[d] If demographics are collected at the prescreening visit, they do not need to be collected again at the Screening Visit.

[e] ECHO: An ECHO performed after the last systemic treatment regimen and within the 28 days prior to the Screening Visit may be used to assess enrollment eligibility and does not need to be repeated at the Screening Visit. If the Investigator determines that an ECHO is not an appropriate assessment of ejection fraction, other methods may be used (eg, MRI, radionuclide angiography) with the approval of the Sponsor Medical Monitor.

[f] Brain MRI: A CT of the brain may be performed if MRI is contraindicated. Brain MRI may be done as part of evaluation of new onset ICANS that occur after ABC-01 treatment.

[g] Screening brain MRI: A brain MRI to confirm eligibility for enrollment will be performed for subjects who are considered high-risk CNS metastasis (eg, NSCLC, breast, colon, or any other histology considered high-risk of CNS metastasis by the Investigator), subjects that have a history of CNS metastasis, or if there is clinical suspicion of CNS involvement. A brain MRI performed after the completion of the subject's last systemic treatment and within the 28 days prior to signing of the screening ICF may be used to confirm enrollment eligibility.

[h] Day -7 procedures may be completed the day prior (24 hours) of initiating lymphodepleting chemotherapy.

[i] Vital signs (ABC-01 Treatment Period): Vital signs on Day 0 will be measured before, during, and within approximately 15 minutes after ABC-01 administration. Vital signs during the required inpatient hospitalization for ABC-01 treatment will be measured at a frequency that is compliant with institutional guidelines.

[j] Adverse events prior to enrollment/leukapheresis: AEs prior to enrollment will be reported if the AE meets SAE criteria related to study procedure or the AE is related to study treatment or procedure (eg, biopsy).

[k] Concomitant medications and procedures prior to enrollment/leukapheresis: Concomitant medications and procedures prior to enrollment/leukapheresis will be reported if related to a SAE or an AE related to a study intervention.

[l] Disease assessment procedures: CT or MRI scans will be done during Screening to confirm enrollment eligibility (refer to Section 5.3). A recent CT or MRI scan performed per standard of care within 28 days prior to signing the screening ICF may be used to confirm enrollment eligibility and does not need to be repeated if, in the opinion of the Investigator, the clinical status of the subject is unchanged. CT or MRI scans to establish baseline disease will be done prior to initiating lymphodepleting chemotherapy. Additionally, disease-specific tumor marker assessment will be performed locally to coincide with the imaging assessment as detailed in Section 7.10.2.

[m] Disease assessment procedures: Baseline CT or MRI of the neck (if applicable), chest, abdomen, pelvis, and other sites of disease will be performed to establish baseline disease prior to initiating lymphodepleting chemotherapy. Baseline CT or MRI scan may be performed up to the 15 days prior to initiating lymphodepleting chemotherapy on Day -7 but should be performed as close to Day -7 as feasible. If bridging therapy was administered (see Section 6.1.3), baseline disease assessment will be done after the completion of bridging therapy and prior to initiating lymphodepleting chemotherapy on Day -7.

[n] Recombinant IL-2 will be administered SC at a dose of 300 000 IU/kg once daily for a maximum of 8 doses with first dose being administered approximately 4 hours following ABC-01 treatment. rIL-2 may be administered up to 24 hours after ABC-01 treatment due to AE or other clinical reasons. The same rIL-2 dose will be administered once daily for a maximum of 8 doses as tolerated. Refer to Section 8.5.1.3 for more information on rIL-2 administration.

[o] Archival tumor tissue for translational analysis (FFPE block or approximately 30 unstained slides and 1 H&E slide) will be shipped to the central laboratory and/or specialty laboratory for immunohistochemistry and genomic/transcriptomic studies during screening. If archival tumor tissue is not available, a biopsy to be procured and shipped to the central laboratory and/or specialty laboratory may be done.

[p] Fresh tumor biopsy: The pretreatment (baseline) fresh tumor biopsy will be collected during the Screening Period up to at least 7 days prior to initiating lymphodepleting chemotherapy (Day -7). An attempt should be made to biopsy lesion(s) that will not impact disease response assessment, and the same lesion(s) should be biopsied at all timepoints, if clinically feasible. If the archival tumor tissue at screening was collected within 6 months prior to enrollment the baseline fresh biopsy sample is not required. (refer to Section7.11.5).

[q] Central laboratory sample collection (inpatient hospitalization and hospitalization extension): Whole blood collection timepoints during inpatient monitoring that fall on a Friday, weekend, or observed holiday (refer to the Central Lab Manual) may be missed or collected out of window as permitted by the Sponsor.

[r] Central laboratory sample collection (inpatient hospitalization and hospitalization extension): The samples below are collected 10 days after ABC-01 treatment and then per the Post-treatment Follow-up Period assessment schedule:

- Whole blood: PBMCs and other exploratory analyses, CBC with differential
- Serum/plasma: key analytes (eg, cytokines, chemokines) and other exploratory analyses

Central laboratory sample collection (Grade 3 or higher CRS or ICANS): the samples below are collected at onset or first re-occurrence (if not already collected on the same date as onset) and every 48 hours after onset and until Grade < 3 or at hospital discharge, whichever occurs first.

- Whole blood: PBMCs and other exploratory analyses, CBC with differential
- Serum/plasma: key analytes (eg, cytokines, chemokines) and other exploratory analyses

Central laboratory sample collection (rehospitalization due to ABC-01 adverse event): If subject is rehospitalized after discharging due to a ABC-01-related AE, the sample collection below is done on day of re-admission, weekly, and day of discharge:

- Whole blood: PBMCs, CBC with differential
- Serum/plasma: key analytes (eg, cytokines, chemokines) and other exploratory analyses

[s] Safety laboratory analysis (rehospitalization due to ABC-01 adverse event): If the subject is rehospitalized after discharge because of an ABC-01 related AE, a chemistry panel and CBC with differential will be performed, at a minimum, on the day of readmission, weekly, and on the day of discharge.

[t] If leukapheresis is delayed more than 5 days after eligibility confirmation, baseline CBC with differential, coagulation including fibrinogen and chemistry panel must be repeated and the medical monitor should be contacted for guidance.

[u] Creatinine clearance is estimated by Cockcroft Gault. If the Investigator determines that Cockcroft Gault is not an appropriate assessment of subject renal function, other methods may be used (eg, 24-hour urine creatinine clearance test) with the approval of the Sponsor Medical Monitor.

[v] Pregnancy test: Serum pregnancy test will be collected on subjects who are female of childbearing potential.

[w] Local Safety Laboratory Analysis: Refer toTable 3for a list of expected local laboratory tests.

[x] Serology testing: HIV (1 and 2), hepatitis B (surface antigen and core antibody, if positive for any, a PCR test should be performed), hepatitis C (antibody/PCR, if the HCV antibody is positive, a screening HCV RNA by any RT-PCR or DNA assay should be performed).

[y] Collected CSF should be sent for standard panel including at least CSF color, clarity, protein, glucose, total number of cells and CSF differential cell count. If infection is suspected, CSF Gram stain, routine culture and molecular tests that detect the genetic material of any microbes present, eg. HSV-1, HSV-2, enterovirus, varicella zoster virus, cytomegalovirus and HHV-6 should be conducted. See Section7.9.9for more details.

[z] Post-screening physical examination with neurological examination, including hospitalization for study treatment, will be performed per institutional procedures and/or guidelines.

[aa] An optional fresh tumor biopsy biopsies may be collected at Day 7 (-2/+7 days) for subjects consenting to the additional fresh tumor biopsy procedure.

[bb] Baseline ICE score should be assessed prior to ABC-01 treatment and maintained in the source documents.

Table 11: Schedule of Assessments for Study ABC-01-001: Post-treatment Through Extended Follow-up Period

Period	Post-treatment Follow-up Period				Extended Follow-up Period						
Visit	**Day 14**	**Day 28**	**Week 6**	**Week 12[c]**	**Week 18**	**Month 6**	**Month 9**	**Month 12**	**Month 15**	**Month 18**	**Month 24**
Days	**D14**	**D28**	**D42**	**D84**	**D126**	**D183**	**D274**	**D365**	**D456**	**D548**	**D730**
Visit Window (days)	**±2**	**±3**	**±7**	**±7**	**±7**	**±14**	**±14**	**±14**	**±14**	**±14**	**±14**
Study Procedures											
Vital Signs	X	X	X	X	X	X	X	X	X	X	X
Physical Examination with Neurological Examination	X	X	X	X	X	X	X	X	X	X	X
Adverse Events	X	X	X	X	X	X					
Concomitant Medications and Procedures	X	X	X	X	X	X					
Targeted Adverse Events							X	X	X	X	X
Targeted Concomitant Medications							X	X	X	X	X
Subsequent Therapy for Disease					X	X	X	X	X	X	X
Survival Status[a]					X	X	X	X	X	X	X
Disease Response Assessment Procedures											
Follow-up CT or MRI			X	X	X	X	X	X	X	X	X
Disease Response Assessment			X	X	X	X	X	X	X	X	X
Disease-specific tumor markers			X	X	X	X	X	X	X	X	X
Central Specimen Collection/Analysis											
Fresh Tumor Biopsy		X	X[b]	X[b]				X[b]			
Blood Samples for PBMCs	X	X	X	X	X	X	X	X	X	X	X

Table 11: Schedule of Assessments for Study ABC-01-001: Post-treatment Through Extended Follow-up Period (Continued)

Period	Post-treatment Follow-up Period				Extended Follow-up Period						
Visit	**Day 14**	**Day 28**	**Week 6**	**Week 12[c]**	**Week 18**	**Month 6**	**Month 9**	**Month 12**	**Month 15**	**Month 18**	**Month 24**
Days	**D14**	**D28**	**D42**	**D84**	**D126**	**D183**	**D274**	**D365**	**D456**	**D548**	**D730**
Visit Window (days)	**±2**	**±3**	**±7**	**±7**	**±7**	**±14**	**±14**	**±14**	**±14**	**±14**	**±14**
Blood Sample for CBC with Differential	X	X	X	X	X	X	X	X	X	X	X
Blood Sample for cfDNA		X		X		X		X			X
Serum Sample for Key Analytes (eg, cytokines, chemokines)	X	X	X	X	X	X	X	X	X	X	X
Serum Sample for Immunogenicity Studies		X		X			X			X	
CSF and/or other bodily fluids	Only required if requested by the Sponsor for assessment of new onset ICANS or other clinically significant AEs										
Local Laboratory Analysis[d]											
Chemistry Panel	X	X	X	X	X	X	X	X	X	X	X
CBC with Differential	X	X	X	X	X	X	X	X	X	X	X
Coagulation Panel including fibrinogen	X	X									
CRP[e]	X	X									
Ferritin[e]	X	X									
CSF[f]	If feasible, for any Grade 2 or higher ICANS event										

Abbreviations: CBC, complete blood count; CT, computed tomography; cfDNA, cell-free DNA; D, day; MRI, magnetic resonance imaging; PBMCs, peripheral blood mononuclear cells

[a] Survival status: Sponsor may ask for survival status outside of the scheduled visits (eg, for survival sweep), which may be confirmed by telephone or other type of contact with the subject or subject's caregiver.

[b] Fresh tumor biopsy: A post-treatment tumor biopsy at Day 28 after ABC-01 treatment is required if determined to be clinically feasible by the Investigator. If unable to collect a tumor biopsy at Day 28, the tumor biopsy may be collected at Week 6 or Week 12 if determined to be clinically feasible by the Investigator. In addition, post-treatment tumor biopsy may be collected at Month 12 or disease progression, whichever occurs first, if the subject remains on study and it is determined to be clinically feasible by the Investigator. An attempt should be made to biopsy lesion(s) that will not impact disease response assessment, and the same lesion(s) should be biopsied at all timepoints if clinically feasible.

[c] Subjects who were treated with ABC-01 and early exit from the study prior to Week 12 due to reasons other than Disease Progression, will be asked to complete the Week 12 visit procedures before proceeding to follow-up visits for early exit subjects as described in Section8.6.5. If the subject Early Exits before Week 12 due to Disease Progression, then the Disease Progression visit should be performed.

[d] Local Safety Laboratory Analysis: Refer toTable 3for a list of expected local laboratory tests.

[e] CRP, ferritin, and fibrinogen should be performed beyond 28 days only if abnormally high levels are noted on Day 28 assessment.

[f] Collected CSF should be sent for standard panel including at least CSF color, clarity, protein, glucose, total number of cells and CSF differential cell count. If infection is suspected, CSF Gram stain, routine culture and molecular tests that detect the genetic material of any microbes present, eg. HSV-1, HSV-2, enterovirus, varicella zoster virus, cytomegalovirus and HHV-6 should be conducted. See Section7.9.9for more details.

Table 12: Schedule of Assessments for Study ABC-01-001: Long-Term Follow-up Period (Month 30 through Month 180), Disease Progression Visit, and Early Exit Follow-up

Period	Long-Term Follow-up Period						Unscheduled	Early Exit Follow-up procedures
Visit	**M30**	**M36**	**M42**	**M48**	**M54**	**M60 to M180**	**Disease Progression**	**Early Exit Follow-up[i]**
Days (from D0)	**D913**	**D1095**	**D1278**	**D1460**	**D1643**	**D1825 to D5475 Annual[j]**	**Unscheduled**	**Per Extended and/orLTFU schedule**
Visit Window (days)	**±30 days**	**±30**	**±30**	**±30**	**±30 days**	**±30**	**N/A**	**N/A**
Study Procedures								
Vital Signs							X	
Physical Examination with Neurological Examination							X	
Adverse Events							X[a]	X[b]
Concomitant Medications and Procedures							X[c]	X[d]
Targeted Adverse Events							X[a]	X[b]
Targeted Concomitant Medications and Procedures							X[c]	X[d]
Subsequent Therapy for Disease	X	X	X	X	X	X		X
Survival Status[e]	X	X	X	X	X	X		X
Disease Response Assessment Procedures								
CT or MRI	X[f]	X[f]	X[f]	X[f]	X[f]	X[f]	X[l]	X[f,l]
Disease Response Assessment	X[f]	X[f]	X[f]	X[f]	X[f]	X[f]	X[l]	X[f,l]
Disease-specific tumor markers	X[f]	X[f]	X[f]	X[f]	X[f]	X[f]	X[l]	X[f,l]
Central Specimen Collection/Analysis								
Fresh Tumor Biopsy[g]							X	
Blood Samples for PBMCs	X	X	X	X	X	X	X	X[h]
Blood Sample for CBC with Differential	X	X	X	X	X	X	X	X[h]
Blood Sample for cfDNA							X	

Table 12: Schedule of Assessments for Study ABC-01-001: Long-Term Follow-up Period (Month 30 through Month 180), Disease Progression Visit, and Early Exit Follow-up (Continued)

Period	Long-Term Follow-up Period						Unscheduled	Early Exit Follow-up procedures
Visit	**M30**	**M36**	**M42**	**M48**	**M54**	**M60 to M180**	**Disease Progression**	**Early Exit Follow-up[i]**
Days (from D0)	**D913**	**D1095**	**D1278**	**D1460**	**D1643**	**D1825 to D5475 Annual[j]**	**Unscheduled**	**Per Extended and/orLTFU schedule**
Visit Window (days)	**±30 days**	**±30**	**±30**	**±30**	**±30 days**	**±30**	**N/A**	**N/A**
Serum Sample for Key Analytes (eg, cytokines, chemokines)							X	
Serum Sample for Immunogenicity Studies							X	
Local Laboratory Analysis[k]								
Chemistry Panel							X	
CBC with Differential							X	

Abbreviations: AE, adverse event; CBC, complete blood count; CRP, C-reactive protein; CT, computed tomography; cfDNA, cell-free DNA; D, day; M, month; MRI, magnetic resonance imaging; N/A, not applicable; PBMCs, peripheral blood mononuclear cells.

[a] Adverse events (Unscheduled Visit Disease Progression): Adverse event reporting requirements will follow the requirements applicable to the study period that the visit occurred. For example, if the unscheduled disease progression visit occurred on Month 6 or earlier all AEs are reported, if between Month 6 and Month 24, only targeted AEs are reported.

[b] Adverse events (Early Exit Follow-up): AE assessment until 30 days after the last study procedure was completed or initiation of anticancer therapy, whichever is earliest. Adverse event reporting requirements will follow the requirements applicable to the study period that the early exit occurred. For example, if early exit occurred on Month 6 or earlier all AEs are reported, if between Month 6 and Month 24, only targeted AEs are reported.

[c] Concomitant medications and procedures (Unscheduled Disease Progression Visit): Concomitant medications and procedures reporting requirements will follow the requirements applicable to the study period that the visit occurred. For example, if the unscheduled disease progression visit occurred on Month 6 or earlier all concomitant medications are reported, if between Month 6 and Month 24, only targeted concomitant medications are reported.

[d] Concomitant medications and procedures (Early Exit Follow-up): Concomitant medications and procedures assessment until 30 days after the last study procedure was completed or initiation of anticancer therapy, whichever is earliest. Concomitant medications reporting requirements will follow the requirements applicable to the study period that the visit occurred. For example, if the unscheduled disease progression visit occurred on Month 6 or earlier all concomitant medications are reported, if between Month 6 and Month 24, only targeted concomitant medications are reported.

[e] Survival status: Sponsor may ask for survival status outside of the scheduled visits (eg, for survival sweep) which may be confirmed by telephone or other type of contact with the subject or subject's caregiver.

[f] Disease response assessment procedures (imaging and disease-specific tumor markers): Disease response data after Month 24 will be per standard of care, until initiation of subsequent anticancer treatment.
[g] Fresh tumor biopsy: A fresh tumor biopsy should be collected at disease progression if disease progression occurred after ABC-01 treatment. An attempt should be made to biopsy lesion(s) that will not impact disease response assessment, and the same lesion(s) should be biopsied at all timepoints if clinically feasible.
[h] Subjects who early exit the study and were treated with ABC-01 will continue whole blood sample collection for PBMCs and CBC with differential per the Extended Follow-up and Long-Term Follow-up schedules until Month 180.
[i] Early Exit Follow-up: All subjects who were treated and early exited will complete the assessments/procedures specified below on the Extended Follow-up schedule if early exit occurred prior to Month 24, or on the Long-Term Follow-up schedule if early exit occurred after Month 24. Subjects who undergo leukapheresis and/or lymphodepleting chemotherapy but do not receive ABC-01 will be required to complete safety follow up for at least 30 days or until the resolution of any ongoing procedure or lymphodepleting chemotherapy -related AE(s) and will then come off study.
[j] Annual is considered 365 days. For example, Month 60 is Day 1825 (±30 days) from treatment. Month 72 (Year 6) is Day 2190 (±30 days) from treatment.
[k] Local Safety Laboratory Analysis: Refer to Table 3 for a list of expected local laboratory tests.
[l] Not required at Disease Progression visit or Early Exit Visit if previous assessment completed within the past 6 weeks.

APPENDIX C. ASTCT CRSAND ICANS CONSENSUS GRADING AND IMMUNE EFFECTOR CELL-ASSOCIATED ENCEPHALOPATHY SCORE

Table 13: ASTCT CRS Consensus Grading

CRS Parameter	Grade 1	Grade 2	Grade 3	Grade 4
Fever[a]	Temperature ≥ 38°C	Temperature ≥ 38°C	Temperature ≥ 38°C	Temperature ≥ 38°C
With				
Hypotension	None	Not requiring vasopressors	Requiring a vasopressor with or without vasopressin	Requiring multiple vasopressors (excluding vasopressin)
And/or[b]				
Hypoxia	None	Requiring low-flow nasal cannula[c] or blow-by	Requiring high-flow nasal cannula[d], facemask, nonrebreather mask, or Venturi mask	Requiring positive pressure (eg, CPAP, BiPAP, intubation and mechanical ventilation)

Abbreviations: ASTCT, American Society for Transplantation and Cellular Therapy; BiPAP, bilevel positive airway pressure; CPAP, continuous positive airway pressure; CRS, cytokine release syndrome.
Note: Organ toxicities associated with CRS may be graded according to CTCAE v5.0, but they do not influence CRS grading.
[a] Fever is defined as temperature ≥ 38°C not attributable to any other cause. In subjects who have CRS then receive antipyretic or anti-cytokine therapy such as tocilizumab or corticosteroids, fever is no longer required to grade subsequent CRS severity. In this case, CRS grading is driven by hypotension and/or hypoxia.
[b] CRS grade is determined by the more severe event: hypotension or hypoxia not attributable to any other cause. For example, a subject with temperature of 39.5°C, hypotension requiring 1 vasopressor, and hypoxia requiring low-flow nasal cannula is classified as Grade 3 CRS.
[c] Low-flow nasal cannula is defined as oxygen delivered at ≤ 6 L/minute. Low flow also includes blow-by oxygen delivery, sometimes used in pediatrics.
[d] High-flow nasal cannula is defined as oxygen delivered at > 6 L/minute.

Table 14: ASTCT ICANS Consensus Grading

Neurotoxicity Domain	Grade 1	Grade 2	Grade 3	Grade 4
ICE score[a]	7-9	3-6	0-2	0 (subject is unarousable and unable to perform ICE)
Depressed level of consciousness[b]	Awakens spontaneously	Awakens to voice	Awakens only to tactile stimulus	Subject is unarousable or requires vigorous or repetitive tactile stimuli to arouse. Stupor or coma
Seizure	N/A	N/A	Any clinical seizure focal or generalized that resolved rapidly or nonconvulsive seizures on EEG that resolve with intervention	Life-threatening prolonged seizure (> 5 min); or Repetitive clinical or electrical seizures without return to baseline in between
Motor findings[c]	N/A	N/A	N/A	Deep focal motor weakness such as hemiparesis or paraparesis
Elevated ICP/ cerebral edema	N/A	N/A	Focal/local edema on neuroimaging[d]	Diffuse cerebral edema on neuroimaging; decerebrate or decorticate posturing; or cranial nerve VI palsy; or papilledema; or Cushing's triad

Abbreviation: ASCTC, American Society for Transplantation and Cellular Therapy; EEG, electroencephalogram; ICE, immune effector cell-associated encephalopathy; ICANS, immune effector cell-associated neurotoxicity syndrome; ICP, intracranial pressure; N/A, not applicable.
Note: ICANS grade is determined by the most severe event (ICE score, level of consciousness, seizure, motor findings, raised ICP/cerebral edema) not attributable to any other cause; for example, a subject with an ICE score of 3 who has a generalized seizure is classified as Grade 3 ICANS.
[a] A subject with an ICE score of 0 may be classified as Grade 3 ICANS if awake with global aphasia, but a subject with an ICE score of 0 may be classified as Grade 4 ICANS if unarousable.
[b] Depressed level of consciousness should be attributable to no other cause (eg, no sedating medication).
[c] Tremors and myoclonus associated with immune effector cell therapies may be graded according to CTCAE v5.0, but they do not influence ICANS grading.
[d] Intracranial hemorrhage with or without associated edema is not considered a neurotoxicity feature and is excluded from ICANS grading. It may be graded according to CTCAE v5.0.

Table 15: Immune Effector Cell-associated Encephalopathy Score

Task	Direction	Score[a]
Orientation	State: year, month, city, hospital	4
Naming	Name: 3 objects	3
Following simple command	Follow: simple command	1
Writing	Ability to write a simple sentence	1
Attention	Count: backwards from 100 by 10,s	1

[a] Scoring: 10, no impairment; 7-9, Grade 1 ICANS; 3-6, Grade 2 ICANS; 0-2, Grade 3 ICANS; 0 due to subject unarousable and unable to perform ICE assessment, Grade 4 ICANS.

APPENDIX D. ABC-01 AND RIL-2 TOXICITY MANAGEMENT GUIDELINES

Prompt and urgent intervention to prevent progression of CRS is required; however, other causes of systemic inflammatory response should be ruled out, including infection and malignancy progression. Empiric treatment for infection is warranted in the neutropenic patient. Organ toxicities associated with CRS do not influence CRS grading. Organ toxicities should receive a thorough workup and appropriate management.

Fever is defined as temperature >38°C not attributable to any other cause. In subjects who have CRS and then receive antipyretics or anti-cytokine therapy such as tocilizumab or steroids, fever is not required to grade subsequent CRS severity. In this case, CRS grading is driven by hypotension or hypoxia. CYTOKINE RELEASE SYNDROME (CRS) Management guidelines (adapted from NCCN 2023) are found in Table 16.

Table 16: CRS Toxicity Management Guidelines

CRS Grade	Tocilizumab	Steroids[d, e, f]	Additional Supportive Care
Grade 1 Fever (≥38°C)	For early onset (develops within 24 hours of infusion) or prolonged CRS (>3 days)[c] in patients or those with significant symptoms, comorbidities, and/or are >65 years, consider 1 dose of IV tocilizumab 8 mg/kg over 1 hour (not to exceed 800 mg),†	Consider IV dexamethasone 10 mg every 24 hours for early-onset CRS (<72 hours after infusion)	• Sepsis screen and empiric broad-spectrum antibiotics, consider granulocyte colony-stimulating factor (G-CSF) if neutropenic[i] • Maintenance IV fluids for hydration • Symptomatic management of organ toxicities
Grade 2 Fever with hypotension not requiring vasopressors and/or hypoxia[a] requiring low-flow nasal cannula[b] or blow-by	IV tocilizumab 8 mg/kg over 1 hour (not to exceed 800 mg/dose).[d] Repeat in 8 hours if no improvement; no more than 3 doses in 24 hours, with a maximum of 4 doses total†	For persistent refractory hypotension after 1-2 doses of anti-IL-6 therapy: Consider IV dexamethasone 10 mg every 12-24 hours	• IV fluid bolus as needed • For persistent refractory hypotension after two fluid boluses and anti-IL-6 therapy: Start vasopressors, consider transfer to ICU, consider echocardiogram, and initiate other methods of hemodynamic monitoring. Telemetry, ECG, troponin, and BNP if persistent tachycardia • Manage per Grade 3 if no improvement within 24 hours after starting anti-IL-6 therapy • Symptomatic management of organ toxicities
Grade 3 Fever with hypotension requiring a vasopressor with or without vasopressin and/or hypoxia requiring high-flow cannula[g] face	Anti-IL-6 therapy as per Grade 2[d] if maximum dose not reached within 24-hour period	IV dexamethasone 10 mg every 6-12 hours depending on the product. If refractory, manage as Grade 4	• Transfer to ICU, obtain echocardiogram, and perform hemodynamic monitoring • Supplemental oxygen • IV fluid bolus and vasopressors as needed • Symptomatic management of organ toxicities

CRS Grade	Tocilizumab	Steroids[d, e, f]	Additional Supportive Care
mask, nonrebreather mask, or Venturi mask			

Table 16: CRS Toxicity Management Guidelines (Continued)

CRS Grade	Tocilizumab	Steroids[d, e, f]	Additional Supportive Care
Grade 4 Fever with hypotension requiring multiple vasopressors (excluding vasopressin) and/ or hypoxia requiring positive pressure (eg, continuous positive airway pressure [CPAP], bilevel positive airway pressure [BiPAP], intubation, mechanical ventilation)	Anti-IL-6 therapy as per Grade 2[d] if maximum dose not reached within 24-hour period	IV dexamethasone 10 mg every 6 hours. If refractory, consider 3 doses of IV methylprednisolone 1 - 2 g/day. If refractory, consider dosing every 12 hours[g]. Other lines of therapy may be considered[h]	• ICU care and hemodynamic monitoring • Mechanical ventilation as needed • IV fluid bolus and vasopressors as needed • Symptomatic management of organ toxicities

[a] CRS grade is determined by the more severe event: hypotension or hypoxia not attributable to any other cause. For example, a patient with a temperature of 39.5°C, hypotension requiring 1 vasopressor, and hypoxia requiring low-flow nasal cannula is classified as Grade 3 CRS.
[b] Low-flow nasal cannula is defined as oxygen delivered at ≤6 L/min. Low flow also includes blow-by oxygen delivery, sometimes used in pediatrics. High-flow nasal cannula is defined as oxygen delivered at >6 L/min.
[c] Can consider tocilizumab if CRS symptoms persist for >24 hours, and can consider tocilizumab for Grade 1 CRS that develops <72 hours after infusion and consider adding dexamethasone 10 mg x 1. For CRS developing ≥72 hours after infusion, treat symptomatically.
[d] After each dose, assess need for subsequent dosing.
[e] Antifungal prophylaxis should be strongly considered in patients receiving steroids for the treatment of CRS and/or neurotoxicity.
[f] Consider IV dexamethasone 10 mg every 12-24 hours after initial tocilizumab dosing if early-onset CRS.
[g] For example, IV methylprednisolone 1000 mg/day for 3 days, followed by rapid taper at 250 mg every 12 h for 2 days, 125 mg every 12 h for 2 days, and 60 mg every 12 h for 2 days.
[h] Other agents such as anakinra, siltuximab, ruxolitinib, might be considered after discussion with the medical monitor. For Immune Effector Cell-Associated Hemophagocytic Lymphohistiocytosis-Like Syndrome (IEC-HS), treat as per CRS with tocilizumab and steroids, although the suspicion of IEC-HS should

prompt consideration of early initiation of Anakinra and steroids (Hines, 2023). If no improvement is observed within 24 hours, consider increasing dose of anakinra and steroids, and consider starting ruxolitinib. Low dose etoposide can be considered as a last resort for IEC-HS with CNS involvement.

[i] GM-CSF is not recommended in the setting of CAR T-cell therapy. An FDA-approved biosimilar is an appropriate substitute for filgrastim.

† Under conditions of limited tocilizumab availability, consider one of the following conservation strategies:

- Limit tocilizumab use to a maximum of 2 doses during a CRS episode.
- Consider using steroids more aggressively during a CRS episode.
- If necessary, consider replacing second dose of tocilizumab with anakinra.

Table 17: ICANS Management Guidelines
CAR T-CELL-RELATED NEUROTOXICITY TREATMENT (adapted from (Thompson (NCCN), 2023)

Assessment and Supportive Care Recommendations (all grades)

- Neurologic assessment and grading at least twice a day to include cognitive assessment and motor weakness
- MRI of the brain with and without contrast (or brain CT if MRI is not feasible) for ≥ Grade 2 neurotoxicity
- Neurology consultation at first sign of neurotoxicity
- Conduct EEG for seizure activity for ≥ Grade 2 neurotoxicity
- Aspiration precautions; IV hydration
- Use caution when prescribing medications that can cause central nervous system (CNS) depression (aside from those needed for seizure prophylaxis/treatment)
- Antifungal prophylaxis should be strongly considered in patients receiving steroids for the treatment of CRS and/or neurotoxicity.

Treatment by Grade	No Concurrent CRS	Additional Therapy if Concurrent CRS
Grade 1[a]	• Supportive care	IV tocilizumab 8 mg/kg over 1 hour (not to exceed 800 mg/dose)[c, †]
Grade 2	• Supportive care • 1 dose of IV dexamethasone 10 mg and reassess. Can repeat every 6-12 hours, if no improvement.	Anti-IL-6 therapy as per Grade 1[c] Consider transferring patient to ICU if neurotoxicity associated with Grade ≥2 CRS
Grade 3[b]	• ICU care is recommended • IV dexamethasone 10 mg every 6 hours or IV methylprednisolone, 1 mg/kg every 12 hours[d] • Consider repeat neuroimaging (CT or MRI) every 2-3 days if patient has persistent Grade ≥3 neurotoxicity.	Anti-IL-6 therapy as per Grade 1[c]
Grade 4[b]	• ICU care, consider mechanical ventilation for airway protection. • High-dose steroids[e] • Consider repeat neuroimaging (CT or MRI) every 2-3 days if patient has persistent Grade ≥3 neurotoxicity. • Treat convulsive status epilepticus per institutional guidelines.	Anti-IL-6 therapy as per Grade 1[c]

[a] If ICANS develops <72 hours after infusion, consider IV dexamethasone 10 mg every 12-24 hours x 2 doses and reassess.
[b] Patients should undergo assessment for papilledema or other signs of elevated intracranial pressure. If intracranial pressure is excluded, a diagnostic lumbar puncture may be considered for patients with Grade 3-4 neurotoxicity.
[c] Repeat tocilizumab every 8 hours as needed if not responsive to IV fluids or increasing supplemental oxygen. Limit to a maximum of 3 doses in a 24-hour period; maximum total of 4 doses.
[d] IV methylprednisolone 1 g daily for 3-5 days may be considered.
[e] For example, IV methylprednisolone 1000 mg/day (may consider twice a day) for 3 days, followed by rapid taper at 250 mg every 12 hours for 2 days, 125 mg every 12 hours for 2 days, and 60 mg every 12 hours for 2 days.
† Under conditions of limited tocilizumab availability, consider one of the following conservation strategies:

- Limit tocilizumab use to a maximum of 2 doses during a CRS episode.
- Consider using steroids more aggressively during a CRS episode.

If necessary, consider replacing second dose of tocilizumab with anakinra after discussion with the medical monitor.

Table 18: Management of Cerebral Edema

Management of Cerebral Edema			
Supportive Therapy	**Tocilizumab**	**Corticosteroids**	**Follow-up**
• Follow management of neurologic events Grade 4 • Contact Sponsor Medical Monitor • Intensive care unit supportive therapy • Neuro-intensivist consult • If cerebral edema documented or strongly suspected, recommend neurosurgical consult • Optimal head position with elevation of head of bed and straight neck positioning • Administration of diuretics and osmotherapy per institutional practice guidelines • Early tracheal intubation with controlled mechanical mild hyperventilation and good oxygenation • Maintain cerebral perfusion pressure with mild hypervolemia • Avoid hypertension with use of anti-hypertensives (labetalol, nicardipine) • Avoid potent vasodilators • Pharmacological cerebral metabolic suppression (barbiturates, sedation, analgesia, and neuromuscular paralysis, as indicated) • Maintain rigorous glycemic control	Tocilizumab as above in Grade 4 neurologic event management (tocilizumab should be given only if concurrent CRS)	Methylprednisolone 1000 mg IV twice a day	Improving • Very slow corticosteroid taper recommended • Repeat neuroimaging as indicated • Serial neurologic exams as indicated • Consider early neuro-rehabilitation • Discontinue tocilizumab if started Not Improving • Repeat neuroimaging as indicated • Consider methylprednisolone 1000 mg IV 3 times a day or alternative therapy[a] • Consult Sponsor Medical Monitor

Abbreviations: CRS, cytokine release syndrome; IV, intravenous; IVIG, immunoglobulin
Note: Information is based on a review of treatment for cerebral edema (Rabinstein, 2006).

[a] Initiation of alternative therapy should be discussed with the Sponsor Medical Monitor and includes (but is not limited to): anakinra, siltuximab, ruxolitinib, DrugC, IVIG, ATG, intrathecal chemotherapy/steroids, and neurosurgical intervention.

Table 19: IEC-HS Grading (Hines, 2023)

Adverse Event	Grade				
	1	2	3	4	5
IEC-HS*	Asymptomatic or mild symptoms; requires observation and/or clinical and diagnostic evaluation. Intervention not indicated.	Mild to moderate symptoms, with intervention indicated (eg, immunosuppressive agents directed at IEC-HS, transfusions for asymptomatic hypofibrinogenemia)	Severe or medically significant but not immediately life- threatening (eg, coagulopathy with bleeding requiring transfusion support, or hospitalization required for new-onset acute kidney injury, hypotension, or respiratory distress)	Life-threatening consequences: urgent intervention indicated (eg, life-threatening bleeding or hypotension, respiratory distress requiring intubation, dialysis indicated for acute kidney injury)	Death

* Not attributable to other causes; defined by the development of pathological and biochemical features of macrophage activation/HLH that is attributable to IEC therapy and associated with progression or new onset of cytopenias, hyperferritinemia, coagulopathy with hypofibrinogenemia and hepatic transaminitis (>5 × ULN)). While HLH-like manifestations are frequently seen in in patients with severe CRS (as defined by ASTCT), IEC-HS is often delayed in onset and manifests as CRS is resolved/resolving.

Management of Other Important Risks

Cardiac Events in the Setting of CRS

Cardiac manifestations of CRS may include arrhythmias, decreased ejection fraction/heart failure, cardiomyopathy, myocardial ischemia, and cardiac arrest. Tachycardia is common in the setting of CRS, and medications to slow sinus tachycardia should be avoided. Subjects with persistent hypotension not responsive to parenteral fluids and low-dose vasopressors should be evaluated for decreased ejection fraction/heart failure by echocardiogram. These toxicities should be emergently managed per medical judgment and institutional practice guidelines.

Subjects with ≥ Grade 2 cardiac toxicity should be monitored with continuous cardiac telemetry and pulse oximetry. Tocilizumab and corticosteroids should be administered per Table 16. Follow-up ECGs and echocardiograms are recommended to monitor the course of cardiac toxicity to potential resolution.

Hypotension and Renal Insufficiency in the Setting of CRS

Hypotension and renal insufficiency should be treated as described here or according to medical judgment and institutional practice guidelines. Vigorous IV fluid (IVF) hydration may be needed to manage hypotension and vascular leak in the setting of CRS. Subjects should be closely monitored to prevent fluid overload, and, in some cases, continuous veno-venous hemodialysis may be required. Invasive hemodynamic monitoring, for example with a pulmonary artery catheter, may be helpful to optimize fluid management in settings of concurrent severe capillary leak, aggressive IVF administration, and/or pulmonary edema. Antihypertensive medications should be withheld whenever blood pressure begins to decrease below baseline or desired values. Hypotension should be defined as per the ASTCT guideline (Lee, 2019). The baseline blood pressure is defined for this guideline as the average of all blood pressure readings obtained during the 24 hours prior to the ABC-01 administration. The first treatment for hypotension is administration of IV isotonic (crystalloid) fluid boluses. If the hypotension does not respond adequately within 1 hour, a second bolus at a volume per Investigator discretion should be given. CRS management should follow Table 16.

If hypotension persists despite 2 fluid boluses, consideration should be given for monitoring in the intensive care unit and administering vasopressor support.

These guidelines may need to be modified based on institutional guidelines and the clinical characteristics of individual subjects, such as pulmonary status, cardiac function, renal function, and other factors.

Cytopenias

Many subjects will experience febrile neutropenia, which is an expected event, and should be managed with a thorough evaluation for a source of infection and administration of prophylactic broad-spectrum antibiotics per institutional practice guidelines. Granulocyte-colony stimulating factor (G-CSF) should be given according to published guidelines (eg, Infectious Disease Society of America). Fevers should be treated with supportive measures and antipyretics.

Euvolemia should be maintained with addition of isotonic intravenous fluids (eg, crystalloids) as clinically indicated and per institutional practice guidelines. Prolonged cytopenia beyond 30 days

following administration of ABC-01 may require further clinical investigation, including bone marrow biopsy and aspirate.

Blood Product Support for Anemia and Thrombocytopenia

All blood products should be irradiated and leukocyte reduced. Using CBCs as a guide, the subject should receive platelets and packed red blood cells as needed. Attempts should be made to keep hemoglobin > 8.0 g/dL and platelets > 10000/mm^3. Provide adequate platelet support prior to performing a lumbar puncture (eg, platelet > 50000/mm^3). Leukocyte filters should be utilized for all blood and platelet transfusions to decrease sensitization to transfused WBCs and decrease the risk of cytomegalovirus infection.

Infections

- Infection prophylaxis: Subjects should receive prophylaxis for infection with pneumocystis pneumonia, herpes virus, and fungal infections according to National Comprehensive Cancer Network guidelines or standard institutional practice guidelines.
- Fever and neutropenia: Evaluation for a source of infection should be performed per institutional practice guidelines. Fevers should be treated with acetaminophen and comfort measures. Corticosteroids should be avoided. Subjects who are neutropenic and febrile should receive broad-spectrum antibiotics. Maintenance IVF (isotonic [crystalloid] fluids) should be started for most subjects with high fevers, especially if oral intake is poor or ifthe subject has tachycardia. G-CSF should be given according to published guidelines (eg, Infectious Disease Society of America).

Deep Vein Thrombosis Prophylaxis

Deep vein thrombosis prophylaxis should be utilized in all subjects with reduced mobility during hospitalization per institutional guidelines. Low molecular weight heparin is encouraged as long as there are no contraindications (eg, recent surgery, bleeding diathesis, platelet count <50,000/µL) based on benefit/risk. Non-invasive mechanical intermittent pneumatic compression devices for deep vein thrombosis prophylaxis should be used in those who cannot receive anticoagulants due to increased bleeding risk or other concerns.

Tumor Lysis Syndrome

All subjects with significant malignancy burden and without a contradiction such as allergy should be started on prophylaxis (eg, allopurinol) as per institutional practice guidelines prior to initiation of conditioning chemotherapy. Prophylaxis should be discontinued when the risk of tumor lysis has passed.

rIL-2-Associated Toxicity

Recombinant IL2 toxicities will be managed according to institutional guidelines.

Capillary Leak Syndrome (CLS) Associated with rIL-2

Capillary leak syndrome (CLS) is a recognized risk with rIL-2 treatment that may be life-threatening or fatal. The relevant supportive care should be provided according to institutional guidelines.

The following are common signs and symptoms of CLS:

- Hypoalbuminemia
- Edema, including pulmonary edema
- Acute kidney injury and/or oliguria
- Weight gain
- Hypotension
- Hemodynamic instability

All subjects who are/will be treated with ABC-01 product should be assessed for signs/symptoms of CLS appropriately before and throughout rIL-2 administration.

Before initiating administration with rIL-2

- Ensure subject has:
  - adequate cardiac, pulmonary, and renal function per protocol
  - serum albumin ≥3.0 g/dL
- Weigh subject to establish baseline weight

During administration with rIL-2, including:

- Assess weight change from the previous day's pre-dose weight,
- Monitor serum albumin levels
- Assess for
  - New onset or worsening edema, including pulmonary edema
  - Hypotension or hemodynamic instability

CLS management recommendations

- Use caution with rIL-2 ifa subject shows signs of CLS, particularly significant low blood albumin (<3.0 g/dL) or substantial weight gain (>1.5 kg) compared to the previous day's weight.
- Consider withholding rIL-2 if:
  - Serum albumin falls below 2.5 g/dL.
  - New or worsening swelling, including in the lungs, develops.
  - Blood pressure drops significantly (systolic blood pressure ≤80 mm Hg) or becomes unstable.

Recommended action

- Hold rIL-2 until symptoms improve
- Administer albumin supplement as needed to maintain a level ofat least 3.0 g/dL

- If the predose body weight is increased by ≥1.5 kg over the previous day's predose weight, manage fluid status, as indicated clinically (eg, generally with intravenous fluids and vasopressors if hypotensive and with diuretics if normotensive or hypertensive), until body weight increase has resolved (eg, no longer exceeding 1.5 kg daily increase).
- If with additional CLS signs such as edema, fluid overload, acute kidney injury (AKI) and/or hypotension, aggressively manage fluid status and hypotension, if present, which could include intravenous fluids and/or diuretics or other blood pressure management, until resolution of CLS sign/symptom or as clinically indicated.
- Corticosteroids such as methylprednisolone (or an equivalent) can be provided per institutional guidelines until resolution of CLS sign/symptom or as indicated clinically.
- Monitor oxygen saturation closely
- Monitor creatinine and electrolytes daily

Resuming rIL-2 Therapy

rIL-2 administration may only be restarted when the subject is hemodynamically stable, and critical CLS signs/symptoms have resolved /significantly improved to ensure tolerance with the continuous administration of rIL-2.

Fevers Associated with rIL-2 and Supportive Care

Fever is a known toxicity associated to rIL-2 administration.

At the time of rIL-2 administration, it is likely that the subject will be recovering from pancytopenia due to lymphodepleting chemotherapy. Supportive care should be considered during rIL-2 administration, including empiric treatment for infection and rIL-2 based toxicities, and full workup for infectious etiologies must be performed, preferably in consultation with the institution's infectious disease service if available.

Concomitant Medications to Manage rIL-2 Toxicities

The medications below may be administered to manage known rIL-2 toxicities:

- Acetaminophen PO at 250-500 mg every 4 to 6 hours
- Indomethacin PO at 50-75 mg every 6 hours
- H_2-receptor antagonist PO or IV every 12 hours
- Meperidine at 25 to 50 mg IV should be administered for severe chills if feasible
- Ondansetron PO or IV for nausea or vomiting
- Additional antiemetics as needed for nausea or vomiting

If the subject requires high doses of corticosteroid therapy to manage rIL-2 toxicities, additional rIL-2 doses should be withheld.

APPENDIX E. ACCEPTABLE METHODS OF BIRTH CONTROL

Acceptable methods of highly effective birth control for females of childbearing potential and fertile males are methods that can achieve a failure rate of less than 1% per year when used consistently and correctly. Such methods include the below:

- Oral, inserted, injected, or implanted hormonal methods of contraception, provided it has been used for an adequate period of time to ensure effectiveness.
- Correctly placed copper containing intrauterine device (IUD).
- Male sterilization with confirmed absence of sperm in the post-vasectomy ejaculate.
- Bilateral tubal ligation or bilateral salpingectomy.
- Sexual abstinence
  - In the context of this protocol, sexual abstinence is considered a highly effective method of birth control only if refraining completely from heterosexual intercourse during the entire period of risk (ie, during study treatment, including during temporary breaks from treatment, and for at least 12 months after stopping lymphodepleting agent.

Unacceptable methods of birth control include periodic abstinence (calendar, symptothermal, post ovulation methods), withdrawal (coitus interruptus), spermicides only, and lactational amenorrhea method. A female condom and male condom should not be used together.

APPENDIX F. ACCEPTABLE IN VITRO DIAGNOSTIC ASSAYS FOR CONFIRMATION OF *TP53* XYZ AND HLA-A*02:01 POSITIVITY

FoundationOne®CDx (F1CDx, P170019)

Subjects with historical results available based on this test performed at Foundation Medicine will be eligible to enroll in the study.

All patients with no prior evidence of *TP53* XYZwill undergo mutation testing using the FoundationOne®CDx test during Prescreening.

The proposed CTA is an NGS-based in vitro diagnostic device for detection of substitutions, insertion and deletion alterations, copy number alterations and select gene rearrangements using DNA isolated from formalin-fixed paraffin embedded (FFPE) solid tumor tissue specimens. The technology underlying the CTA is the same as the FDA-approved FoundationOne®CDx (F1CDx, P170019). The proposed CTA will be performed in the laboratories of Foundation Medicine Inc. headquartered at 150 Second Street, Cambridge, Massachusetts, 02141. Starting from 1Q 2025, the testing will be performed at 400 Summer Street, Boston, Massachusetts 002210.

The laboratory process used for the NGS CTA is a robust, multistep protocol to generate targeted deep sequence data from adaptor-ligated sequencing libraries of DNA extracted from FFPE tissue. The workflow begins with sample receipt and accessioning into a laboratory information management system, incorporates the generation of targeted sequencing data on an Illumina® sequencer, and culminates with a customized analysis pipeline capable of highly accurate detection of genomic alterations. Multiple steps of the process are run on automated liquid handlers to ensure consistency and efficiency. The assay includes process-matched controls in each run, and defined quality control metrics are used throughout the process to monitor sample and assay performance.

The technology underlying the CTA is the same as the FDA-approved FoundationOne®CDx (F1CDx, P170019) which has been robustly validated for the detection of genetic alterations.

Supporting documentation of the FDA-approved F1CDx device is available at f1cdxlabel.com.

Caris Tissue NGS Assay

Subjects with historical results available based on this test performed at Caris Life Sciences will be eligible to enroll in the study.

Caris's MI Exome is a next-generation sequencing-based test that uses DNA isolated from tumor tissue to provide tumor mutational profiling using a custom human exome panel to detect variants including single nucleotide polymorphisms (SNPs) and insertions and deletions (INDELs). The sequencing results are also used to determine copy number alterations (CNA) and genomic signatures including microsatellite instability (MSI), tumor mutational burden (TMB) and loss of heterozygosity (LoH) via a proprietary bioinformatic pipeline developed and validated by Caris.

The MI TumorSeek Comprehensive Genomic Profile (CGP) panel is validated with an overall accuracy of ≥ 95% when detecting Single Nucleotide Variants (SNV), Insertions and Deletions by looking at over 7500 variants in DNA from 500+ samples, including a comparison to Sanger

Sequencing calls for 20 SNV target areas and 20 Indel target areas across a total of 400 formalin-fixed, paraffin-embedded (FFPE) analyses with ≥ 95% concordance. All precision studies demonstrated ≥ 95% concordance. MSI Analysis by CGP from DNA on 1196 colorectal carcinoma samples, including 75 MSI-High samples, demonstrated accuracy, sensitivity, and specificity metrics ≥ 95%.

Supporting documentation of the Caris NGS assay is available at https://www.carislifesciences.com/products-and-services/molecular-profiling/profiling-technologies/whole-exome-sequencing.

Tempus xT CDx

Subjects with historical results available based on this test performed at Tempus will be eligible to enroll in the study.

The Tempus CDx Assay (xT CDx) is a single site next generation sequencing (NGS) assay. The assay includes reagents, software, instruments, and procedures for testing DNA extracted from formalin-fixed, paraffin embedded (FFPE) tumor specimens and matched normal saliva or blood specimens. The assay employs DNA extraction methods from routinely obtained FFPE tissue samples and matched normal saliva or blood samples. Extracted DNA undergoes whole-genome shotgun library construction and hybridization-based capture of specified regions from 648 cancer-related genes (including intronic overhangs and selected promoter regions), and 239 loci for MSI. Using the Illumina NovaSeq 6000 platform, hybrid-capture-selected libraries are sequenced to highly uniform depth (targeting >500x median coverage of tumor samples, with >95% of exons at >150x coverage and ≥98% of exons at ≥100x coverage). Sequence data is processed using a customized analysis pipeline designed to detect substitutions (SNVs and MNVs), insertions, and deletions in coding and noncoding genomic regions targeted by the assay. Additionally, MSI status is reported based on a genomic signature.

Supporting documentation of the Tempus NGS assay is available at https://www.accessdata.fda.gov/cdrh_docs/pdf21/P210011C.pdf.

Dana Farber OncoPanel Test

Subjects with historical results available based on this test performed at DFCI using OncoPanel will be eligible to enroll in the study.

The Dana Farber OncoPanel test has been validated and performed in a clinical laboratory that is certified by CLIA (CLIA certificate: 22D2040971), under CLIA guidelines for clinical testing.

OncoPanel is a cancer genomic assay designed to detect somatic mutations, copy number variations and structural variants in tumor DNA extracted from fresh, frozen or formalin-fixed paraffin-embedded samples. This assay surveys exonic DNA sequences of 447 cancer genes and 191 regions across 60 genes for rearrangement detection. DNA is isolated from tissue containing at least 20% tumor nuclei and analyzed by massively parallel sequencing using a solution-phase Agilent SureSelect hybrid capture kit and an Illumina Novaseq 6000 sequencer.

OncoPanel has achieved 100% sensitivity with PPA over 99% for the detection of single-nucleotide variants and multi-nucleotide variant (MNV) detection of insertions and deletions compared with single-gene assays and mass spectrometry-based genotyping.

MD Anderson Mutation Analysis Precision Panel (MDA MAPP) Test

Subjects with historical results available based on this test performed at MD Anderson Cancer Center using MD Anderson Mutation Analysis Precision Panel will be eligible to enroll in the study.

The MD Anderson Mutation Analysis Precision Panel has been validated and performed in the clinical laboratory that is certified by CLIA (CLIA certificate: 45D0994328), under CLIA guidelines for clinical testing.

The MDA MAPP assay is a custom high-throughput next generation sequencing-based CLIA assay that uses targeted hybridization-based capture technology for detection of sequence variants/mutations in 610 genes (single nucleotide variants [SNVs] and insertion/deletion alterations [indels]), copy number variants (CNVs) in 583 genes, select gene rearrangements in 34 genes (Fusions), and selected genomic immuno-oncology signatures including microsatellite instability (MSI) and tumor mutational burden (TMB) in DNA isolated from formalin-fixed paraffin embedded (FFPE) tumor tissue and cytology specimens. The assay employs DNA extracted from both tumor tissue and paired normal (blood or tissue) specimens in CLIA-certified molecular diagnostic laboratory. MDA MAPP assay uses the NovaSeq 6000 next generation sequencing platform and bidirectional paired-end sequencing to identify nucleic acid variants for all coding regions from most genes in the panel, the TERT promoter, 1 non-coding RNA gene, and clinically relevant rearrangements.

MDA MAPP has achieved 99.2% sensitivity and 100% specificity for Single Nucleotide variations.

BostonGene Tumor Portrait™ Test

Subjects with historical results available using the BostonGene Tumor Portrait test will be eligible to enroll in the study.

The BostonGene Tumor Portrait test has been validated and is performed in a CLIA certified laboratory (CLIA certificate: 22D2182613), under CLIA guidelines for clinical testing.

BostonGene Tumor Portrait test uses DNA whole exome (WES) and RNA whole transcriptome (RNA-seq) sequencing to detect genomic alterations, such as single nucleotide variants (SNVs), insertions and deletions (Indels), copy number alterations (CNAs), tumor mutational burden (TMB), microsatellite instability (MSI), expressed fusions, frameshifts, as well as interrogating the expression levels of more than 20,000 genes. The BostonGene's Tumor Portrait test report integrates genomic and transcriptomic analyses and in concert with the patient's clinical history provides information regarding biomarkers associated with response and resistance to therapies or therapeutic combinations, suitable for use with NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®).

The BostonGene Tumor Portrai test has demonstrated high specificity (99.9%, 99.9%, >94.8%) and sensitivity (>98.4%, >99.9%, 86.3%) for the detection of SNVs, indels, and CNAs, respectively, with low false-positive and false-negative rates.

Providence St. Joseph Health Molecular Genomics Laboratory ProvSeq

Subjects with historical results available using the ProvSeq 523 test will be eligible to enroll in the study.

The ProvSeq test has been validated and is performed in a CLIA certified laboratory (CLIA certificate: 38D2032720), under CLIA guidelines for testing. The state-of-art ProvSeq 523 panel is based on the Illumina TruSight Oncology 500 assay. This comprehensive genomic profiling assay utilizes next-generation sequencing (NGS) to detect tumor mutational burden (TMB), microsatellite instability (MSI) and somatic variants in 523 cancer-related genes in a single workflow.

This test has been shown to detect >95% of Single Nucleotide Polymorphisms within the reportable range at allelic frequencies of 5%. The limit of detection for the method to accurately call insertions and deletions is 10% allelic frequency. The limit of detection for gene copy number amplification is 8 copies.

The ProvSeq test has demonstrated high specificity (100% for SNV, indel and CNV detection) and high sensitivity (99.85% for SNV, 100% for indel and 100% for CNV detection) with low false-positive and low false-negative rate.

HopeSeq Solid tumor comprehensive panel (City of Hope)

Subjects with historical results available using the HopeSeq solid tumor comprehensive test panel will be eligible to enroll in the study.

The HopeSeq test has been validated and is performed in a CLIA certified laboratory (CLIA certificate: 05D2302026), under CLIA guidelines for testing. The HopeSeq panel utilizes the TruSight Oncology 500 High-Throughput hybrid-capture NGS method and sequenced on the Illumina NovaSeq 6000 System. DNA sequencing assay detects mutations in 523 genes (full exon), copy number variations (CNV) in 500 genes, microsatellite instability (MSI) and tumor mutation burden (TMB).

The lower limit of detection for sequence variations is at 2% allelic frequency at 100X coverage for cases with no previously known molecular results. These assays have been validated to detect known mutations down to 0.3% (3×10^{-3}) allele frequency.

The HopeSeq test has demonstrated high specificity (100% for SNV and indels, 99% for CNV) and sensitivity (100% for SNV and indels, 99% for CNV).

American Red Cross- HLA-A*02:01 Test

The test for the determination of HLA-A*02:01 to be used in this study is the HLA A eRes kit, which is based on the LIFECODES HLA-SSO Typing kits, using EDTA whole blood collected from study subjects.

The LIFECODES HLA-SSO Typing kits utilize sequence-specific oligonucleotides (SSOs) to identify which HLA alleles are present in a PCR-amplified sample. HLA typing by SSO methods is accomplished using SSO probes. Each SSO is challenged with known positive and negative reacting alleles to establish and ensure expected reactivity. The set of SSOs determines the ability to distinguish among the various alleles present in the PCR amplification. The LIFECODES assay is a homogenous multiplex system, in that all SSOs are analyzed

simultaneously, and the entire assay is carried out in a single reaction vessel with the addition of a single reagent.

The HLA A eRes kit shows 98.46% agreement for HLA-A alleles (95.24% lower boundary using one-sided exact method at 95% confidence interval) when comparing HLA typing results at 2 fields in 130 well-characterized samples. Reproducibility is 98.9% to 100%, and percent failure ranged from 0% to 1.04%.

The proposed test will be performed in the laboratories of American Red Cross Blood Services Pacific Northwest Regional Lab at 3131 North Vancouver Avenue, Portland, Oregon 97227-1560. This is a Clinical Laboratory Improvement Amendments (CLIA)-accredited laboratory (CLIA ID 38D875080) that routinely performs this test to support the appropriate selection of interventions for patients based on their HLA type in standard clinical practice. Therefore, patients selected to enroll in this study using the American Red Cross CLIA -accredited HLA assay should be reliably identified as bearing HLA-A*02:01 and be at low risk of inappropriate exposure to ABC-01.

APPENDIX A. INVESTIGATOR'S AGREEMENT

ABC-01-001 Protocol, Amendment 5, 11 Dec 2024

ABC Pharm LP Pharmaceuticals LP

This protocol has been approved by ABC Pharm LP .

Electronically signed by

Reason: I approve this document
Date: Dec 11, 2024 PST

Sponsor Medical Monitor Name (Printed)

Signature

11-Dec-2024

Date

INVESTIGATOR STATEMENT

I have read the protocol in its entirety and agree that it contains all the necessary information for me and my staff to conduct the study as described. I will conduct the study as outlined herein and will make a reasonable effort to complete the study within the time designated.

I agree to comply with the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) Harmonised Tripartite Guidelines on GCP and applicable national or regional regulations and guidelines. I will ensure all study personnel under my supervision has access to the protocol and all relevant information provided by ABC Pharm LP . I will discuss this material with them to ensure they are fully informed about the IP and the study.

I agree and will ensure that financial disclosure statements are completed by me (as the principal Investigator), all sub-Investigators, and, if applicable, our spouses, legal partners, or and dependent children for up to one year after the study is completed.

I agree to ensure that the confidential information contained in this document will not be used for any other purpose other than conduct of the clinical investigation without prior written consent from ABC Pharm LP .

Principal Investigator Name (Printed)

Signature

Date

Site Number